GENERATIVE AI · IMAGE & VIDEO

COURSE SLIDES · WSQ

Generative AI for Image and Video Creation


AI-generated course reference image · see courseware/assets/imagegen-provenance.json

WSQ Course Code: TGS-2020505925 · 2 Days · 16 Programme Hours

Skills Framework TSC: Computer Vision Technology (ICT-DIT-4022-1.1)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v11.0 · 6 September 2026

COURSE ADMINISTRATION

Welcome and Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator displays the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance on the SSG record is required to be eligible for assessment and for funding.
- Attendance is taken three times: AM, PM and again immediately before the assessment block on Day 2.

About the Trainer

?

Your Trainer

General Trainer template — to be completed by the trainer delivering this run

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING AND INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Curriculum Developer and Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Curriculum Developer and Subject Matter Expert for this course (Course Proposal CA-WSQ-2020-013290).

BACKGROUND
PhD; two decades of engineering, data science and AI practice across industry and training.

DELIVERS
WSQ courses in computer vision, machine learning, deep learning, generative AI and data analytics.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Any experience you have with image or video editing, photography, or generative AI tools.
- Any experience you have with Python (the labs use it, but every script is supplied and explained).
- One image or video task at your workplace that you would like to be able to specify, automate or quality-check by the end of Day 2.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for funding eligibility.

Download Course Material

https://lms-tms.tertiaryinfotech.com/

Tertiary Infotech LMS/TMS

My Courses · Attendance · Assessment

Generative AI for Image and Video Creation

Trainer Slides (PDF) ⬇

Learner Guide (PDF) ⬇

Hands-On Labs (folder) ⬇

Assessment Submission ↗

1. **Sign in**
   lms-tms.tertiaryinfotech.com — log in with your registered email (OTP or password).

2. **Open your course**
   Select this course under My Courses.

3. **Download materials**
   Slides, Learner Guide and the labs folder — your open-book references for the assessment.

4. **Submit and survey**
   Upload your completed assessment answers here, and complete the TRAQOM survey.

All course material is downloaded from the LMS/TMS portal. Keep it open: both assessment instruments are open book.

Skills Framework Alignment

Element	Detail
TSC Title	Computer Vision Technology
TSC Code	ICT-DIT-4022-1.1
Proficiency Level	Level 4 — set up and deploy video analytics algorithms and perform system performance evaluations
TSC Description	Develop and deploy vision analytics algorithms and spatial sensing and/or reasoning systems
Assessed knowledge	K1 to K13 — assessed in the Written Assessment
Assessed abilities	A1 to A8 — assessed in the Practical Performance

Every topic, lab and assessment item in this course maps to these TSC knowledge and ability statements. Nothing is assessed that is not taught.

WRITTEN ASSESSMENT · 13 QUESTIONS · 60 MINUTES

Assessed Knowledge — K1 to K13

Code	Knowledge statement	Taught in
K1	Vision system concepts	Topic 01
K2	Business applications of vision systems	Topic 01
K3	Methods to represent image and video data	Topic 02
K4	Image and video processing, filtering and transformation methods	Topic 02
K5	Feature extraction and representation techniques	Topic 03
K6	Local feature descriptions, edge, colour, texture and motion	Topic 03
K7	Global feature descriptions, statistical and geometrical methods	Topic 03
K8	Deep learning concepts	Topic 04
K9	Object segmentation, detection and recognition	Topic 04
K10	Activity tracking, generative models, scene understanding and event discovery	Topic 04
K11	Vision system architecture	Topic 05
K12	Vision communication protocols	Topic 05
K13	Real-world design constraints and solution options	Topic 05

The Written Assessment carries exactly one question per knowledge statement, in this order.

PRACTICAL PERFORMANCE · 5 TASKS · 90 MINUTES

Assessed Abilities — A1 to A8

Code	Ability statement	Practised in
A1	Identify the needs of vision systems technology in industrial applications	Lab 01
A2	Apply the principles of processing, filtering and analysis methods for video data	Lab 02, Lab 03, Lab 04, Lab 05
A3	Analyse global feature descriptions	Lab 07
A4	Design and implement feature extraction and representation methods	Lab 06, Lab 07
A5	Design and apply machine-learning based methods for object detection, object tracking and activity recognition	Lab 08
A6	Design and apply video analytics algorithms for high-level video analytics tasks	Lab 09, Lab 10, Lab 11
A7	Design the architecture of applied vision systems	Lab 12
A8	Design, develop and evaluate edge-based and cloud-based systems	Lab 12

PP Task 1 covers A1; Task 2 covers A2 and A3; Task 3 covers A4; Task 4 covers A5 and A6; Task 5 covers A7 and A8.

BY THE END OF THIS COURSE YOU WILL BE ABLE TO

Learning Outcomes

1. **LO1**
   Understand basic vision systems concepts and applications
2. **LO2**
   Apply image processing
3. **LO3**
   Implement feature extraction
4. **LO4**
   Apply machine learning based computer vision methods
5. **LO5**
   Implement video analytics algorithms
6. **LO6**
   Evaluate edge vs cloud-based computer vision systems

THE JOURNEY

Course Outline — Five Delivery Topics

Day 1

- **Topic 01 — Generative AI Fundamentals for Image and Video Creation**
  – LU1 · ELO1 · K1, K2, A1
- **Topic 02 — AI Image Generation, Editing and Visual Enhancement**
  – LU2 · ELO2 · K3, K4, A2
- **Topic 03 — AI-Based Visual Features, Styles and Consistency**
  – LU3 · ELO3 · K5, K6, K7, A3, A4

Day 2

- **Topic 04 — Generative AI Video Creation, Animation and Editing**
  – LU4 + LU5 · ELO4 + ELO5 · K8, K9, K10, A5, A6
- **Topic 05 — Evaluating Cloud and Edge AI Creative Workflows**
  – LU6 · ELO6 · K11, K12, K13, A7, A8
- **12 hands-on labs**
  – Every topic is practised on data you measure yourself, with an evidence checklist per lab

Five delivery topics mapped onto the six approved Learning Units. Topic 04 consolidates LU4 and LU5 concept delivery; both units keep their full practical allocation.

Lesson Plan — Day 1

Time	Min	Componen t	Session
08:45–09:00	15	ADMIN	Course welcome, attendance, introductions and LMS access; outside programme minutes
09:00–10:15	75	CR	Topic1 / LU1: vision and generative pipelines; facilitated Lab01 needs-analysis discussion (K1,K
10:15–10:30	15	BREAK	Tea break; excluded
10:30–11:45	75	CR	Topic2 / LU2: image representation, filtering, geometric transforms and controlled generation/ed
11:45–12:45	60	PP	LU2 practical: Labs02,03,04,05 core exercises, 15 minutes each; extensions completed after class
12:45–13:45	60	LUNCH	Lunch; excluded
13:45–15:00	75	CR	Topic3 / LU3: local/global descriptors, visual style and consistency (K5,K6,K7)
15:00–15:15	15	BREAK	Tea break; excluded
15:15–16:15	60	PP	LU3 practical: Labs06 and07, 30 minutes each (A3,A4)
16:15–17:30	75	CR	Topic4 / LU4: learned vision models, segmentation/detection/recognition and evaluation (K8,K9)
17:30–18:30	60	PP	LU4 practical part1: Lab08 object detection/evaluation (A5)

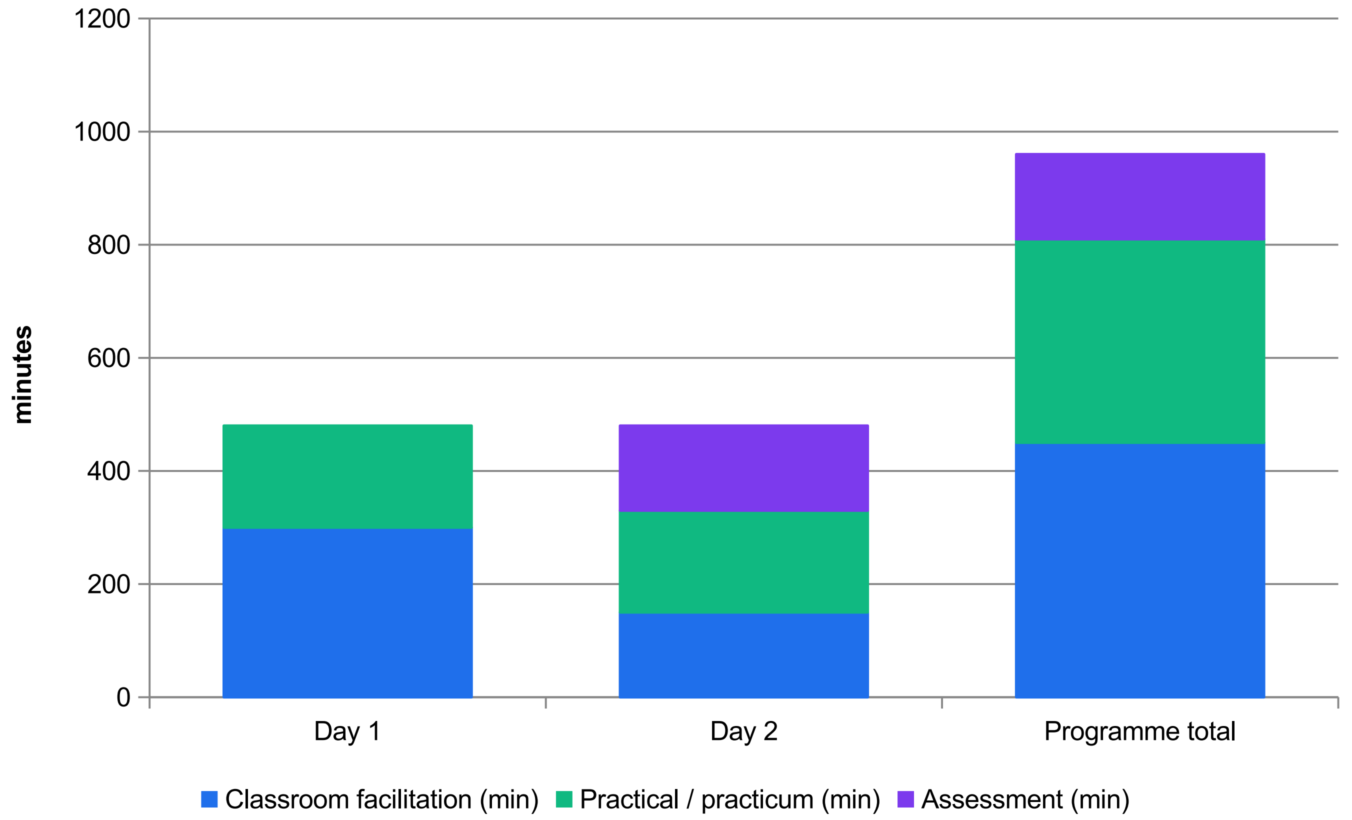
Programme minutes on Day 1: 480 (classroom facilitation + practical + assessment). Lunch, tea breaks and administration are additional, non-instructional time.

Lesson Plan — Day 2

Time	Min	Component	Session
08:45–09:00	15	ADMIN	Attendance and technical setup; outside programme minutes
09:00–10:00	60	PP	LU4 practical part2: Labs09 storyboard/video prompts and11 captions/export, 30 minutes each (A5,
10:00–10:15	15	BREAK	Tea break; excluded
10:15–11:30	75	CR	Topic4 / LU5: temporal consistency, tracking, generative video and event analysis (K10)
11:30–12:30	60	PP	LU5 practical: Lab10 motion, tracking and temporal continuity (A6)
12:30–13:30	60	LUNCH	Lunch; excluded
13:30–14:45	75	CR	Topic5 / LU6: architectures, protocols and measured deployment trade-offs (K11,K12,K13)
14:45–15:00	15	BREAK	Tea break; excluded
15:00–16:00	60	PP	LU6 practical: Lab12 cloud-edge architecture and evaluation (A7,A8)
16:00–16:10	10	ADMIN	Assessment briefing and attendance; outside programme minutes
16:10–17:10	60	AS	Written Assessment: 13 open-ended questions, K1-K13, individual/open book
17:10–18:40	90	AS	Practical Performance: five tasks, A1-A8, individual/open book; submission and sign-off

Programme minutes on Day 2: 480 (classroom facilitation + practical + assessment). Lunch, tea breaks and administration are additional, non-instructional time.

Programme Hours — 16 Hours Reconciled



450 min classroom
7.5 hours across the five topics, 75 min per approved Learning Unit.

360 min practical
6 hours across 12 labs: 0 / 60 / 60 / 180 / 60 by topic.

150 min assessment
WA 60 min + PP 90 min = 2.5 h, per the registered Assessment Plan.

Breaks are additional
1-hour lunch, two 15-minute tea breaks and the administration block sit outside the 960 programme minutes.

MEASURED

Column totals taken from the approved course proposal: 7.5 h classroom facilitation + 6 h practical + 2.5 h assessment = 16 h (960 min). The Lesson Plan schedule sums to exactly these figures.

EVERY FOLDER IS INDEPENDENTLY USABLE

The 12 Hands-On Labs

Lab	Title	Topic	A	K	In class
01	Vision-System Needs Analysis and Generative Pipeline…	01	A1	K1, K2	in LU1
02	Image and Video Data Representation	02	A2	K3	15 min
03	Prompt Experiment Matrix for Controlled Image Generation	02	A2	K3, K4	15 min
04	Masking, Inpainting, Outpainting and Background Swap	02	A2	K3, K4	15 min
05	Filtering, Enhancement and Restoration Measured with PSNR…	02	A2	K4	15 min
06	Local Features: Colour Segmentation, Canny Edges and Corner…	03	A4	K5, K6	30 min
07	Global Descriptors, Template Matching and Brand-Consistency…	03	A3, A4	K7	30 min
08	Object Detection and Quantitative Evaluation	04	A5	K8, K9	60 min
09	Storyboard to Video Prompt Pack and Deterministic Animatic	04	A6	K10	30 min
10	Motion, Tracking and Temporal Continuity Measurement	04	A6	K10	60 min
11	Captions, Overlays, Safe Areas and Export Profiles	04	A6	K10	30 min
12	Cloud-Edge Architecture Design and Measured Trade-offs	05	A7, A8	K11, K12, K13	60 min

Each lab folder carries its own numbered steps, mock data, local media, prompt pack, expected outputs, evidence checklist and troubleshooting. No lab needs a paid service.

TARGET LEARNER PROFILE

Entry Requirements and Who This Is For

1

Knowledge and skills

Computer literacy at ICAS Level 2; minimum 3 GCE 'O' Levels including English, or WPL Level 5.

2

Experience

Minimum 1 year of working experience; minimum 18 years of age.

3

Technical

Windows or Mac laptop, a modern browser, and Python 3 with opencv-python and NumPy installed.

4

Coding

Basic Python is helpful but not assumed — every lab script is supplied, run and explained line by line.

5

Typical roles

Content creator, digital marketer, social media manager, graphic designer, video editor, multimedia producer, instructional designer, marketing communications executive.

6

What you leave with

12 runnable labs, a versioned prompt library, and measurement gates you can put in front of any delivered asset.

Briefing for Assessment

- Place phones and all other materials under the table or on the floor.
- No photographs and no recording of assessment scripts.
- No discussion during the assessment.
- Use a black or blue pen for any hard-copy response.
- No liquid paper and no correction tape.
- Assessment scripts are collected when time is up.
- Take the Assessment digital attendance before the paper starts.

Final Assessment

Instrument	Covers	Duration	Format
Written Assessment WA(Q&A)	K1 to K13 — 13 open-ended short-answer questions, one per knowledge statement	60 min	Individual, summative, open book
Practical Performance (PP)	A1 to A8 — 5 scenario-based tasks (Task 1: A1 · Task 2: A2, A3 · Task 3: A4 · Task 4: A5, A6 · Task 5: A7, A8)	90 min	Individual, summative, open book
Total	Full TSC coverage; assessor-to-candidate ratio 1:3 to 1:10	2.5 hours	Day 2, from 16:10

Open book means the slides, the Learner Guide and approved course materials only. A minimum 75% attendance record and a 'Competent' outcome are both required for funding. An appeal process is available.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA(Q&A) 60 min, then PP 90 min — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

Both instruments are open book and both are individual. The assessor records C or NYC against every knowledge and ability statement.

How Every Figure in This Deck Is Labelled

1

MEASURED

Timed or counted on stated hardware or data, with the sample size and the percentile given. Most numbers in this deck come from the labs you will run.

2

MODELLED

Derived from a documented relationship, with the relationship shown — for example diffusion latency proportional to S steps, or to S x F for F video frames.

3

PUBLISHED

Reported in a cited source, with the source and its page range named on the slide and in the source register.

4

ILLUSTRATIVE

Assumed values used to show the shape of a relationship. Never a quotation, never a forecast, and never to be lifted into a client document.

LABEL THE ARTEFACT, NOT JUST THE INTENT

Provenance Rules for Every Asset You Touch

1

SIMULATED

A deterministic OpenCV rendering made for this course. Never described as a photograph, never as model output. Every such file carries a burned-in banner.

2

AI-GENERATED

Copied unchanged with a PROVENANCE.md recording the tool, the date and the exact prompt. Current Gemini image outputs also carry a SynthID watermark.

3

REAL PRERECORDED

A real captured sequence, recorded with its source URL, retrieval date and SHA-256 so the copy can be verified.

4

ANIMATIC

A deterministic render of storyboard frames, labelled on every frame. It exists to fix timing before generation spend — it is never submitted as a generated film.

01

TOPIC 01

Generative AI Fundamentals for Image and Video Creation

Vision-system concepts and business applications · the analysis-to-synthesis pipeline · artificial neurons, activation functions and CNNs · VAE, GAN and diffusion architectures · what each model can and cannot guarantee

LU1 · ELO1 · K1, K2, A1

Key Concepts — Topic 01

1

A vision system is a five-stage pipeline

Sense (sensor + optics) -> represent (pixel array, colour space) -> analyse (features, detection, tracking) -> decide (thresholds, policy) -> act or synthesise. Generative AI adds the synthesise stage; it does not remove the first four.

2

Classification and generation solve different conditional problems

A discriminative classifier learns p(label | image) or a decision boundary. A conditional generator models p(image | condition). These have different objectives: running a classifier backward does not generally generate an image.

3

Three architectures, three failure signatures

VAE: a regularised latent space; some variants lose fine detail. GAN: potentially sharp output, with risks of mode collapse and unstable training. Diffusion: potentially high fidelity, with cost from iterative sampling.

4

A neuron is a weighted sum plus a non-linearity

y = f(sum(w_i x_i) + b). Without the non-linearity f, any depth of stacked layers collapses to a single linear map. ReLU, sigmoid, tanh, softmax and GELU are the ones you will see in vision models.

5

A CNN learns the filters that OpenCV hand-codes

A 3x3 Sobel kernel is a hand-designed edge detector; a CNN's first-layer kernels can learn edge and colour-blob responses from training data. Related local filtering, learned instead of specified.

6

Business value comes from throughput, not novelty

Costed use cases: variant generation for catalogue and social, background replacement for product shots, format repurposing, storyboard previsualisation, synthetic data for detector training, and automated QC of the output.

A Vision System Is a Five-Stage Pipeline

1 SENSE

Sensor and optics. Resolution, frame rate, exposure, mounting. Everything downstream inherits the limits set here.

→

2 REPRESENT

Pixel array, dtype, channel order, colour space, storage. H x W x C uint8 in BGR is what OpenCV hands you.

→

3 ANALYSE

Features, segmentation, detection, tracking. Turns pixels into measurements a decision can be made on.

→

4 DECIDE

A numeric threshold and what happens either side of it: publish, queue for review, or reject and regenerate.

→

5 SYNTHESISE

Create or edit a new asset. This is the stage generative AI adds — it does not remove the first four.

SO WHAT Generative AI extends the pipeline; it does not replace it. Every asset a model produces still has to be represented, analysed against a measurement and decided on before it ships.

K1 · WHY THIS COURSE TEACHES BOTH

Analysis and Synthesis Are the Same Maths, Reversed

Pixels → meaning

- **Analytical computer vision**
  - Input: pixels. Output: a label, a box, a mask or a number.
  - Classifier: p(label | image)
  - Trained by minimising error against labelled truth.
  - Evaluated with accuracy, precision, recall, IoU, mAP.
  - Fails visibly: a wrong label is checkable.

Meaning → pixels

- **Generative computer vision**
  - Input: a prompt, an image, a mask. Output: pixels.
  - Sampler: draws from p(image | condition)
  - Trained by learning to reverse a corruption process.
  - Evaluated with FID, KID, LPIPS, CLIP faithfulness, FVD.
  - Fails plausibly: a wrong image still looks like an image.

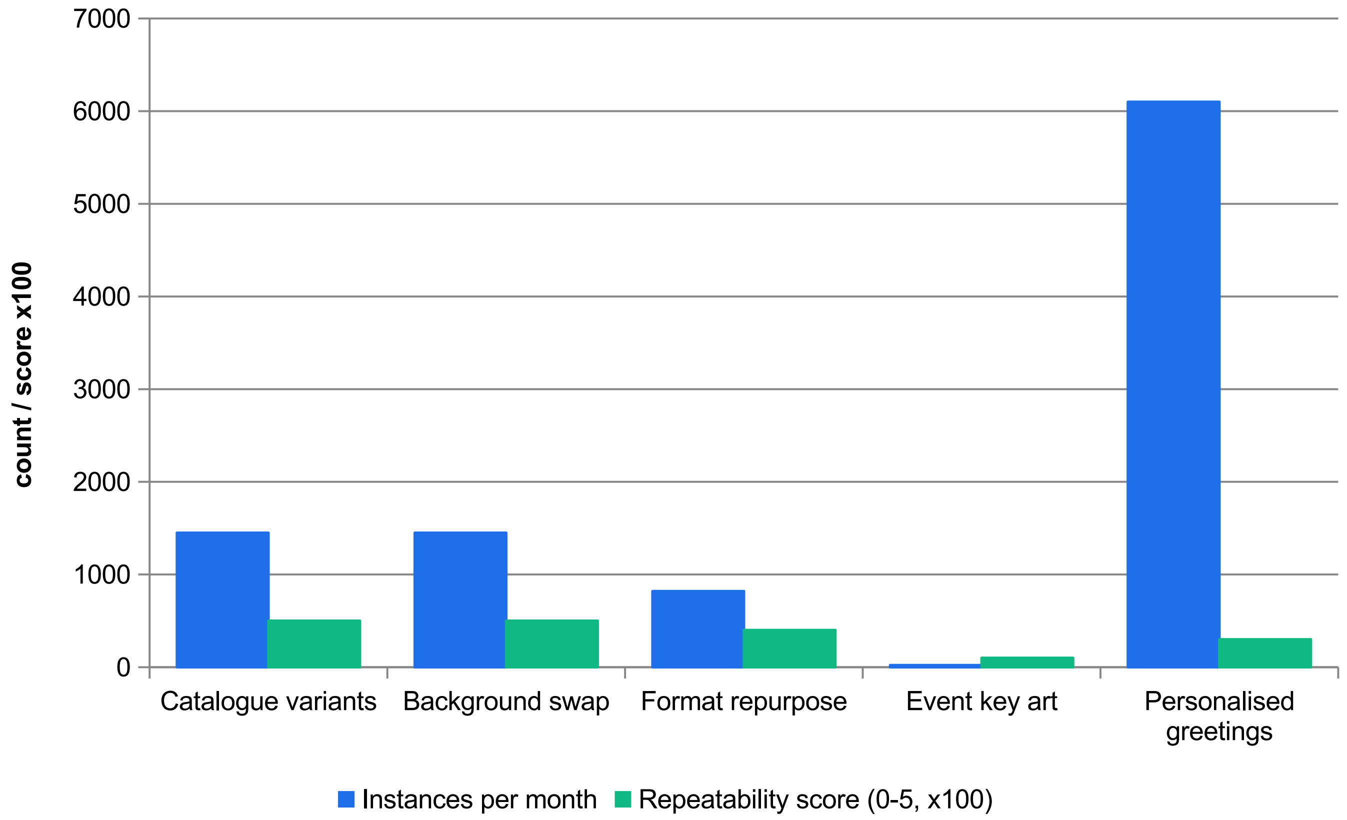
That last row is the reason the analytical half of this course exists: a generated asset that is wrong is not obviously wrong, so it has to be measured.

Business Applications of Vision Systems

Application	What the system does	What it replaces
Catalogue variant generation	Produce N approved variants of one master shot against a fixed brief	Repeat studio shoots
Background replacement	Mask the subject and re-render the setting	Location hire and travel
Format repurposing	Outpaint or crop one master into a delivery ladder	Manual re-layout per platform
Storyboard previsualisation	Turn a shot list into reviewable frames before any shoot	Whole-crew concept days
Automated QC	Score every delivered asset against a brand reference	Eyeball review of every file
Shelf and stock monitoring	Detect gaps and count facings from fixed cameras	Manual store walks
Synthetic training data	Generate labelled examples for a rare defect class	Waiting for the defect to occur

Every row is a task that is visual, repeated, specifiable and currently expensive. Those four properties are the need test you apply in Lab 01.

Where the Value Actually Comes From



Volume alone is not enough

Personalised greetings are the highest volume case and still fail — they carry personal data.

Repeatability is the killer

Event key art scores 1/5: every instance is bespoke, so there is nothing to specify.

The winners score on both

Catalogue work is high volume AND fully specifiable — that is what pays back.

Constraints are gates

Privacy and latency eliminate cases before any cost argument is made.

ILLUSTRATIVE

Figures are the synthetic NorthQuay case data used in Lab 01 (labs/lab-01-.../data/candidate-use-cases.csv). They are classroom values, not market data.

The Six Need Criteria You Will Score

1

VOLUME

5 if the task repeats 500+ times a month, 3 for 50–499, 1 below 50. A vision system earns its build cost through throughput.

2

REPEATABILITY

5 if framing, subject class and output format are identical every time. Low repeatability is the most common reason a vision project fails to pay back.

3

LATENCY CLASS

Scored inverted against difficulty: batch = 5, interactive = 3, hard real-time = 1. A high score means the requirement does not constrain your architecture.

4

ACCURACY TOLERANCE

5 where a human reviews every output, 1 where a wrong output reaches a customer or a safety decision unreviewed.

5

PRIVACY RISK

Inverted: 5 where no identifiable person is involved, 1 where customer faces or personal data are unavoidable. This one is a gate, not a weight.

6

COST GAP

(manual − automated) / manual. 5 above 0.6, 3 for 0.3–0.6, 1 below. A classroom scoring boundary, not a measured payback point.

A1 · HOW TO READ A SCORING MATRIX

A privacy classification is a gate, not a weighting.

Eliminate on constraint first, then optimise on cost among what survives. A case that scores 30 out of 32.5 and moves customer faces off the store network is still rejected.

Three Generations of Computer Vision

Hand-designed

- You write the kernel: Sobel, Canny, Harris.
- Deterministic and inspectable.
- Fast, no training data, no GPU.
- Breaks when the scene changes.
- Still the right tool for a fixed rig.

Learned (discriminative)

- The network learns the kernels from labels.
- CNNs, Viola-Jones cascades, HOG + SVM.
- Generalises across scenes.
- Needs labelled data and evaluation.
- Evaluated with IoU, precision, recall.

Learned (generative)

- The model learns the data distribution.
- VAE, GAN, diffusion, diffusion transformer.
- Creates new pixels from a condition.
- Needs distributional and perceptual metrics.
- Fails plausibly, so it needs a gate.

K8 FOUNDATION · INTRODUCED HERE, ASSESSED IN TOPIC 04

An Artificial Neuron

The unit

$$y = f\left(\sum w_i x_i + b \right)$$

Weighted sum of the inputs, plus a bias, passed through a non-linearity f. The weights and the bias are what training changes.

Why f matters

$$W_2(W_1x) = (W_2W_1)x$$

Without f, any depth of stacked linear layers collapses to a single linear map. The non-linearity is what makes depth mean anything.

A layer

$$y = f(Wx + b)$$

W is a matrix, so one layer is many neurons at once. A 4096 x 4096 layer holds 16,777,216 weights — a number you will meet again in Topic 05.

You do not train a network in this course. You do need to explain what the parts are, because Topic 05's memory and adapter arithmetic is built on them.

Activation Functions You Will Meet

Function	Definition	Range	Where it shows up
ReLU	max(0, x)	[0, ∞)	Hidden layers of most CNNs — cheap and non-saturating
Leaky ReLU	x if x>0 else αx	(−∞, ∞)	GAN discriminators, where a dead ReLU kills the gradient
Sigmoid	$1 / (1 + e^{-x})$	(0, 1)	Binary output — a GAN discriminator's real/fake score
Tanh	$(e^{x} - e^{-x})/(e^{x} + e^{-x})$	(−1, 1)	Generator output layers, matching images scaled to [−1, 1]
Softmax	$e^{x_i} / \Sigma\, e^{x_j}$	(0, 1), sums to 1	Multi-class output — the class probabilities of a classifier
GELU	$x \cdot \Phi(x)$	(−∞, ∞)	Transformer blocks, including diffusion transformers

The Written Assessment asks you to name three activation functions used by an artificial neuron. Naming them is the minimum; saying where each is used is what a competent answer looks like.

Convolution Is the Whole Idea — Worked by Hand

```
# a 3x3 kernel slides over the image; each output pixel is a weighted sum
#
#   dst(x,y) = SUM over i,j of  K(i,j) * src(x+i, y+j)

import numpy as np, cv2

src = np.array([[ 10,  10,  10, 200, 200],
                [ 10,  10,  10, 200, 200],
                [ 10,  10,  10, 200, 200]], np.float32)

sobel_x = np.array([[-1, 0, 1],
                    [-2, 0, 2],
                    [-1, 0, 1]], np.float32)

# centre pixel of column 2: the left column is 10s, the right column is 200s
#   = (-1*10) + (1*10) + (-2*10) + (2*200) + (-1*10) + (1*200)
#   = -10 + 10 - 20 + 400 - 10 + 200 = 570   -> a strong vertical edge

out = cv2.filter2D(src, -1, sobel_x)
print(out[1, 2])            # 570.0
```

A kernel is a small matrix

Change the numbers and you change what the filter detects. Nothing else changes.

Sum to 1 to preserve brightness

A 5x5 box blur is 25 cells of 1/25. A kernel that does not sum to 1 changes the image's mean level.

A CNN learns these numbers

The first-layer kernels of a trained CNN converge to edge and colour-blob detectors on their own.

Same operation, both worlds

cv2.filter2D and a convolutional layer do the same arithmetic; one is specified, one is learned.

VERIFY You can compute this cell by hand and get 570. Do it once — it is the only way the rest of the course stops being magic.

Convolutional Neural Network — What Each Layer Does

INPUT	H x W x 3 pixel array, usually normalised to [0,1] or [−1,1].
CONV + ACTIVATION	Learned kernels convolve the input; a non-linearity follows each. Early layers find edges and colour blobs; later layers find parts and objects.
POOLING	Downsamples spatially (max or average), which shrinks the feature map and buys a little translation tolerance.
FLATTEN / GLOBAL POOL	Collapses the spatial map into a feature vector — the image's learned representation.
FULLY CONNECTED + SOFTMAX	Maps the vector to class scores, then to probabilities that sum to one.

Depth buys abstraction
Stacked convolutions grow the receptive field, so later layers see more of the image.

Weight sharing buys efficiency
One kernel is reused across the whole image instead of a weight per pixel.

The feature vector is reusable
It is what a classifier head consumes, and what an FID or CLIP score is computed in.

This is the encoder half
A generator runs the same idea in reverse: vector in, pixels out.

The encoder/decoder split you see here is exactly the structure a VAE, a GAN generator and a diffusion U-Net all reuse.

Variational Autoencoder — Encoder, Latent Space, Decoder

ENCODER	Compresses the image to the parameters of a distribution: a mean μ and a log-variance $\log \sigma^2$, not a single point.
LATENT SPACE	A low-dimensional, smooth, continuous space. Nearby points decode to similar images, which is what makes interpolation work.
SAMPLE	Draw z from that distribution. This is what makes the model generative rather than a plain autoencoder.
DECODER	Maps z back to a full image, trying to reconstruct the original.

Strength
A smooth latent space — good for interpolation and for compression.

Weakness
Blurry output. The reconstruction loss rewards averaging over possible pixel values.

Still used
As the autoencoder inside latent diffusion, where the blur does not matter.

Cited
S6 · Multimodal Generative AI, Ch.5 pp.56–58

The blur is not a bug to be tuned away: it is what a loss that rewards average correctness produces.

The Reparameterisation Trick

The problem

$$z \sim N(\mu, \sigma^2)$$

Sampling is a random step. Gradients cannot flow back through it, so the encoder never learns whether a bad output was its fault or the sampler's.

The fix

$$z = \mu + \sigma \odot \varepsilon$$
$$\varepsilon \sim N(0, 1)$$

Move the randomness outside the path. ε is drawn from a fixed standard normal; μ and σ are ordinary differentiable outputs.

Why it works

$$\partial z/\partial \mu = 1$$
$$\partial z/\partial \sigma = \varepsilon$$

The path from encoder to output is now unbroken, so backpropagation reaches the encoder and both halves learn together.

Source: S6 · Multimodal Generative AI in the Enterprise, Ch.5 pp.57–58.

Generative Adversarial Network — Two Networks in a Zero-Sum Game

GENERATOR G

Takes a random noise vector and transforms it into an image that could plausibly have come from the training set.

DISCRIMINATOR D

A binary classifier. Outputs a probability: 1 means 'real', 0 means 'fake'.

ADVERSARIAL LOOP

D is trained to catch fakes; G is trained to fool D. They alternate, and both improve.

EQUILIBRIUM

In theory, G's output is so realistic that D is no better than chance. In practice, reaching it is hard.

Strength

Sharp, high-fidelity images — far better than a VAE at the same size.

Weakness

Training is notoriously unstable and hard to balance.

Signature failure

Mode collapse: G finds a few outputs that fool D and stops producing variety.

Cited

S6 · Multimodal Generative AI, Ch.5 pp.59–60

Introduced by Goodfellow et al. in 2014. Still the right choice where single-pass sampling speed matters more than diversity.

Mode Collapse — What It Looks Like and Why It Happens

Symptom

- **What you see**

– Every sample looks like a handful of near-identical images.

– Variety collapses even though quality stays high.

– Example: the generator discovers that dark-haired faces are reliably scored 'real', and stops producing any other hair colour.

– Sample-quality metrics such as FID can still look acceptable.

Mechanism

- **Why it happens**

– G is rewarded for fooling D, not for covering the data distribution.

– A narrow set of outputs that reliably fools D is a local optimum.

– The adversarial signal is indirect: G only learns whether D was fooled by the whole sample.

– This is exactly why precision-and-recall metrics for generators exist — they detect mode dropping when FID does not.

Sources: S6 Ch.5 p.60 (mode collapse); S17 Ch.7 pp.96–97 (precision and recall for generators detect mode dropping even when FID is similar).

Diffusion — Forward Corruption, Reverse Generation

FORWARD (FIXED)

Add a small amount of Gaussian noise, repeatedly, as a Markov chain. No learning happens here at all — it is a known, fixed procedure.

AFTER T STEPS

The image has become indistinguishable from pure noise. Every intermediate state was recorded.

TRAIN

Show the network a noisy state and its timestep; train it to predict the noise that was added. One clear, stable objective.

REVERSE (LEARNED)

Start from pure noise. Predict the noise, subtract a step's worth, repeat. Each step makes the image slightly cleaner.

SAMPLE

After S steps a coherent image emerges that was never in the training set.

SO WHAT The training objective is direct and stable at every step — predict the noise — which is why diffusion scales where GANs did not. The price is inference speed: you must walk back through every step.

The Diffusion Objective

Forward step

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \varepsilon$$

Any noisy state can be produced directly from the clean image and a noise draw, so training does not have to simulate the whole chain.

Training loss

$$L = \mathbb{E}[\ \lambda_t \cdot \|\varepsilon - \varepsilon_\theta(x_t, t, c)\|^2\ ]$$

Predict the noise ε that was added, given the noisy image, the timestep t and the condition c. A plain regression — that is the whole objective.

Sampling cost

```
lat ∝ S              (image)
lat ∝ S · F          (video)
```

S denoising steps for an image; S x F for F video frames. This single relationship drives the whole cloud-versus-edge argument in Topic 05.

Sources: S17 Ch.7 pp.96, 101 (objective); S18 Ch.7 pp.98–99 (latency proportional to S, and to S x F for video).

K1 · PUBLISHED COMPARISON

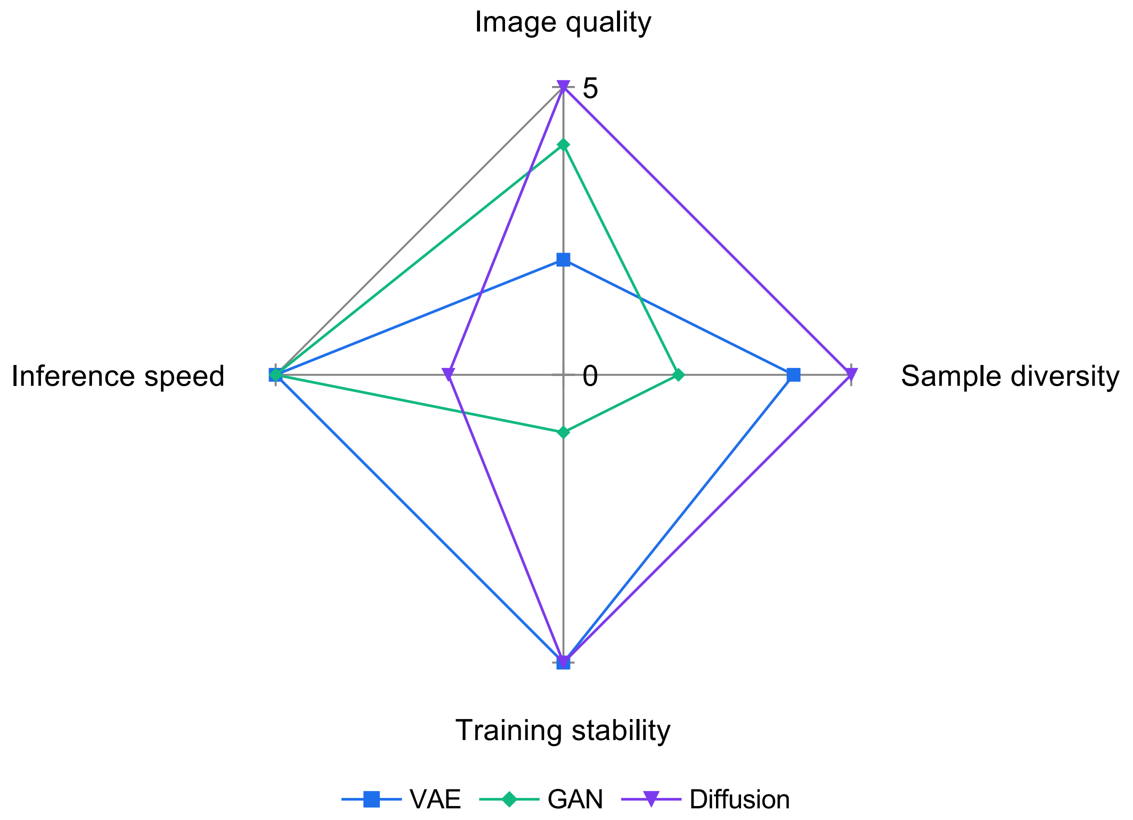
Comparing the Three Architectures

Property	VAE	GAN	Diffusion
Image quality	Lower fidelity, blurry	High fidelity, sharp	State of the art
Sample diversity	High — covers the distribution	Low — prone to mode collapse	High — avoids mode collapse
Training stability	Stable, tractable loss	Notoriously unstable	Very stable, simple loss
Inference speed	Fast — one forward pass	Fast — one forward pass	Slow — many iterative steps
Primary use	Latent representation, compression	High-resolution synthesis, style transfer	High-quality text-to-image and video

Source: S6 · Multimodal Generative AI in the Enterprise, Ch.5 p.62, Table 5.1. Reproduced as a comparison of properties, not as a recommendation.

K1 · SAME FOUR PROPERTIES, ORDINAL SCORES

The Trade-off, Drawn



No architecture wins everywhere
Diffusion trades inference speed for everything else.

That trade is attackable
Step distillation and latent-space modelling exist precisely to buy the speed back — Topic 05.

GANs are not obsolete
Single-pass sampling still matters where latency dominates.

VAEs did not disappear
They became the autoencoder inside latent diffusion.

ILLUSTRATIVE

Ordinal scores from 1 to 5 assigned by the course author to the qualitative comparison published in S6 Ch.5 p.62, Table 5.1. They show the SHAPE of the trade-off; they are not measurements and must not be quoted as such.

K1 · THE HONEST CAPABILITY STATEMENT

What Each Model Can and Cannot Guarantee

1 **Can: plausibility**
A diffusion model reliably produces something that looks like the training distribution.

2 **Cannot: factual correctness**
Nothing in the objective rewards being right. A rendered label, a count or a mechanism can be confidently wrong.

3 **Can: conditioning**
Text, an image, a mask, a depth map or a reference set can all steer the sample.

4 **Cannot: exact reproduction**
Same prompt, different sample. Repeatability comes from pinning the configuration, not from hoping.

5 **Can: local edits**
With a mask, the preserved region is guaranteed by construction rather than by the model's good behaviour.

6 **Cannot: knowing it is wrong**
There is no confidence output you can threshold. That is what the measurement gates in Topics 02–04 are for.

K1 · A PRACTICAL RULE

Generated text inside an image is the canary.

If a model renders a caption, a logo or a price wrongly — and it usually does — that is a visible instance of the same failure that is invisible everywhere else in the frame. Put text on in post, and measure the rest.

Worked Example — Scoring One Candidate Use Case

Criterion	Evidence from the case file	Score	Weight	Contribution
Volume	1,450 instances / month (≥ 500)	5	1.00	5.00
Repeatability	Studio rig, identical framing every time	5	1.50	7.50
Latency class	Batch, 48-hour turnaround	5	0.75	3.75
Accuracy tolerance	Every output reviewed before publication	5	0.75	3.75
Privacy risk	No identifiable person in frame	5	1.25	6.25
Cost gap	(9800 − 2600)/9800 = 0.735 (≥ 0.6)	5	1.25	6.25
TOTAL	Maximum possible is 32.5			32.50

Scoring 32.5 out of 32.5 is not approval. The case still has to pass the cost gate: 2600/1450 = S$1.79 per asset against a S$1.20 ceiling. Score then gates — never the other way round.

A1 · THE DELIVERABLE FROM LAB 01

The Five-Stage Pipeline Specification You Will Write

1

1 SENSE

Source (catalogue asset, store camera, or text prompt), resolution and rate. For a text prompt, mark the pixel fields not applicable and state the intended output dimensions.

2

2 REPRESENT

Colour space, working resolution, and where the asset is stored.

3

3 ANALYSE

The specific named measurement that will gate quality — an HSV histogram distance against the brand reference, an edge density, a detector recall, a PSNR or SSIM value.

4

4 DECIDE

The numeric threshold, and what happens on each side of it: auto-publish, human review queue, or reject and regenerate. A pipeline without a threshold is a demo.

5

5 ACT / SYNTHESISE

The output artefact, its formats and aspect ratios, who consumes it, and whether the stage is ANALYTICAL or GENERATIVE.

Image Classification, Worked End to End

INPUT

A 224x224x3 array, scaled to [0,1] or [-1,1]. Every model states the normalisation it was trained with; using a different one silently degrades accuracy.

FEATURE EXTRACTION

Stacked convolutions and pooling reduce the array to a feature vector — the image's learned representation.

CLASSIFIER HEAD

A fully connected layer maps that vector to one score per class. Raw scores, not probabilities.

SOFTMAX

$e^{x_i} / \Sigma e^{x_j}$ turns the scores into probabilities that sum to 1. '85% cat' means 85% of the probability mass, not 85% confidence.

DECISION

argmax gives the label; the runner-up margin tells you how close the call was. A 0.51/0.49 split and a 0.99/0.01 split both report the same label.

SO WHAT Softmax always produces a confident-looking distribution, even for an input from a class the model has never seen. Always look at the margin, not just the top label.

Latent Space — Why Modern Models Do Not Work on Pixels

Working on pixels

- **Pixel space is expensive**
  - A 1024x1024 RGB image is 3,145,728 values.
  - Diffusion has to process that array once per denoising step.
  - At 25 steps that is 25 full passes over three million values, per sample.
  - Memory traffic, not arithmetic, becomes the bottleneck.

Working on latents

- **Latent space is compressed**
  - An autoencoder maps the image to a much smaller array first.
  - Diffusion runs entirely inside that compressed representation.
  - The decoder expands the final latent back to full resolution once.
  - This is why the VAE did not disappear — it became the encoder and decoder inside latent diffusion.

Source S17 Ch.7 p.95: latent diffusion is listed as the scaling strategy for resolution, with lower bandwidth cost at higher resolutions as the typical gain. It returns in Topic 05 as a deployment lever.

How a Prompt Actually Reaches the Pixels

1

1 TOKENISE

The prompt is split into tokens and embedded as vectors. A misspelled brand name becomes different tokens and therefore a different condition.

2

2 TEXT ENCODER

A language model turns the token sequence into a conditioning representation — the semantic content of what you asked for.

3

3 CROSS-ATTENTION

At each denoising step the image model attends to that representation, so the text steers every step, not just the first.

4

4 GUIDANCE

Classifier-free guidance amplifies the difference between the conditioned and unconditioned prediction. Higher guidance means more literal, and less diverse.

5

5 DECODE

The final latent is expanded to pixels by the decoder.

6

Why word order matters

Attention weights are not a bag of words. Repetition and contradictory instructions measurably degrade the result — which is what Lab 03 sets out to quantify.

K1 · SOURCES S6 CH.5 PP.56–61, S14 CH.1 PP.1–5

How the Field Got Here

When	What changed	Why it mattered
1970s	AARON — a symbolic, rule-based art program	Creativity bounded by the rules its author coded by hand
2012	AlexNet wins the ImageNet challenge	Proved deep networks learn visual patterns directly from data; the field moved from hand-coded rules to learned realism
2014	Generative adversarial networks introduced	Sharp, high-fidelity synthesis for the first time — at the cost of unstable training
2015	Diffusion first proposed	Largely overlooked for six years
~2021	Diffusion becomes practical at scale	Stable training plus high fidelity; becomes the backbone of modern text-to-image
Now	Diffusion transformers and natively multimodal models	Longer coherent video, and text-and-image reasoning inside a single model

The dates matter less than the pattern: each architecture was introduced to fix the specific failure of the one before it.

TOPIC 01 · ELO1 · A1

Lab 01 — Vision-System Needs Analysis and Generative Pipeline Specification

INPUTS

Synthetic reference frame representing a NorthQuay store-shelf camera view (deterministically rendered with OpenCV — labelled SIMULATED, not a photograph and not generative-model output)

TECHNICAL QUESTION

Which use case meets both the weighted criteria and the hard cost/privacy constraints?

EVIDENCE TO PRODUCE

A completed needs-analysis scoring matrix, a written rejection rationale, and a labelled five-stage pipeline specification for the two selected use cases.

LAB 01 · IN-CLASS · Spreadsheet or text editor · Python 3 (optional scorer) · no paid service required

Detailed procedures and prompts: Learner Guide / labs/lab-01-vision-needs-analysis/README.md

Recap — Topic 01

- A vision system is sense → represent → analyse → decide → act or synthesise. Generative AI adds the fifth stage; it does not remove the first four.
- Analysis and synthesis learn the same joint distribution and differ only in the direction of sampling — which is why a generated asset still has to be measured.
- VAE, GAN and diffusion trade image quality, sample diversity, training stability and inference speed against each other; no architecture wins on all four.
- A neuron is a weighted sum plus a non-linearity, and a CNN learns the kernels that classical computer vision hand-codes.
- A vision system is needed where the task is visual, repeated at volume, specifiable, and currently expensive — and where no constraint gate blocks it.
- You can now score a candidate use case on six criteria and specify its five-stage pipeline with a named measurement and a numeric threshold. (A1, K1, K2)

02

TOPIC 02

AI Image Generation, Editing and Visual Enhancement

How image and video data is represented · convolution, filtering and geometric transformation · prompt structure and controlled variance · masking, inpainting, outpainting and background swap · enhancement measured with PSNR and SSIM

LU2 · ELO2 · K3, K4, A2

Key Concepts — Topic 02

1 An image is a typed 3-D array

OpenCV returns H x W x C uint8 in BGR order. shape, dtype and channel order are the three facts that cause most integration bugs; matplotlib expects RGB.

2 A video is frames plus a clock

cap.get(cv2.CAP_PROP_FRAME_COUNT) / FPS gives duration; a 1080p 30 fps RGB stream is 1920*1080*3*30 = 186.6 MB/s uncompressed. That number drives every cloud-vs-edge decision later.

3 HSV separates hue from saturation and brightness

For uint8 OpenCV HSV, H is 0-179 (degrees/2), S and V are 0-255. Hue can be more stable than raw RGB under brightness changes, but low saturation, shadows and colour casts still cause failures. Test the actual lighting range.

4 Kernel filtering combines neighbouring pixels

dst(x,y) = sum over the kernel of K(i,j)*src(x+i, y+j). A 5x5 box kernel of 1/25 blurs; Sobel approximates a first derivative; Laplacian a second derivative. Sharpening combines a scaled derivative response with the original. cv2.filter2D performs correlation; flip an asymmetric kernel for convolution.

5 Geometric transforms are matrix multiplies

Translation and rotation are 2x3 affine matrices used by warpAffine; perspective is a 3x3 homography used by warpPerspective. Affine preserves parallel lines; perspective preserves only straightness.

6 Generative editing is masked, conditioned denoising

These labs use white = edit and black = preserve for binary masks; provider interfaces may differ. Check their contract and verify unmasked output regions. Inpainting fills inside the frame; outpainting extends the canvas; background replacement targets the complement of the subject.

An Image Is a Typed 3-D Array

```
import cv2

img = cv2.imread('reference/nq-product-flatlay.png')
if img is None:
    raise SystemExit('cannot read the file')   # imread does NOT raise

print(img.shape)      # (720, 1280, 3)   -> height, width, channels
print(img.dtype)      # uint8            -> 0..255 per channel
print(img[10, 10])    # [255   0   0]    -> B, G, R  (NOT R, G, B)

# indexing is [row, column] = [y, x] — the opposite order to a coordinate
print(img[100, 200])  # the pixel at x=200, y=100
```

shape is (H, W, C)

Height first. Getting this backwards is the single most common OpenCV bug.

dtype is uint8

Add a float array without clipping and values wrap modulo 256 — the image fills with speckle.

Channel order is BGR

Pixel (10,10) in the supplied calibration patch is pure blue, so a correct read shows a high FIRST element.

imread returns None on failure

It does not raise. Always guard, or you get a NoneType error fifty lines later.

VERIFY Lab 02 verifies all four of these on the supplied file before anything else is done to it.

K3 · WHY THE CHANNELS ARE NOT EQUAL

Grayscale Is a Weighted Sum, Not an Average

OpenCV's conversion

```
Y = 0.299·R + 0.587·G + 0.114·B
```

Green dominates because human vision is most sensitive to it. An unweighted mean of the three channels is a different, wrong image.

Check it yourself

```
0.587 × 255 = 149.7
```

The pure-green calibration patch in the Lab 02 reference file converts to a grayscale value of about 150. Read the pixel and confirm it.

What you lose

```
3 channels → 1
```

Two-thirds of the data, and all hue information. A red and a green object of the same luminance become indistinguishable.

Every detector in Topic 04 takes grayscale input, so this conversion sits under work you will do two topics from now.

Three Colour Models, Three Jobs

For capture and display

- **RGB / BGR — additive light**
  - How a display emits colour and how a sensor records it.
  - Three channels that all change when the light changes.
  - Bad for thresholding: an exposure drop moves all three.
- **CMYK — subtractive ink**
  - How a printer lays down colour. Not used in this course, but it is why a print proof never matches a screen exactly.

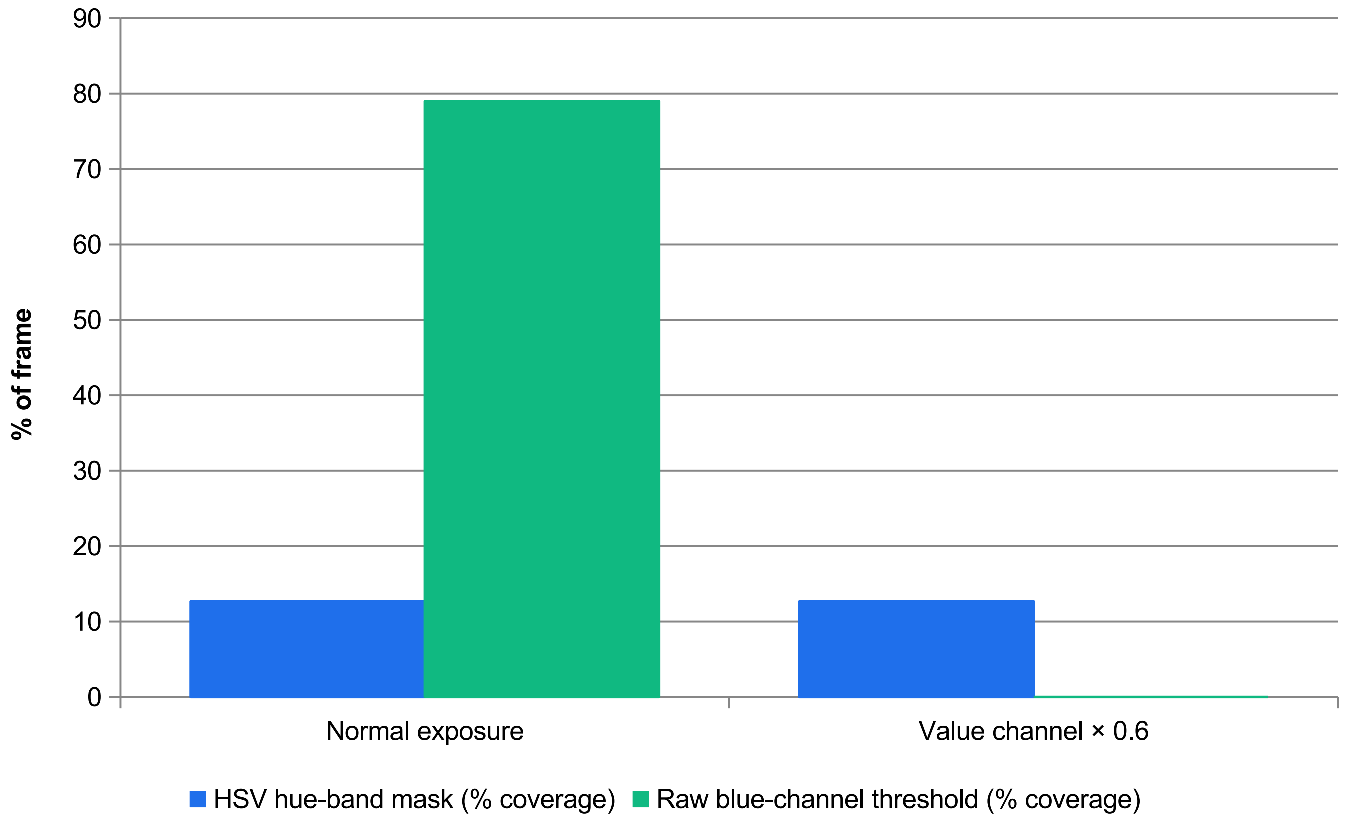
For analysis

- **HSV — how people describe colour**
  - Hue: which colour. OpenCV 8-bit range is 0–179 (degrees halved so the value fits a byte).
  - Saturation: how much grey is mixed in. 0–255.
  - Value: how bright. 0–255.
  - Good for thresholding: hue survives an exposure change, so a mask built on it does not collapse when the lighting shifts.

Lab 06 measures this claim: under a 40% exposure drop the HSV brand mask held at 12.66% coverage while a raw blue-channel threshold collapsed from 79.0% to 0.0%.

K3 · THE MEASUREMENT BEHIND THE CLAIM

HSV Versus a Raw-Channel Threshold Under an Exposure Drop



HSV: no change at all

12.66% before, 12.66% after. Hue is unchanged by a uniform brightness scale.

Raw channel: total collapse

79.0% to 0.0%. Every pixel fell below the fixed threshold.

Not a universal law

Clipping, noise and a white-balance shift can still move hue. Test on your own deliveries.

Why it matters here

A brand-colour gate built on a raw channel will silently pass or fail everything after a lighting change.

MEASURED

Measured in Lab 06 on the supplied synthetic master with opencv-python 4.13. The V channel was scaled by 0.6 to simulate an under-exposed delivery; the same threshold was then re-run.

Video Is Frames Plus a Clock

```
cap = cv2.VideoCapture('reference/nq-shelf-pan.mp4')
w   = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))     # 640
h   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))    # 360
fps = cap.get(cv2.CAP_PROP_FPS)                  # 25.0
n   = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))     # 150  (from the container)

decoded = 0
while True:
    ok, frame = cap.read()
    if not ok:
        break
    decoded += 1
print(n, decoded)        # trust the LOOP, not the property

# uncompressed byte rate = W * H * channels * bytes-per-channel * fps
#   640 x 360 x 3 x 25  =  17,280,000 B/s  =  17.3 MB/s
# 1920 x 1080 x 3 x 30  = 186,624,000 B/s  = 186.6 MB/s
```

Properties are metadata

CAP_PROP_FRAME_COUNT is read from the container index and is backend-dependent. It can disagree with what decodes.

Duration = frames / fps

Not from the filename, and not from what the client said.

17.3 MB/s for one small clip

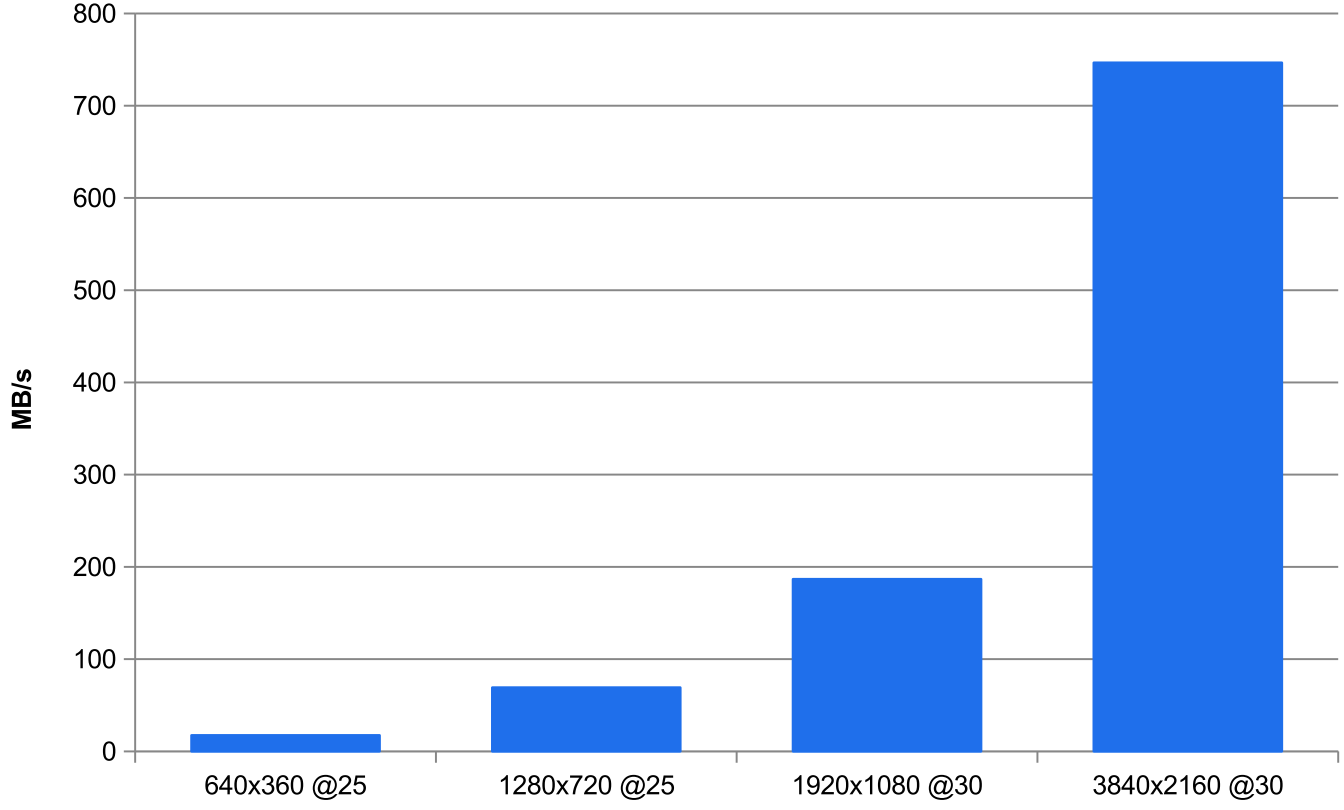
Multiply by 41 stores in Topic 05 and the raw-frame architecture becomes arithmetically impossible.

Compression ratio measured

The supplied clip encodes to about 284 kB for 6 s — roughly 365:1 against uncompressed.

VERIFY Lab 02 writes all of these into out/representation-report.json and compares them against a supplied expected-value file.

Uncompressed Byte Rate by Format



Scaling is brutal
4K at 30 fps is 43x the small clip. Resolution enters squared.

Compression is not optional
The measured encode of the course clip is roughly 365:1.

It is per camera
One store camera at 1080p30 exceeds a 20 Mbit/s uplink by more than 70x.

This is Topic 05's split A
Sending raw frames to the cloud is included there precisely so it can be shown to be infeasible.

MODELLED

Computed directly from width × height × 3 bytes × frame rate. Arithmetic, not a measurement — but the arithmetic is what eliminates a whole architecture in Topic 05.

K4 · IMAGE AND VIDEO PROCESSING, FILTERING AND TRANSFORMATION

Filtering is convolution. Transformation is a matrix multiply.

Two operations account for almost everything OpenCV does to an image before a model ever sees it — and both are things you can compute by hand on a small example.

K4 · SIX OPERATIONS, SIX DIFFERENT JOBS

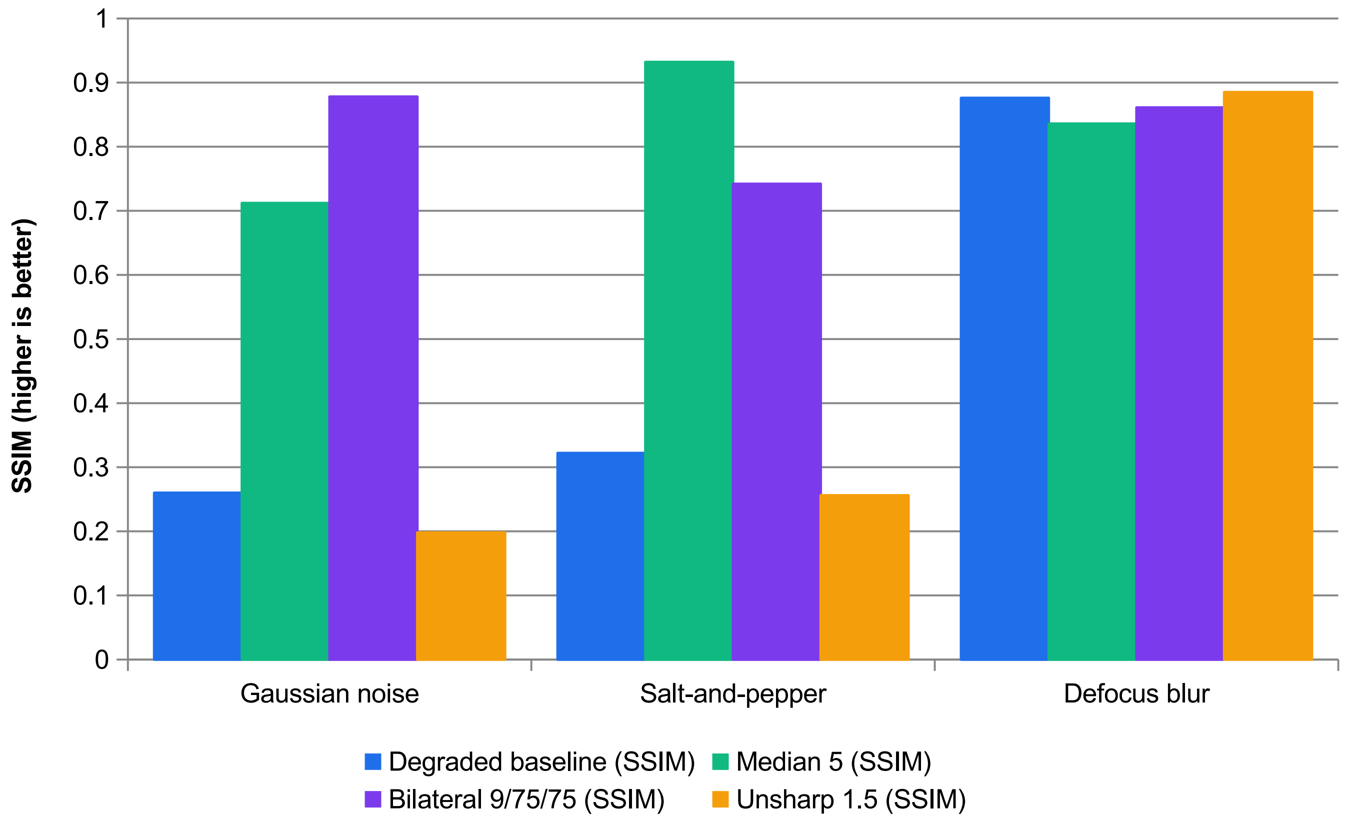
The Filter Bank — What Each One Is For

Filter	Kernel / mechanism	Best against	Cost
Box blur 5x5	25 cells of 1/25 — a plain mean	Nothing in particular; the reference case	Very low
Gaussian blur	Weights fall off with distance from the centre	Gaussian sensor noise	Low
Median	Rank filter — replaces each pixel with the neighbourhood median	Salt-and-pepper impulse noise	Low
Bilateral	Weights by spatial distance AND intensity difference	Noise, while keeping edges	High
Unsharp mask	original × 1.5 + blurred × (−0.5) — adds the high-frequency residual back	Soft focus on a clean image	Low
Morphology	Erode, dilate, open, close with a structuring element	Cleaning a binary mask	Very low

A median is immune to an outlier that a mean is not — that single property is why it dominates on impulse noise. Lab 05 measures the margin.

K4, A2 · LAB 05 RESULTS ON THE SUPPLIED REFERENCE FRAME

Measured — Which Filter Actually Wins, by Degradation



Median owns impulse noise

0.932 against a 0.322 baseline — the largest single margin in the matrix.

Bilateral owns Gaussian noise

0.878, because it smooths within regions without crossing edges.

Unsharp slightly improves this blur

Unsharp reaches0.885 versus0.876 baseline on this fixture. This small gain does not restore lost detail.

Unsharp makes noise worse

It amplifies exactly the high-frequency content that noise lives in.

MEASURED

Measured in Lab 05 with opencv-python 4.13 on the supplied 800x600 reference frame, using the NumPy SSIM implementation shipped with the course. Values are SSIM against the clean master, so higher is better.

Two Metrics, Two Different Questions

PSNR

```
MSE = mean((a − b)²)
PSNR = 10·log₁₀(255² / MSE)
```

A per-pixel error average, in decibels. Cheap, and it catches gross error. It rewards aggressive smoothing, because smoothing lowers mean squared error.

SSIM

$$\frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$$

Luminance, contrast and structure, over a sliding Gaussian window. $C_1=(0.01\cdot255)^2$, $C_2=(0.03\cdot255)^2$. It punishes the lost texture that smoothing cost you.

When they disagree

```
inspect the task
```

Neither metric guarantees perceived quality. Lab05 asks you to inspect a cell where the two rank filters differently and explain which you would gate a customer-facing asset on.

cv2.PSNR ships in the base wheel. The cv2.quality SSIM module does NOT — it is contrib-only — so the course ships its own NumPy implementation in ssim.py rather than adding a dependency for one function.

Geometric Transformation — Affine and Perspective

```
import numpy as np, cv2
h, w = img.shape[:2]

# TRANSLATION — a 2x3 affine matrix, applied by warpAffine
M = np.float32([[1, 0, 100],
                [0, 1,  50]])          # shift +100 x, +50 y
shifted = cv2.warpAffine(img, M, (w, h))

# ROTATION about an arbitrary centre, with scale
M = cv2.getRotationMatrix2D((w/2, h/2), angle=30, scale=1.0)
rotated = cv2.warpAffine(img, M, (w, h))

# AFFINE from three point correspondences
M = cv2.getAffineTransform(src_pts3, dst_pts3)      # 2x3

# PERSPECTIVE from four point correspondences
M = cv2.getPerspectiveTransform(src_pts4, dst_pts4)  # 3x3
flat = cv2.warpPerspective(img, M, (300, 300))
```

Affine is 2x3

Scaling, translation, rotation, shear. Parallel lines stay parallel.

Perspective is 3x3

A homography. Straight lines stay straight, but parallel lines converge.

Three points vs four

An affine map is determined by 3 correspondences; a homography needs 4.

Interpolation matters

INTER_AREA for shrinking, INTER_CUBIC or INTER_LINEAR for enlarging.

VERIFY The Written Assessment asks you to name three geometric transformations OpenCV provides. Scaling, translation, rotation, affine and perspective are all acceptable — say which matrix each needs.

K4, K5 · NAME THREE THRESHOLDING OPTIONS FOR SEGMENTATION

Thresholding — Five Options, One Decision Rule Each

1

THRESH_BINARY

dst = maxVal if src > thresh, else 0. The default: a hard two-level mask.

2

THRESH_BINARY_INV

dst = 0 if src > thresh, else maxVal. The same mask, inverted — used when the object is darker than the background.

3

THRESH_TRUNC

dst = thresh if src > thresh, else src. Caps the bright end without discarding the dark detail.

4

THRESH_TOZERO

dst = src if src > thresh, else 0. Keeps bright detail, discards everything below.

5

THRESH_TOZERO_INV

dst = 0 if src > thresh, else src. The complement.

6

Adaptive + Otsu

Adaptive computes a threshold per neighbourhood, for uneven illumination. Otsu picks one global threshold by maximising between-class variance.

Segmentation by Colour — the HSV Route

```
hsv  = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)

# a slate-blue band. NOTE the OpenCV hue scale: 210 degrees is H = 105
mask = cv2.inRange(hsv, (98, 90, 60), (116, 255, 255))

# RED WRAPS AROUND — it needs two bands ORed together
m1   = cv2.inRange(hsv, (0,   120, 80), (8,   255, 255))
m2   = cv2.inRange(hsv, (172, 120, 80), (179, 255, 255))
red  = cv2.bitwise_or(m1, m2)

# clean the mask: close pinholes, then grow past the object boundary
k    = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5))
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, k)
mask = cv2.dilate(mask, k, iterations=2)

print('coverage %.2f%%' % (100.0 * (mask > 0).mean()))
```

Halve the degrees

A 210-degree blue is H=105 in OpenCV, not 210. This is the most common HSV error.

Red needs two bands

It spans H near 179 and H near 0. One inRange call silently misses half of it.

Close, then dilate

Closing fills compression pinholes; dilation grows the mask a few pixels so a fill can blend.

Always report coverage

A tag mask should be low single digits. If it is 30%, the band is catching the product.

VERIFY Lab 04 builds exactly this mask and then proves the edit was contained by measuring change inside and outside it.

Masked Generative Editing — the Mechanism

BUILD THE MASK

A single-channel 8-bit image. WHITE (255) marks the pixels the model may rewrite; BLACK (0) marks the pixels it must preserve.

STATE THE SEMANTICS

Record which colour is editable as an explicit field. Semantics are not universal across tools, and an inverted mask does not error — it produces a confident wrong image.

CONDITION THE SAMPLER

The prompt plus the unmasked pixels condition the denoising. The model only writes inside the mask.

MEASURE CONTAINMENT

Mean absolute difference inside the mask (clearly non-zero) and outside it (effectively zero). That is the numeric proof.

SO WHAT Verify, do not assume. Lab 04's audit is the difference between 'the model said it only changed the tag' and 'the unmasked region is provably unchanged'.

Four Masked Edit Modes

Inpaint

- Fills a masked region INSIDE the frame.
- Removes an object, or replaces it.
- Mask from colour, from a semantic segmentation class, or by hand.
- Dilate a few pixels or you get a halo.
- Classical baseline: cv2.inpaint (Telea, Navier-Stokes).

Outpaint

- Extends the frame BEYOND its original edges.
- Also called generative expand.
- Changes aspect ratio without cropping.
- Needs a padded canvas AND a mask.
- Acceptance: the original columns stay bit-identical.

Background swap

- Masks the complement of the subject.
- Re-renders the setting around a kept subject.
- A composite cannot fix the matte edge or colour spill; a re-render can.
- Acceptance: subject histogram distance stays inside threshold.

Mask-Free Editing — Faster to Author, Weaker to Guarantee

Mechanism

- **How it works**

– You supply a text instruction and no mask. The model infers both WHAT to change and WHERE.

– One documented mechanism, DiffEdit, compares the denoising for a source prompt against a target prompt and derives its own mask from the difference.

– Natively multimodal models go further: text and image are reasoned over in a single model, which is what makes conversational editing possible.

Consequence

- **What it costs you**

– The preserved region is INFERRED, not guaranteed.

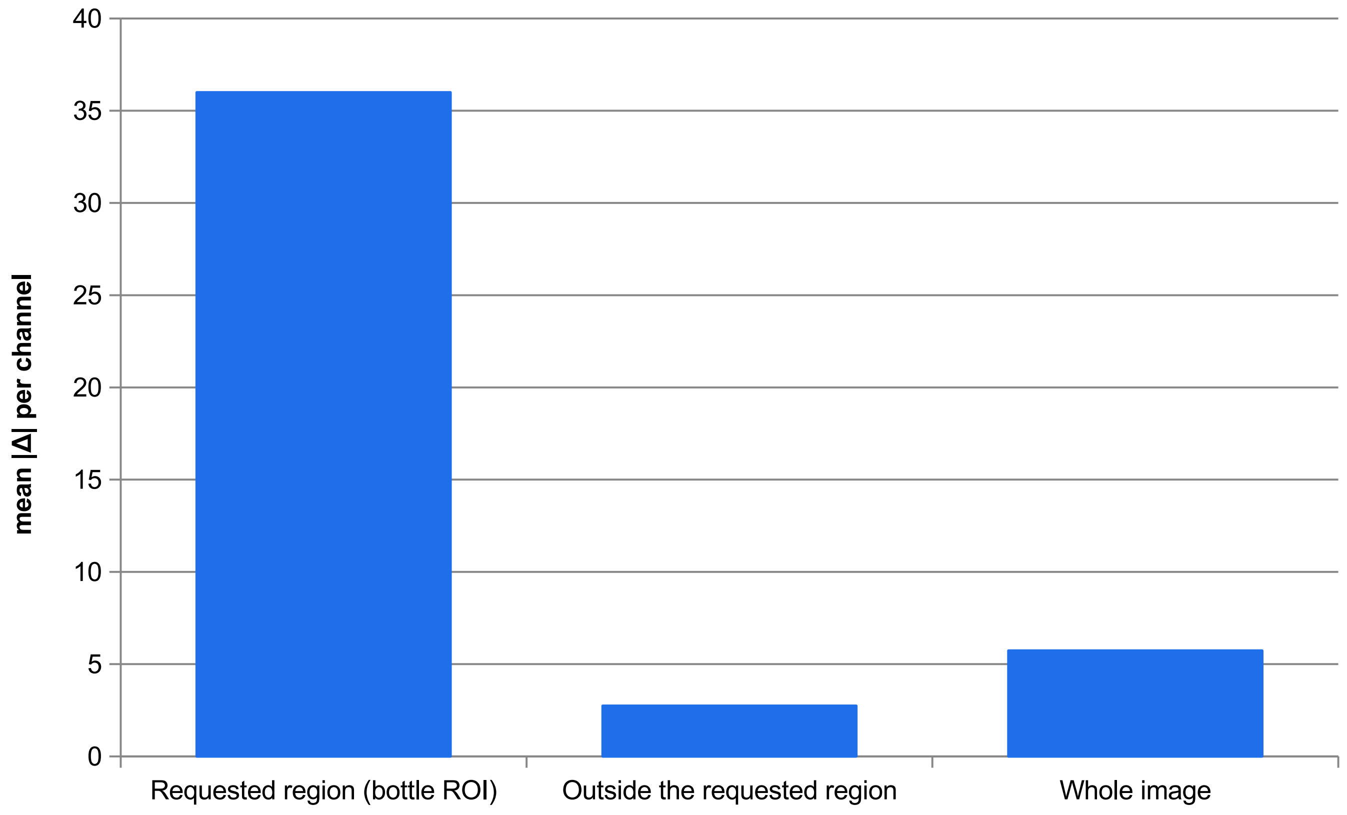
– The supplied AI-edited pair in Labs 03 and 04 measures this: the requested region changed by a mean absolute difference of 36.0/255, but pixels OUTSIDE the requested region still moved by 2.74/255.

– Fine for ideation and social mock-ups. Not what you ship as a commercial master.

Source for the measurement: courseware/assets/edit-measurements.json, supplied with the AI-edited bottle pair. Method: mean absolute RGB channel difference on an approximate rectangular ROI — not a segmentation mask.

K4, A2 · THE SUPPLIED AI-EDITED PAIR

Measured — A Mask-Free Edit Changes More Than You Asked



The edit landed

36.0/255 inside the region the instruction named — the colour change happened.

But it leaked

2.74/255 outside it. The instruction said 'keep non-bottle pixels unchanged'. They moved.

Small is not zero

For a commercial master, 'small' is not the acceptance criterion — 'provably contained' is.

This is why masks exist

A mask states intended edit support; verify whether the provider preserves outside pixels.

MEASURED

From courseware/assets/edit-measurements.json, supplied with the coral/teal bottle pair. Method: mean absolute RGB channel difference on an approximate rectangular ROI, not a segmentation mask. Pixel difference is not perceptual or semantic quality.

Three Mask Geometries, Three Editing Contracts

1

OBJECT REGION

Input: colour-selected tag region. White pixels designate the repair region; black pixels preserve the surrounding product. Morphology changes the boundary, so inspect coverage.

2

BACKGROUND REGION

Input: subject silhouette. Its complement designates background replacement. Subject containment is checked separately from visual similarity.

3

NEW CANVAS PANELS

Input: centred1024px square on an1820px-wide canvas. Two398px panels contain815,104 editable pixels; the original centre stays protected.

A2 · THE ACCEPTANCE GATE, NOT AN OPINION

Auditing a Masked Edit

Inside the mask

```
mean |orig − result|
over mask > 0
```

Must be clearly non-zero. If it is near zero, the edit did not happen — the mask was empty or inverted.

Outside the mask

```
mean |orig − result|
over mask == 0
```

Must be effectively zero. The Lab 04 acceptance threshold is 0.5/255. Anything above it means the edit was not contained.

Subject preservation

```
Bhattacharyya(H,S)
≤ 0.15
```

For a background swap, the subject's hue-saturation histogram must stay within 0.15 of the source. This checks colour-distribution similarity, not complete subject identity.

cv2.inpaint only writes inside the mask, so a non-zero outside value on the classical baseline means you compared against the wrong source image.

Classical Repair Versus Generative Fill

	cv2.inpaint (Telea / Navier-Stokes)	Generative inpainting
Mechanism	Propagates surrounding texture inward from the mask boundary	Samples new pixels conditioned on the surrounding context and a prompt
Invents content	No — it can only continue what is already there	Yes — it can add an object that was never in the frame
Determinism	Fully deterministic; same input, same output	Stochastic; the same prompt gives a different sample
Good for	Small blemishes, dust, a tag on a flat backdrop	Replacing an object, changing a scene, filling a large region
Fails by	Smearing when inpaintRadius pulls texture from too far away	Producing something plausible and wrong
Cost	Milliseconds, on a CPU	Seconds, on an accelerator, per sample

Lab 04 runs the classical repair first so the mechanism is visible, then writes the generative specification for the same three jobs.

A2 · THE METHOD BEHIND LAB 03

Prompting is experimental design, not wording.

Hold every factor constant except the one under test, generate several samples per condition, and score the spread. Then you are measuring the prompt instead of arguing about it.

A2 · SEPARATE THEM, THEN VARY ONE AT A TIME

The Four Prompt Factors

1 **SUBJECT**
Who or what, and in what state. 'Locked' means an explicit description of colour, construction, fit and pose — not a product name.

2 **CONTEXT**
Setting, time of day, ground surface, mid-ground content. This is what a model invents most freely when you leave it out.

3 **STYLE**
The photographic or illustrative register, plus an explicit colour and contrast rule. 'Cinematic' is a named style; it is not a locked one.

4 **TECHNICAL**
Lens, depth of field, lighting direction, aspect ratio, output size. Explicit values, not adjectives.

5 **NEGATIVE**
What must not appear. At minimum: no rendered text or logos in frame, no additional people, no change to the product colour.

6 **VERSION**
A prompt without a version number cannot be audited when an asset is later challenged. This is the field people forget.

Structured Prompting — One Key Per Factor

```
{
  "prompt_version": "v3",
  "brief_reference": "NQ-HARBOUR-LINE-AW26",
  "subject":   "the Harbour Line jacket: slate-blue shell, single chest seam,
               stand collar, worn open over a plain cream tee by an adult
               model, three-quarters to camera, hands at sides",
  "context":   "harbour boardwalk at first light; mooring bollards in the
               mid-ground; wet timber underfoot",
  "style":     "documentary product photography, restrained colour palette,
               natural contrast",
  "technical": {"lens": "35mm", "depth_of_field": "shallow",
               "lighting": "low-angle warm key from camera left, soft fill",
               "aspect_ratio": "4:5", "image_size": "2K"},
  "negative_constraints": ["no text or logos rendered in the image",
                           "no lens flare", "no additional people",
                           "no colour cast on the jacket shell"]
}
```

Change one value, not a sentence

To test a tighter framing you change `lens` alone, and the other three factors are provably unchanged.

It forces completeness

An empty key is visible. A missing clause in a paragraph is not.

It versions and diffs

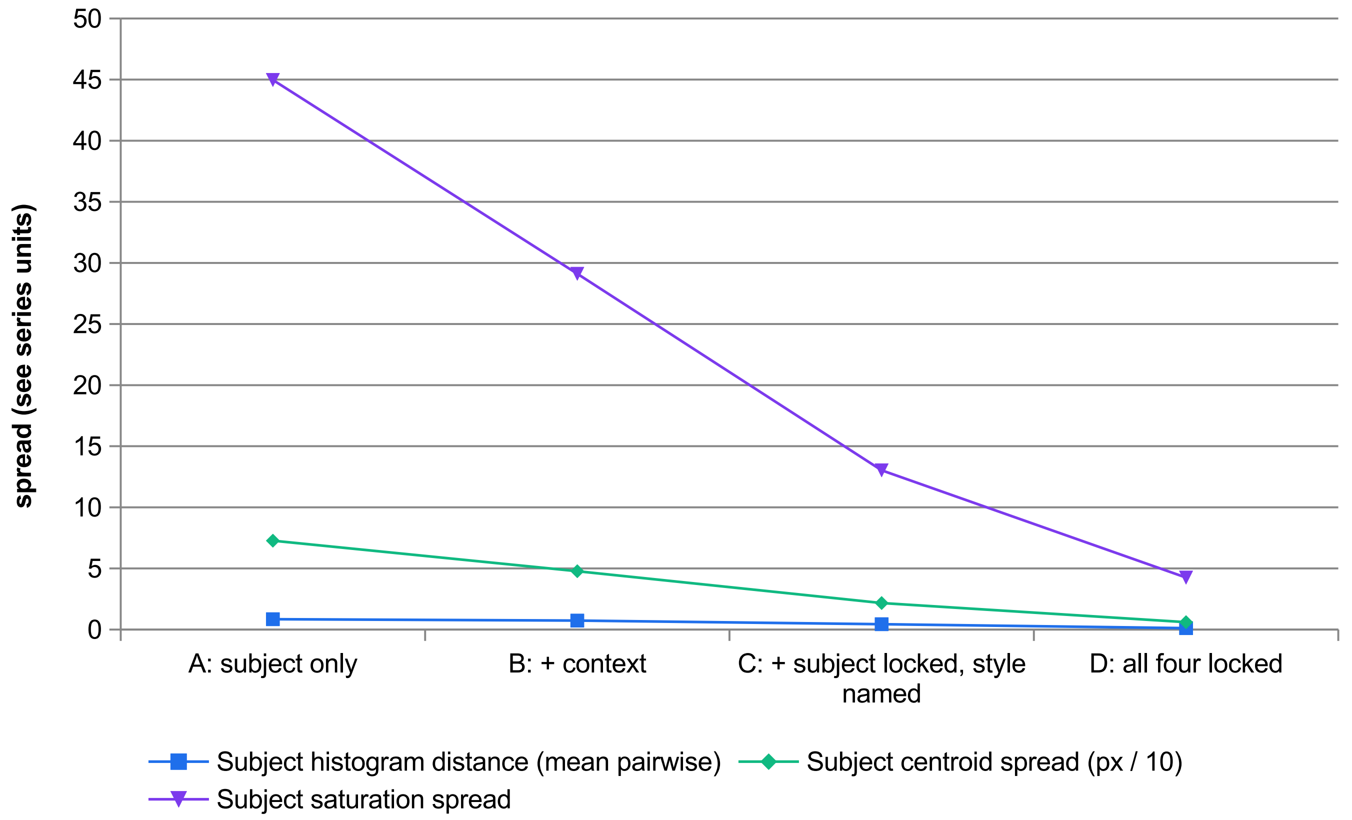
Two prompts can be compared character-by-character, which is how you find the word that leaked between cells.

It is auditable

When an asset is challenged, the exact prompt that produced it can be produced.

VERIFY Lab 03 freezes the winning prompt as out/prompt-v3.json with these exact fields, then records which generation path was used.

Measured — Prompt Specificity Versus Output Variance



The trend is the lesson

These three selected metrics decline on the designed simulation; edge-density spread does not.

Do not infer a causal effect

Several prompt factors change progressively. This simulation is not a controlled one-factor experiment.

Measure the subject, not the frame

A backdrop gradient dominates a whole-frame histogram. Mask the subject first.

Smooth a sparse histogram

A flat-coloured product puts its mass in one bin; without smoothing every pair reads as maximally different.

MEASURED

Measured in Lab 03 with opencv-python 4.13 on the supplied simulated variant sets. The SIMULATION is designed to reproduce the characteristic spread of each condition — real model output will be noisier and the ordering is not guaranteed. Centroid spread is divided by 10 so the three series share an axis.

K4, A2 · MODEL IDS CHANGE — RE-VERIFY BEFORE YOU RELY ON ONE

Current Image-Generation Models — Verified 6 September 2026

Model ID	Position	Sizes	Notes
gemini-3.1-flash-image	Versatile generalist	0.5K / 1K / 2K / 4K	The documented workhorse for generation and editing
gemini-3.1-flash-lite-image	Fastest and most economical	1K only	Adds a wider aspect-ratio set including 3:2, 4:5, 21:9
gemini-3-pro-image	Premium, complex visual tasks	1K / 2K / 4K	For work where sample quality dominates cost
gemini-2.5-flash-image	Legacy pioneer model	1K / 2K / 4K	Still documented; superseded by the 3.1 family
imagen-4.0-generate-001	DEPRECATED	—	Documented shutdown 17 August 2026. Appears in older tutorials, including our reference ebook. Do not specify it.

Source S20/S21: ai.google.dev/gemini-api/docs/image-generation and /imagen, verified 6 September 2026. All generated images carry a SynthID watermark. The deprecated row is kept deliberately so you recognise it in older material.

A2 · OPTIONAL IN LAB 03 — THE LAB COMPLETES WITHOUT IT

The Current Documented Call Pattern

```
from google import genai
import base64, json

client = genai.Client()
spec   = json.load(open('out/prompt-v3.json'))

# text to image
interaction = client.interactions.create(
    model='gemini-3.1-flash-image',        # NOT imagen-4.0-* (deprecated)
    input=json.dumps(spec),
    response_format={'type': 'image', 'aspect_ratio': '4:5'},
)
open('out/gen/cellD_01.png','wb').write(
    base64.b64decode(interaction.output_image.data))

# multi-turn conversational edit — carries the previous result forward
refined = client.interactions.create(
    model='gemini-3.1-flash-image',
    input='warm the key light slightly; change nothing else',
    previous_interaction_id=interaction.id,
)
```

Optional, never required

Lab 03 ships a simulated variant set so the deliverable never depends on a paid or rate-limited service.

previous_interaction_id

This is what makes conversational editing possible — the model keeps the context.

SynthID on every output

Record it in the provenance note alongside the model ID and prompt version.

Record which path you took

Generated or simulated. Never label a simulated stand-in as model output.

VERIFY Verified against the official documentation on 6 September 2026. Re-verify before you rely on it.

Enterprise Practices That Make This Repeatable

1. **A brand style guide FOR AI**
   The locked subject, style and technical values, written once and reused as the continuity block in every prompt.

2. **Integration, not a side tool**
   Generation sits inside the existing creative workflow, or it produces assets nobody can find.

3. **An explicit review gate**
   Named measurement, numeric threshold, and what happens on each side of it.

4. **Centralised asset management**
   One place where the asset, its prompt version, its model ID and its review record live together.

5. **Prompt-engineering literacy**
   The creatives write the prompts. Structured prompting is a skill to train, not a tool to buy.

6. **Provenance on every derivative**
   The label travels: a crop of a simulated asset is still a simulated asset.

Histogram Equalisation and CLAHE

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# GLOBAL equalisation: remap intensities so the histogram is roughly flat
eq = cv2.equalizeHist(gray)          # one mapping for the whole image

# CLAHE: contrast-limited adaptive histogram equalisation
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
loc   = clahe.apply(gray)            # a mapping PER 8x8 tile, then interpolated

# the clip limit is what stops CLAHE amplifying noise in flat regions:
# histogram bins above the limit are clipped and redistributed before the
# cumulative distribution is built.

# for colour, equalise LUMINANCE only — never the three BGR channels separately
lab      = cv2.cvtColor(img, cv2.COLOR_BGR2LAB)
lab[:,:,0] = clahe.apply(lab[:,:,0])
out      = cv2.cvtColor(lab, cv2.COLOR_LAB2BGR)
```

Global fails on uneven light

One mapping cannot serve a frame that is bright on one side and dark on the other.

CLAHE is local

A separate mapping per tile, bilinearly interpolated so tile edges do not show.

clipLimit controls noise

Without it, adaptive equalisation amplifies sensor noise in flat regions into visible grain.

Equalise luminance, not colour

Equalising B, G and R separately shifts hue. Convert to LAB or YCrCb and touch only the luminance channel.

VERIFY Enhancement changes the image. Measure the result against the reference with PSNR and SSIM, exactly as in Lab 05 — 'it looks better' is not a deliverable.

Morphology — Cleaning a Mask Before You Trust It

1. **EROSION**
   Shrinks the foreground. Removes isolated white specks and thin bridges between blobs that should be separate.

2. **DILATION**
   Grows the foreground. Fills small holes and reconnects a broken outline. It is what gives an inpaint mask its blending margin.

3. **OPENING**
   Erode then dilate. Removes small objects while leaving the size of the large ones roughly unchanged.

4. **CLOSING**
   Dilate then erode. Fills small holes and gaps while leaving the overall size roughly unchanged — the usual first step on a compression-noisy mask.

5. **THE KERNEL DECIDES**
   cv2.getStructuringElement with MORPH_RECT, MORPH_ELLIPSE or MORPH_CROSS. An ellipse is the safe default for organic shapes.

6. **ORDER MATTERS**
   Close then dilate, as Lab 04 does: fill the pinholes first, then grow past the boundary. Dilating first just grows the pinholes too.

Regions, Drawing and Blending

```
# a region of interest is just NumPy slicing — no copy unless you ask for one
roi = img[120:180, 120:180]          # [y0:y1, x0:x1]
img[300:360, 300:360] = roi          # paste it elsewhere

# annotate — every overlay in Lab 08 and Lab 11 is built from these
cv2.rectangle(img, (x, y), (x+w, y+h), (0, 200, 0), 2)
cv2.circle(img, (cx, cy), r, (0, 90, 255), -1)         # -1 = filled
cv2.line(img, (x1, y1), (x2, y2), (255, 0, 0), 3, cv2.LINE_AA)
cv2.putText(img, txt, (x, y), cv2.FONT_HERSHEY_SIMPLEX, 0.6,
            (20, 20, 20), 1, cv2.LINE_AA)

# blending — the watermark and cross-fade operation
#   g(x) = (1 - alpha)*f0(x) + alpha*f1(x)
out = cv2.addWeighted(img1, 0.7, img2, 0.3, 0)     # weights + a scalar offset

# alpha compositing with a real alpha channel (Lab 11's logo overlay)
a   = logo[:, :, 3:4].astype('float32') / 255.0
roi[:] = (a * logo[:, :, :3] + (1 - a) * roi).astype('uint8')
```

Slicing is [y, x]
Rows before columns. The opposite order to the (x, y) that every drawing function takes.

addWeighted is a line
Two weights and an offset. Vary alpha from 0 to 1 and you have a cross-fade.

Alpha needs IMREAD_UNCHANGED
Read a PNG without it and the alpha channel is silently dropped — that is the black box around your logo.

LINE_AA for anything shown
Anti-aliased strokes; the default is jagged and looks like a bug in a screenshot.

VERIFY Lab 11 composites the logo with exactly this alpha blend and then proves the result sits inside the safe rectangle.

Super-Resolution — and the Claim You Must Not Make

Mechanism

- **What it does**

– Maps a low-resolution image to a higher-resolution one, inventing plausible detail that was not captured.

– Trained on pairs, so it learns what detail USUALLY accompanies a given low-resolution pattern.

– Reported alongside restoration and relighting as a distinct translational task.

– Evaluated with PSNR and SSIM against a true high-resolution capture, plus FID and LPIPS for perceptual quality.

What you are allowed to claim

- **The honest acceptance clause**

– Where a true high-resolution capture EXISTS, compute PSNR and SSIM against it.

– Where it does not, a human reviews the output and it is labelled UPSCALED.

– No numeric fidelity claim may be made against a reference that does not exist.

– Scoring an upscale against its own downscaled input is a meaningless number — and it is the one people quote.

The same discipline applies to any restoration: state what the output is a restoration OF, and label the derivative.

Aspect Ratios — Say Pixels, Not Names

Delivery	Ratio	Pixels	Why the name alone is not enough
Feed post	1:1	1080 x 1080	Unambiguous, but state it anyway for the safe-area calculation
Story / Reel	9:16	1080 x 1920	9:16 is also 720x1280 and 1440x2560 — a 3x difference in caption legibility
Portrait post	4:5	1080 x 1350	The default crop many platforms apply if you supply 1:1
Hero / landscape	16:9	1920 x 1080	Also 1280x720 and 3840x2160
Outpaint canvas	~16:9	1820 x 1024	1.7773:1 — chosen so a 1024 source sits on whole-pixel boundaries with 398 columns each side

Current image models document a wide aspect-ratio set including 1:1, 3:2, 2:3, 3:4, 4:3, 4:5, 5:4, 9:16, 16:9 and 21:9. Ask for the ratio AND the pixel dimensions.

Safety Controls You Will Meet on an Image API

1. **Person generation**
Documented values dont_allow, allow_adult and allow_all. Deactivating person generation removes a whole class of risk for externally accessible use cases.

2. **Safety filter level**
A threshold on the output classifier. Setting it permissively is a decision that has to be recorded, not a default to be nudged.

3. **Banned-word filters**
One provider's primary mechanism is an extensive banned list plus community moderation — cheap, and easy to route around.

4. **Configurable tolerance**
Another exposes a user-set safety slider, with permissively generated images forced private.

5. **Watermarking**
SynthID on current Gemini image output; C2PA Content Credentials metadata elsewhere. Different mechanisms, different failure modes.

6. **Red-teaming**
Adversarial prompt sets probing bias, impersonation and unsafe content — and re-run after every model or guardrail change, not once at launch.

Lab 02 — Image and Video Data Representation

INPUTS

Synthetic product flat-lay reference still, 1280x720 (deterministically rendered with OpenCV — SIMULATED classroom asset, not a photograph)

TECHNICAL QUESTION

How do frame dimensions, channel order, FPS and compression change the payload?

EVIDENCE TO PRODUCE

A measurement report (`out/representation-report.json`) recording shape, dtype, channel means, HSV ranges, frame count, FPS, duration and the computed uncompressed byte rate for the supplied clip.

LAB 02 · 15 MIN · Python 3 · opencv-python · NumPy · offline after dependencies are installed; no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-02-image-video-representation/README.md

TOPIC 02 · ELO2 · A2

Lab 03 — Prompt Experiment Matrix for Controlled Image Generation

INPUTS

16 supplied variant images — four cells of four variants each, deterministically rendered with OpenCV to reproduce the characteristic spread of an under-specified, partially specified, subject-locked and fully specified prompt. Every file carries a SIMULATED banner: these are classroom stand-ins for model output, NOT generative-model output.

TECHNICAL QUESTION

How do specification changes affect measured variation across the supplied outputs?

EVIDENCE TO PRODUCE

A completed factor-level table and four-run comparison, a versioned `prompt-v3.json` structured prompt, and `out/variance-report.csv` scoring the measured spread of each cell.

LAB 03 · 15 MIN · Text editor · Python 3 · opencv-python · NumPy · OPTIONAL free-tier image model. The lab completes fully offline using the supplied simulated variant sets.

Detailed procedures and prompts: Learner Guide / labs/lab-03-prompt-experiment-matrix/README.md

TOPIC 02 · ELO2 · A2

Lab 04 — Masking, Inpainting, Outpainting and Background Swap

INPUTS

Synthetic 1024x1024 product shot with a subject, a plain backdrop and a stray price-tag rectangle (deterministically rendered with OpenCV — SIMULATED)

TECHNICAL QUESTION

Which pixels changed inside and outside each edit mask?

EVIDENCE TO PRODUCE

Three masks (`out/mask_tag.png`, `out/mask_bg.png`, `out/mask_outpaint.png`), the classical inpaint results, `out/mask-audit.csv` proving the unmasked region was preserved, and `out/edit-spec.json` specifying the generative equivalent.

LAB 04 · 15 MIN · Python 3 · opencv-python · NumPy · offline with preinstalled dependencies; no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-04-masking-inpainting-outpainting/README.md

Lab 05 — Filtering, Enhancement and Restoration Measured with PSNR and SSIM

INPUTS

Synthetic 800x600 clean reference frame with flat regions, hard edges, fine texture and a smooth gradient — the four content types that separate filters (deterministically rendered with OpenCV — SIMULATED)

TECHNICAL QUESTION

Which enhancement improves the stated metric, and what visual defect remains?

EVIDENCE TO PRODUCE

`out/enhancement-matrix.csv` scoring 5 filters against 3 degradations with PSNR and SSIM, plus a one-paragraph recommendation naming the winning filter per degradation and whether their rankings differ.

LAB 05 · 15 MIN · Python 3 · opencv-python · NumPy · offline with preinstalled dependencies; no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-05-filtering-enhancement-metrics/README.md

TOPIC 02 · ELO2 · K3, K4, A2

Recap — Topic 02

- An image is H x W x C uint8 in BGR order; a video is frames plus a frame rate, and the uncompressed byte rate that follows drives every later architecture decision.
- HSV separates hue from brightness, which is why a colour gate built on it survived a 40% exposure drop unchanged while a raw-channel threshold collapsed to zero.
- Filtering is convolution with a kernel; geometric transformation is a 2x3 affine or a 3x3 perspective matrix.
- Median dominates impulse noise and bilateral dominates Gaussian noise — measured, not asserted — and nothing recovers defocus, because deblurring is an inverse problem.
- PSNR is a per-pixel average and SSIM is structural; when they disagree, SSIM is the one to gate a customer-facing asset on.
- A binary mask is what makes a generative edit auditable: white is editable, black is preserved, and containment is proved by measuring inside and outside.
- Prompting is experimental design. Lock one factor at a time, measure the spread, and freeze the winner as versioned JSON. (A2, K3, K4)

03

TOPIC 03

AI-Based Visual Features, Styles and Consistency

Feature extraction and representation · local descriptors — edge, colour, texture, corner and motion · global descriptors — statistical and geometrical · measuring style and composition consistency across a campaign

LU3 · ELO3 · K5, K6, K7, A3, A4

Key Concepts — Topic 03

1

A feature is a repeatable, discriminative measurement

Good features are repeatable under rotation, scale and illumination change, and different enough between classes to separate them. Everything downstream inherits the quality of this choice.

2

Local versus global is a decision about scope

Local descriptors answer 'what is at this point' (Canny edge, ORB keypoint, optical-flow vector). Global descriptors answer 'what is this whole image like' (colour histogram, Hu moments, edge density). Consistency work needs global.

3

Corner detectors differ in what they are invariant to

Harris: rotation-invariant, not scale-invariant. Shi-Tomasi: Harris with a min-eigenvalue score. FAST: segment test, very fast, no descriptor. SIFT/ORB: scale- and rotation-invariant with a matchable descriptor.

4

Canny is a four-step algorithm, not a filter

Gaussian smooth -> Sobel gradient magnitude and direction -> non-maximum suppression -> hysteresis thresholding with two thresholds. The 1:2 to 1:3 low:high ratio is the standard starting point.

5

Global statistics are how you audit a brand look

Normalised HSV histograms compared with Bhattacharyya distance, mean saturation, edge density and Hu moments give a numeric style fingerprint you can threshold and put in a review gate.

6

Faithfulness needs a metric, not an opinion

Text-image alignment via CLIP-style cosine similarity, object/attribute recall via a detector, and masked-region consistency for edits — each catches a different way a generated asset can be wrong.

What Makes a Feature Worth Extracting

1

REPEATABLE

The same physical point yields the same measurement under rotation, scale change and different lighting. Without this, nothing downstream can match anything.

2

DISCRIMINATIVE

Different enough between classes to separate them. A feature that is identical for every object carries no information.

3

LOCAL ENOUGH

Computed from a bounded neighbourhood, so partial occlusion does not destroy every feature at once.

4

COMPACT

A descriptor you can store and compare cheaply. ORB is 32 bytes per keypoint; SIFT is 512. At 500 keypoints per frame that is a 16x bandwidth decision.

5

EFFICIENT

Extractable inside your latency budget. FAST exists because detection had to run in real time on modest hardware.

6

MEANINGFUL TO THE TASK

A texture descriptor will not catch a recolour, and a colour histogram will not catch an over-sharpened delivery. Match the feature to the failure.

Local Versus Global — a Decision About Scope

PIXEL

One value, or one BGR triple. Carries no context at all.

LOCAL FEATURE

'What is at this point?' A Canny edge pixel, an ORB keypoint with its 32-byte descriptor, an optical-flow vector.

REGION FEATURE

'What is in this area?' A masked histogram, a texture statistic, a connected component's shape.

GLOBAL DESCRIPTOR

'What is this whole image like?' A normalised colour histogram, mean saturation, edge density, Hu moments.

EMBEDDING

'What does this image mean?' A learned vector from a CNN or a vision-language model — the space FID and CLIP scores live in.

SO WHAT Brand-consistency work needs GLOBAL descriptors: the question is 'does this whole asset look like the approved master?', not 'is there a corner at pixel (412, 88)?'

Canny Is an Algorithm, Not a Filter

Stage 1–2

```
Gaussian smooth
then Sobel:
G = √(Gx² + Gy²)
θ = atan2(Gy, Gx)
```

Smooth first or you detect noise. Sobel gives both the gradient magnitude and its direction at every pixel.

Stage 3

```
non-maximum
suppression
along θ
```

Keep a pixel only if it is the local maximum in the gradient direction. This is what turns a thick gradient ridge into a one-pixel line.

Stage 4

```
hysteresis:
strong  > high
weak    > low
keep weak if
connected to strong
```

Two thresholds. Pixels above `high` seed an edge; pixels above `low` continue one only if they connect to a seed. A 1:2 to 1:3 low:high ratio is the standard starting point.

You cannot tune Canny sensibly without knowing which stage each parameter touches. Lab 06 sweeps the ratio at a fixed high threshold and asks you to pick the one that keeps the silhouette continuous without lighting up the texture panel.

Edge Detection, Measured Rather Than Eyeballed

```python
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = cv2.GaussianBlur(gray, (5, 5), 1.0)      # smooth BEFORE Canny

for lo, hi in [(200, 200), (100, 200), (66, 200)]:   # ratios 1:1, 1:2, 1:3
    edges   = cv2.Canny(gray, lo, hi)
    density = np.count_nonzero(edges) / edges.size
    print(f'{lo:3d}:{hi:3d}  edge density {density:.5f}')

# Sobel on its own, if you want the gradient rather than the binary map
gx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
gy = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
mag, ang = cv2.cartToPolar(gx, gy, angleInDegrees=True)

# Laplacian — second derivative, one call, no direction information
lap = cv2.Laplacian(gray, cv2.CV_64F, ksize=3)
```

Edge density is the metric
Non-zero pixels divided by total pixels. It is what catches an over-smoothed or over-sharpened delivery in Lab 07.

Lower low-threshold, more edges
Density is non-decreasing as the low threshold falls on the same smoothed image.

Sobel keeps direction
Canny throws the angle away. Keep Sobel when you need orientation — HOG is built on exactly this.

Laplacian is isotropic
One kernel, no direction. Cheaper, noisier, and the basis of a common focus measure.

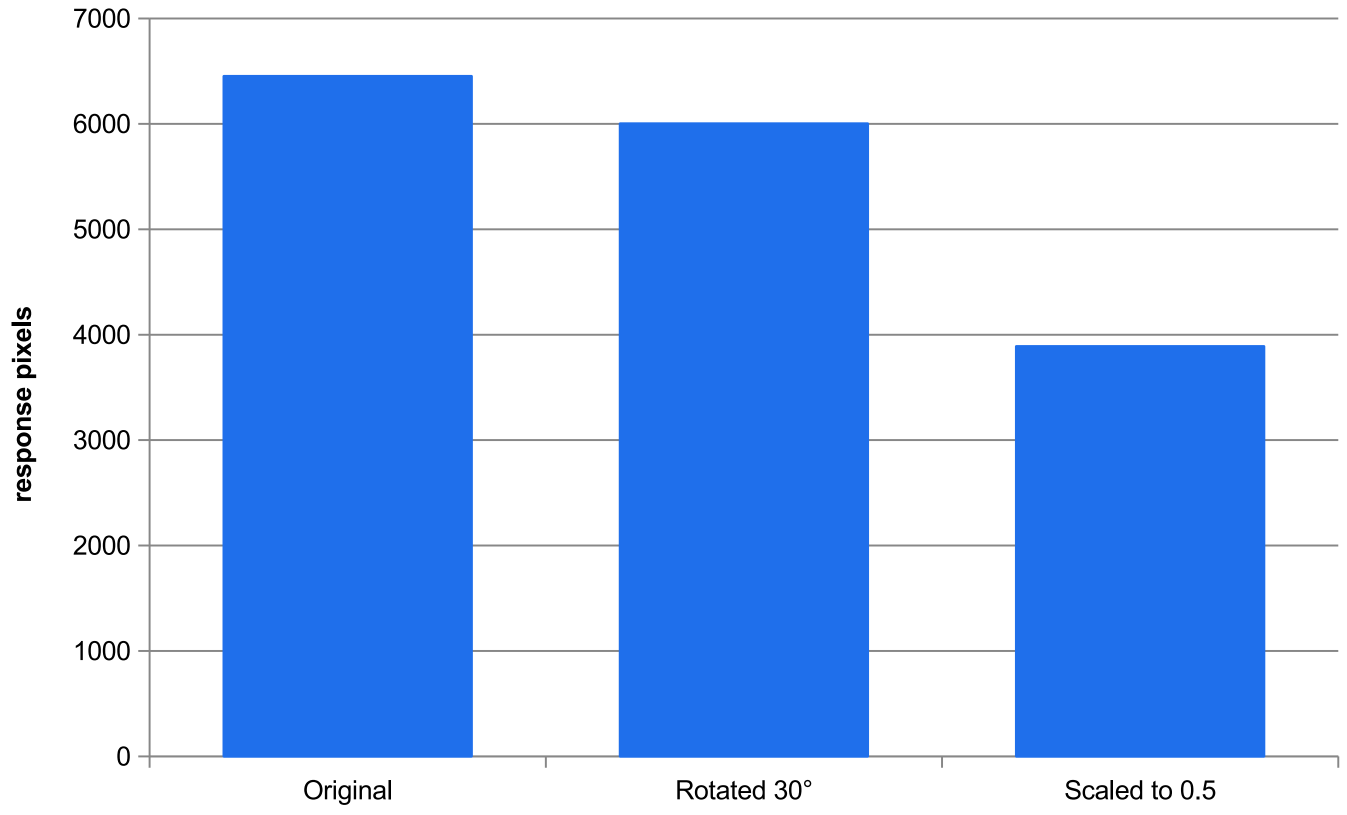
VERIFY Lab 06 writes every one of these densities into out/features-report.json so the choice of ratio is recorded, not remembered.

Corner and Keypoint Detectors — What Each Is Invariant To

Detector	Mechanism	Rotation	Scale	Descriptor
Harris	Intensity variation in ALL directions, via the structure tensor	Covariant	No	None — a response map
Shi-Tomasi	Harris with a minimum-eigenvalue score and minimum spacing	Covariant	No	None — a point set
FAST	Segment test on a ring of 16 pixels around the candidate	Partially	No	None — detector only
ORB	FAST keypoints + oriented BRIEF descriptor, over a scale pyramid	Yes	Yes	32 bytes, binary
SIFT	Difference-of-Gaussian extrema + gradient orientation histograms	Yes	Yes	128 floats = 512 bytes
SURF	NOT in the standard opencv-python wheel — non-free	—	—	Do not specify it

A detector without a descriptor cannot match. FAST finds points very fast and then you still need something to describe them — which is exactly what ORB adds.

Measured — Harris Under Rotation and Under Scale



Rotation: roughly held

6,452 to 6,002, a 7% change. Harris is rotation-covariant in theory; interpolation and cropping account for the rest.

Scale: 40% lost

6,452 to 3,889. A fixed-window Harris has no scale pyramid, so half-size features fall out of its window.

This is the whole argument

Scale invariance is why SIFT builds a difference-of-Gaussian pyramid and why ORB inherits one.

Count is a weak proxy

Repeatability of CORRESPONDING keypoints is the proper measure. Say so when you report counts.

MEASURED

Measured in Lab 06 with opencv-python 4.13 on the supplied 900x700 synthetic master. The count is RESPONSE PIXELS above a threshold, not distinct corners, and count equality does not prove the same corners were found.

K6, A4 · MEASURED: 15 FILTERED MATCHES ACROSS A 20° ROTATION AND 0.8 SCALE

Matching a Delivery Against the Approved Master

```
orb = cv2.ORB_create(nfeatures=500)
kp1, des1 = orb.detectAndCompute(master_gray,  None)   # des1.shape -> (n, 32)
kp2, des2 = orb.detectAndCompute(variant_gray, None)

# HAMMING for a binary descriptor. L2 is for SIFT and is wrong here.
bf      = cv2.BFMatcher(cv2.NORM_HAMMING, crossCheck=True)
matches = sorted(bf.match(des1, des2), key=lambda m: m.distance)

# cross-checked matches may still contain outliers; verify geometric consistency
median = float(np.median([m.distance for m in matches]))
kept   = [m for m in matches if m.distance <= 0.75 * median]
print(len(matches), '->', len(kept), 'after the median-distance heuristic')

vis = cv2.drawMatches(master_gray, kp1, variant_gray, kp2, kept[:40], None,
                      flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS)
```

Match the metric to the descriptor

NORM_HAMMING for ORB, BRIEF and BRISK; NORM_L2 for SIFT. Mixing them raises a type error at best.

crossCheck is a cheap filter

A match must be mutually best in both directions.

Then filter on distance

At most0.75x the median; a heuristic, not the Lowe ratio test. Ten good matches beat two hundred bad ones.

State what it cannot do

ORB matching absorbs rotation and moderate scale. It does not absorb a heavy crop, a style transfer or a recolour.

VERIFY Lab 06 reports both the raw and the filtered match count, and saves the drawn correspondence image as evidence.

Motion Is a Local Feature Too

Track chosen points

- **Sparse optical flow**

– Lucas-Kanade: track a chosen set of keypoints from frame to frame.

– Cheap, because you only compute at the points you care about.

– Needs good points to start from — Shi-Tomasi is the usual choice.

– Fails when the point leaves the frame or is occluded.

Measure every pixel

- **Dense optical flow**

– Farnebäck: a displacement vector for EVERY pixel.

– Returns an H x W x 2 array; convert with cv2.cartToPolar to magnitude and angle.

– Expensive — measured at 54.2 ms per 1280x720 frame in Lab 12, the single most costly stage in the pipeline.

– Gives you a whole-frame motion statistic, which is how Lab 10 detects a continuity break automatically.

Both are developed properly in Topic 04. They appear here because motion is a local feature description in the same sense that an edge or a corner is.

K7 · GLOBAL FEATURE DESCRIPTIONS

Global descriptors are how a brand gate stops being an opinion.

A colour histogram distance, a saturation delta, an edge-density ratio and a shape-match score turn 'it feels off-brand' into four numbers, a threshold and a named change.

Four Global Descriptors and What Each Catches

Descriptor	Computed as	Catches	Good direction
Hue-saturation histogram	2-D normalised histogram, 30 x 32 bins, compared with Bhattacharyya distance	Recolour, palette drift	Low
Mean saturation	Mean of the S channel over the subject mask	A desaturated or oversaturated treatment	Close to the master
Edge density ratio	Canny non-zero fraction, divided by the master's	Over-sharpening, over-smoothing	Near 1
Hu moments / matchShapes	Seven values from normalised central moments; compared with CONTOURS_MATCH_I1	Silhouette or crop change	Low

Hu moments are invariant to translation, scale and rotation, so they describe SHAPE independently of where the shape sits in the frame. Take $-\mathrm{sign}(h)\cdot\log_{10}|h|$ before reading them: the raw values span many orders of magnitude.

Comparing Histograms — Know Which Way Is Good

```
def hs_hist(img, mask):
    hsv  = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
    hist = cv2.calcHist([hsv], [0, 1], mask, [30, 32], [0, 180, 0, 256])
    hist = cv2.GaussianBlur(hist, (0, 0), 2.0)   # smooth a SPARSE histogram
    cv2.normalize(hist, hist, 0, 1, cv2.NORM_MINMAX)
    return hist

h = hs_hist(master, subject_mask(master))
for name, m in [('CORREL',       cv2.HISTCMP_CORREL),        # self = 1.0
                ('CHISQR',       cv2.HISTCMP_CHISQR),        # self = 0.0
                ('INTERSECT',    cv2.HISTCMP_INTERSECT),     # self = sum
                ('BHATTACHARYYA', cv2.HISTCMP_BHATTACHARYYA)]:# self = 0.0
    print(name, cv2.compareHist(h, h, m))

# BHATTACHARYYA is the gate: bounded in [0,1], 0 is identical, well behaved
```

Compare with itself first
Knowing which direction 'good' points in for each method is what prevents an inverted gate shipping.

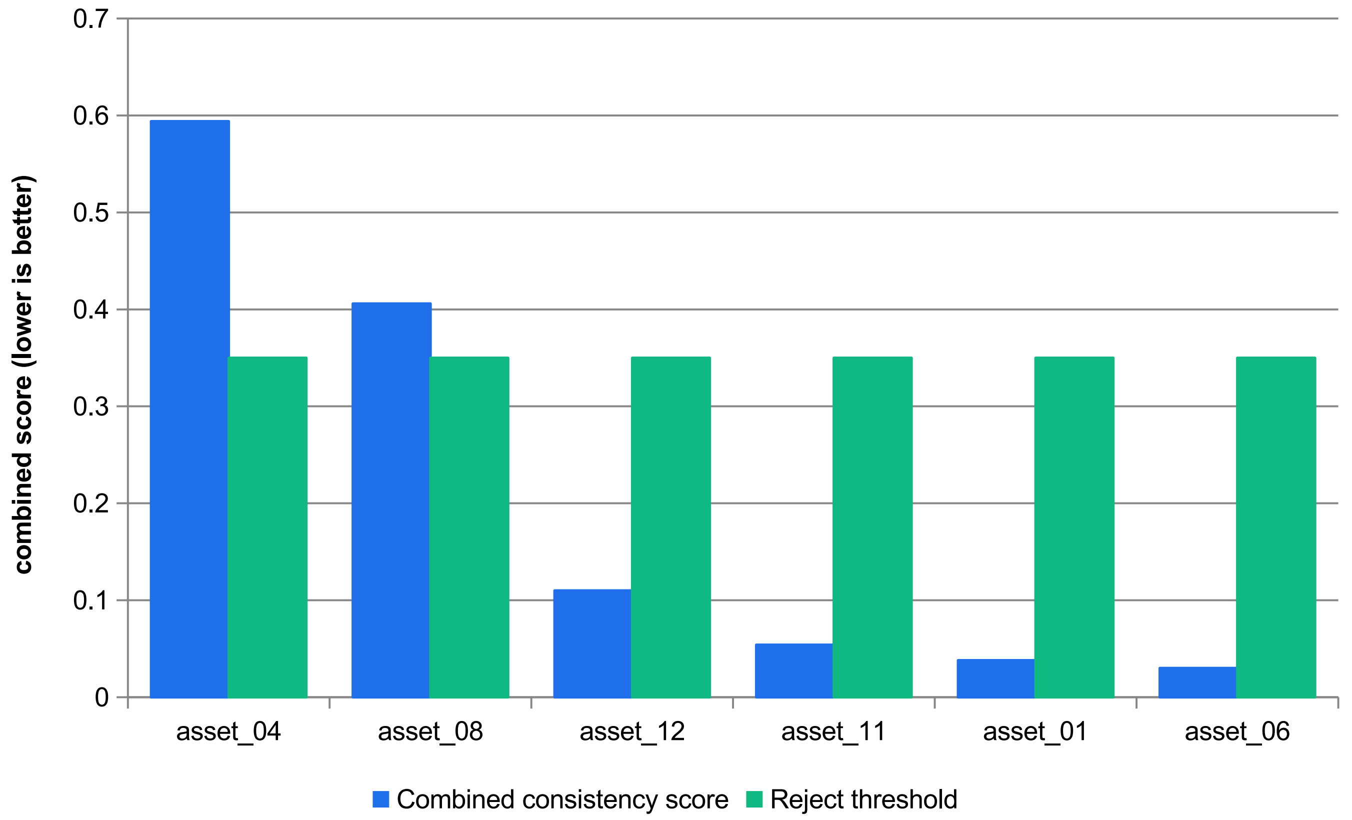
Mask the subject
A backdrop gradient dominates a whole-frame histogram and the number then describes the background.

Smooth a sparse histogram
A flat-coloured product puts its mass in one bin; without smoothing, a 3-degree hue shift and a 60-degree one both read as 'completely different'.

Normalise both sides
Comparing a normalised histogram against an un-normalised one makes every asset an outlier.

VERIFY Lab 07 asks you to print all four self-comparison values before you use any of them as a gate.

Measured — Auditing 12 Deliveries Against the Approved Master



asset_04 — recoloured

0.594, driven by the hue-saturation histogram (0.514) and a 16-point saturation delta.

asset_08 — over-sharpened

0.406, driven by an edge-density ratio of 1.35 against the master.

asset_11 — logo misplaced

Passes every descriptor at 0.054, and is caught only by the template-matching placement gate.

The margin matters

The next asset sits at 0.110 — a 3.7x gap below the lowest rejection. A gate with no margin is a coin toss.

MEASURED

Measured in Lab 07 with opencv-python 4.13 on the supplied 12-asset synthetic campaign set. The combined score is a weighted, normalised sum of the four global descriptors; the weights and normalisation constants are in data/consistency-thresholds.json.

Template Matching — Present, and in the Right Place

```
tpl = cv2.cvtColor(cv2.imread('reference/nq-logo-template.png'), cv2.COLOR_BGR2GRAY)
res = cv2.matchTemplate(asset_gray, tpl, cv2.TM_CCOEFF_NORMED)

# the result is a RESPONSE MAP, not a box: every position gets a score
minv, maxv, minloc, maxloc = cv2.minMaxLoc(res)

#   TM_SQDIFF and TM_SQDIFF_NORMED  ->  take minLoc  (LOW is best)
#   everything else                 ->  take maxLoc  (HIGH is best)

score_pass = maxv >= 0.70
box        = {'x': [620, 880], 'y': [20, 140]}      # required placement
place_pass = (box['x'][0] <= maxloc[0] <= box['x'][1] and
              box['y'][0] <= maxloc[1] <= box['y'][1])

verdict = 'ACCEPT' if (score_pass and place_pass) else 'REJECT'
```

It slides a patch and scores

A similarity value at every position. The peak is your best candidate.

Two methods invert

SQDIFF and SQDIFF_NORMED use minLoc. Getting this backwards puts your detection in the opposite corner of the frame.

Score AND placement

asset_11 passes on score and fails on placement. Presence is not correctness.

State the limits

Not scale invariant, not rotation invariant, and it degrades under illumination change unless you use a normalised method.

VERIFY When the score fails but the logo is visibly present, re-test with ORB matching before rejecting — that escalation is written into the gate.

K7, A3 · SOURCES S9, S17, S22

Consistency in a Generative Workflow

1

The continuity block

Write the locked subject, wardrobe, lighting and lens ONCE, then paste it byte-for-byte into every prompt. Retyping is how drift enters.

2

Reference images

Current video models accept up to three reference images to guide content. That is a documented mechanism, not a hope.

3

Image-to-video from a fixed frame

Starting every shot from the same conditioning frame is the strongest continuity lever available.

4

Natively multimodal models

Reasoning over text and image in one model is what makes holding a character or product across several images tractable at all.

5

Retrieval conditioning

Cross-attention to exemplar images gives style control without growing the model — a documented scaling strategy.

6

Then measure it anyway

Every one of the above improves the odds. None of them guarantees the result, so the descriptor gate still runs.

Evaluation Axes for Generated Visual Media

Axis	Metrics	The pitfall to state
Fidelity and diversity	FID, KID, precision/recall for generators	FID is encoder-dependent and sensitive to sample size; it can miss mode dropping that precision/recall catches
Text-image faithfulness	CLIP-based cosine similarity; object and attribute recall from a detector	Compositional and grounding errors — 'a red cube on a blue sphere' scores well with the objects swapped
Editing fidelity	Masked-region consistency; identity preservation	Sensitive to local artefacts, and blind to whether the edit was the right edit
Video quality	FVD, optical-flow agreement	Temporal drift and motion artefacts that no per-frame score can see
Robustness	Corruption suites, adversarial tests, calibration error	Report performance ACROSS distribution shifts, not on one clean set
Provenance	Watermark detectability under common edits	Test under compression, resizing and cropping — not on the pristine file

Reproduced as an evaluation checklist. The third column is the part people omit — a metric quoted without its pitfall is a claim, not evidence.

K7 · YOU WILL NOT COMPUTE THESE HERE — YOU WILL CITE THEM CORRECTLY

The Two Distributional Metrics, Briefly

FID

$$\|\mu_r - \mu_g\|^2 + \mathrm{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g\Sigma_r)^{1/2})$$

Fits a Gaussian to real and to generated features from a fixed image encoder, then measures between them — an approximation of a 2-Wasserstein distance in feature space. Encoder-dependent.

KID

```
MMD² with a
polynomial kernel
```

Maximum mean discrepancy in the same feature space. Unbiased at finite sample sizes, which is exactly where FID is least trustworthy.

LPIPS

```
learned perceptual
distance between
deep features
```

A pairwise perceptual distance for two specific images, where FID and KID compare two whole distributions. Different question, different tool.

Source S17 · Ch.7 pp.96–97. Lab 05 and Lab 07 use PSNR, SSIM and histogram distances because those run offline on a laptop. Know which metric answers which question, and never quote a single FID as if it settled the matter.

Contours — Geometry You Can Measure

```
_, binimg = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
contours, hierarchy = cv2.findContours(binimg, cv2.RETR_EXTERNAL,
                                       cv2.CHAIN_APPROX_SIMPLE)
c = max(contours, key=cv2.contourArea)      # the subject

area  = cv2.contourArea(c)
per   = cv2.arcLength(c, closed=True)
circ  = 4 * np.pi * area / (per ** 2)        # 1.0 = a perfect circle
x, y, w, h = cv2.boundingRect(c)
extent = area / float(w * h)                 # how much of its box it fills
hull   = cv2.convexHull(c)
solid  = area / float(cv2.contourArea(hull)) # convexity

hu = cv2.HuMoments(cv2.moments(binimg)).flatten()
log_hu = [-np.sign(v) * np.log10(abs(v)) for v in hu]   # readable form
```

RETR_EXTERNAL for outlines

Outer contours only. RETR_TREE when you need the hole structure as well.

Circularity is shape-only

$4\pi A/P^2$ is dimensionless, so it does not change with size.

Extent and solidity

How much of its bounding box and of its convex hull the shape fills — cheap descriptors that separate a jacket from a bollard.

Take log of Hu

Raw Hu values span many orders of magnitude. $-\mathrm{sign}(h)\cdot\log_{10}|h|$ is the standard readable form.

VERIFY Lab 07 uses matchShapes with CONTOURS_MATCH_I1, which is built on exactly these moments, to detect a silhouette or crop change.

Texture — the Descriptor Family That Catches Material Change

1. **Why texture at all**
   Colour catches a recolour and edges catch a sharpen. Neither catches 'the fabric now looks like plastic'.

2. **Statistical measures**
   Local mean, standard deviation, entropy and skewness over a window. Cheap, and they separate flat from busy.

3. **Local binary patterns**
   Threshold each neighbour against the centre and read the ring as a binary number. Compact and largely illumination-invariant.

4. **Gradient statistics**
   Histograms of oriented gradients — the same construction HOG detection uses, applied as a texture description rather than a detector.

5. **Frequency measures**
   The variance of a Laplacian response is the classic focus measure: it collapses on a defocused frame.

6. **Where this course uses it**
   Edge density stands in for texture in the Lab 07 gate. It is a proxy — say so when you report it.

Text-Image Faithfulness, and Where It Breaks

CLIP-style score

```
cos(E_text(t),
    E_image(x))
```

Embed the prompt and the image into a shared space with a vision-language model, then take the cosine similarity. Higher means the image better matches the text.

The compositional gap

```
'a red cube on
a blue sphere'
```

A high similarity score does not guarantee the RELATIONSHIP is right. Swap the colours and the score barely moves — the objects and attributes are all present.

The fix

```
object recall
via a detector
+ attribute recall
```

Count the objects a detector finds and check their attributes, rather than trusting one similarity number. Two measurements, two different failure modes.

This is the same discipline as everywhere else in the course: one metric answers one question. Faithfulness needs at least two.

Feature Matching Beyond QC

Real uses

- **Where matching earns its keep**

– Panorama stitching: match keypoints between overlapping frames, fit a homography, warp and blend.

– Document rectification: four correspondences give the perspective transform that flattens a photographed page.

– Duplicate and near-duplicate detection across an asset library.

– Verifying a delivery IS a variant of the approved master, not a different shoot.

Honest limits

- **What it will not do for you**

– It will not survive a style transfer or a recolour — the descriptors change.

– It will not survive a heavy crop that removes the matched region.

– It will not tell you the image is on-message, legally cleared or well-composed.

– State the limits in the gate specification, or someone will assume it detects everything.

cv2.findHomography with RANSAC is the standard next step after matching: it fits the transform while rejecting the outliers your ratio filter missed.

Turning Four Numbers Into a Review Gate

1 **Normalise before you weight**
You cannot add a Bhattacharyya distance to a saturation difference in raw units. Divide each by a stated 'worst' value and clip to [0,1].

2 **Weight by what matters**
Lab 07 weights the histogram at 0.40 and the other three at 0.20 each, because palette drift is the failure the client actually complains about.

3 **Set the threshold with margin**
0.35, with the lowest rejection at 0.406 and the highest acceptance at 0.110. A gate with no margin is a coin toss.

4 **Report the WORST descriptor**
Not just the combined score. 'Rejected, edge-density ratio 1.35 against a 1.0 reference' is actionable; a combined score is not.

5 **Escalate, do not just reject**
If the logo score fails but the logo is visibly present, re-test with ORB before rejecting. Write the escalation into the gate.

6 **State what it does NOT assess**
Copy, legal claims, whether the asset is on-message. Visual consistency only.

Lab 06 — Local Features: Colour Segmentation, Canny Edges and Corner Keypoints

INPUTS

Synthetic 900x700 campaign asset containing the slate-blue brand block, a hard-edged product silhouette, a fine-texture panel and a chequered calibration corner (deterministically rendered with OpenCV — SIMULATED)

TECHNICAL QUESTION

Which feature measurements survive the supplied lighting, rotation and scale changes?

EVIDENCE TO PRODUCE

`out/features-report.json` with the HSV segmentation coverage, the Canny threshold sweep, response/count measurements for five detectors, and an ORB match count between the asset and its rotated copy.

LAB 06 · 30 MIN · Python 3 · opencv-python · NumPy · no network, no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-06-local-features-edges-keypoints/README.md

TOPIC 03 · ELO3 · A3, A4

Lab 07 — Global Descriptors, Template Matching and Brand-Consistency Scoring

INPUTS

The approved campaign master, 900x700 (deterministically rendered — SIMULATED)

TECHNICAL QUESTION

Which descriptor or logo gate identifies each deliberately altered asset?

EVIDENCE TO PRODUCE

`out/consistency-scores.csv` ranking all 12 assets against the master with four global descriptors and a combined score, plus `out/logo-placement.csv` from template matching, and a reviewed list of fixture deviations with the descriptor or logo gate that flagged each, including any misses.

LAB 07 · 30 MIN · Python 3 · opencv-python · NumPy · no network, no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-07-global-descriptors-consistency/README.md

TOPIC 03 · ELO3 · K5, K6, K7, A3, A4

Recap — Topic 03

- A good feature is repeatable, discriminative, local enough, compact, efficient, and matched to the failure you are trying to catch.
- Local descriptors answer 'what is at this point'; global descriptors answer 'what is this whole image like'. Brand-consistency work needs global.
- Canny is four stages — smooth, gradient, non-maximum suppression, hysteresis — and you cannot tune it without knowing which stage each parameter touches.
- Harris is rotation-covariant and not scale-invariant; measured, its response fell 40% at half scale. That is why SIFT and ORB build scale pyramids.
- ORB is 32 bytes per keypoint and SIFT is 512 — a 16x storage and bandwidth decision that returns in Topic 05.
- Four global descriptors caught three planted outliers in a 12-asset campaign by three different mechanisms, with a 3.7x margin below the lowest rejection.
- Template matching answers 'is it present and in the right place', and it is neither scale- nor rotation-invariant — state that limit whenever you use it. (A3, A4, K5, K6, K7)

04

TOPIC 04

Generative AI Video Creation, Animation and Editing

Deep learning concepts · object segmentation, detection and recognition · detector evaluation with IoU and precision/recall · storyboard-to-prompt workflow · temporal coherence, tracking, activity and event discovery · captions and export profiles

LU4 + LU5 · ELO4 + ELO5 · K8, K9, K10, A5, A6

Key Concepts — Topic 04

1

Classification, detection and segmentation differ by output shape

Classification returns one label per image; detection returns a list of boxes with labels and scores; segmentation returns a label per pixel. The evaluation metric changes with the output shape.

2

Viola-Jones is a real machine-learning detector

Haar-like rectangle features + integral image + AdaBoost feature selection + a cascade of stages that rejects background windows early. It ships inside OpenCV, so it runs with no downloaded weights.

3

IoU is the contract between a prediction and the truth

IoU = area(intersection) / area(union). At a stated confidence threshold, match detections one-to-one to ground truth with the correct class and an agreed IoU threshold, for example >=0.5. Duplicates/unmatched predictions are false positives; unmatched ground-truth objects are false negatives.

4

Video adds temporal coherence to spatial coherence

Each frame must look right AND be consistent with its neighbours. Flicker, identity drift and warping backgrounds are temporal failures that a per-frame quality score cannot see.

5

3-D U-Nets and Diffusion Transformers buy that coherence

Extending the 2-D U-Net to 3-D adds the time axis so the model learns spatiotemporal features; treating video as a sequence of spatiotemporal patches lets a transformer model long-range dependencies across many seconds.

6

Motion analysis is how you verify continuity numerically

Background subtraction isolates the moving foreground; mean shift and CAMShift climb the back-projected histogram to track it; Farneback optical flow gives a dense displacement field. Centroid jitter and scale drift then become numbers.

Three Tasks, Three Output Shapes, Three Metrics

Classification

- Output: one label per image.
- 'This frame contains a person.'
- Metric: accuracy, precision, recall, F1.
- Cheapest to label and to evaluate.
- Says nothing about where or how many.

Detection

- Output: a list of boxes, each with a label and a score.
- 'Two people, at these coordinates.'
- Metric: IoU, then precision/recall/mAP.
- Needs a matching rule between prediction and truth.

Segmentation

- Output: a label per PIXEL.
- 'These pixels are person, those are shelf.'
- Metric: IoU per class, mean IoU, Dice.
- Most expensive to label; what masked editing consumes.

K8, K9 · IT SHIPS INSIDE opencv-python — NO WEIGHTS TO DOWNLOAD

Viola-Jones — a Real Machine-Learning Detector, Built Into OpenCV

HAAR-LIKE FEATURES

Differences of summed rectangle intensities: an eye band is darker than the cheeks below it, a nose bridge is brighter than its sides.

INTEGRAL IMAGE

Precompute a running sum so ANY rectangle total is four lookups regardless of its size. This is what makes tens of thousands of features affordable.

ADABOOST

Train weak classifiers on single features and keep only the small subset that actually discriminates, weighting the hard examples more each round.

CASCADE OF STAGES

Apply the features in stages. A window failing any stage is rejected immediately, so most background is discarded in the first few stages.

Why it still matters

It runs on a CPU, offline, with no model file. That is why this course can teach detection without a download.

What it is not

It is not deep learning. It is boosted feature selection over hand-designed features.

Two parameters

scaleFactor steps the image pyramid; minNeighbors sets how many overlapping hits are required.

Cascade scores

detectMultiScale3 can return levelWeights; they are ranking scores, not calibrated probabilities.

cv2.data.haarcascades points inside the installed package and holds the trained frontal-face and eye cascade XML files.

Running Both Built-In Detectors

```
# 1 — Viola-Jones cascade (intensity-difference features)
cas   = cv2.CascadeClassifier(cv2.data.haarcascades +
                              'haarcascade_frontalface_default.xml')
gray  = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)      # MUST be grayscale
faces = cas.detectMultiScale(gray, scaleFactor=1.05,
                             minNeighbors=2, minSize=(24, 24))

# 2 — HOG + linear SVM (gradient-statistics features)
hog = cv2.HOGDescriptor()
hog.setSVMDetector(cv2.HOGDescriptor_getDefaultPeopleDetector())
boxes, weights = hog.detectMultiScale(img, winStride=(8, 8),
                                      padding=(8, 8), scale=1.05)

# optional dataset-specific HOG box adjustment; evaluate raw and adjusted boxes
x, y, w, h = boxes[0]
pw, ph     = int(0.15 * w), int(0.05 * h)
corrected  = [x + pw, y + ph, w - 2*pw, h - 2*ph]
```

Cascade: intensity differences

Rectangle sums via an integral image, selected by AdaBoost, applied in stages.

HOG: gradient statistics

Cell histograms of oriented gradients, block-normalised, classified by a linear SVM at many scales.

HOG returns weights

Those SVM decision values are what you sweep to build a real precision-recall curve.

Declare post-processing

This fixture uses15%/5% box shrink. Compare raw and adjusted boxes; it is not universally beneficial.

VERIFY Both detectors are evaluated in Lab 08 against exact ground truth that was generated WITH the frames, so every box is exact rather than annotated.

IoU — the Contract Between a Prediction and the Truth

Definition

$$\text{IoU} = \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)}$$

Intersection over union. 1.0 is a perfect box; 0.0 is no overlap at all. Everything in detection evaluation hangs off this one ratio.

Worked by hand

```
A=[0,0,100,100]
B=[50,0,100,100]
∩ = 50×100 = 5000
∪ = 20000−5000
  = 15000
IoU = 0.333
```

Two boxes overlapping on half their width. Compute it once by hand — Lab 08's script asserts exactly this value as a self-test before it evaluates anything.

The matching rule

```
greedy, by score
same class only
each truth box
used at most ONCE
```

That last clause is what stops a detector scoring well by emitting the same box ten times.

A prediction that matches an unused ground-truth box above the threshold is a true positive; an unmatched prediction is a false positive; an unmatched ground-truth box is a false negative.

K9, A5 · WRITE THE THREE FORMULAS, NOT JUST THE NUMBERS

Precision, Recall and F1

Precision

$$\frac{TP}{TP + FP}$$

Of everything the detector flagged, what fraction was real? Low precision means a reviewer wades through noise.

Recall

$$\frac{TP}{TP + FN}$$

Of everything that was really there, what fraction did the detector find? Low recall means things never reach a human at all.

F1

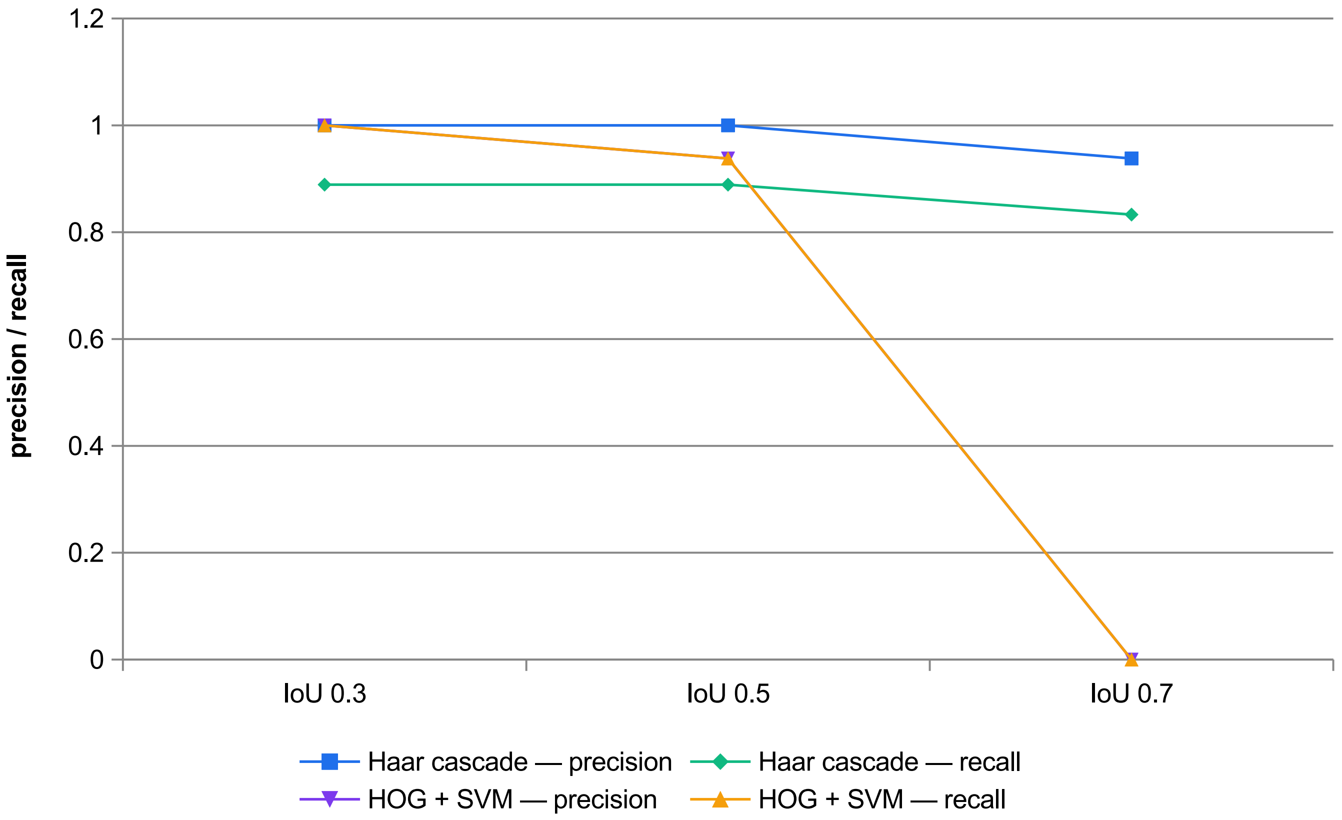
$$\frac{2 \cdot P \cdot R}{P + R}$$

The harmonic mean. Useful as a single summary, and misleading as an objective — the cost of a false positive and a false negative are rarely equal.

Where the denominator is zero, report 0.0 AND say it is a zero-denominator case rather than a measured score. A silent 0.0 reads as a result.

K9, A5 · LAB 08 · 12 FRAMES, 18 FACE BOXES, 16 PERSON BOXES

Measured — Both Detectors Across Three IoU Thresholds



Both are strong at 0.5

Cascade P=1.00 R=0.89; HOG P=R=0.94. On this easy, synthetic distribution.

HOG collapses at 0.7

Precision and recall both hit zero. Its boxes are correct in POSITION but not tight enough in EXTENT.

That is a localisation finding

It is not 'HOG is bad'. It is 'HOG's localisation does not meet a 0.7 contract'.

Choose the threshold first

A review queue can live with 0.3. An auto-crop pipeline cannot.

MEASURED

Measured in Lab 08 with opencv-python 4.13 on the 12 supplied SYNTHETIC frames, whose ground truth was generated with the frames. These numbers describe the detectors on THIS distribution and do not transfer to photographic data.

A5 · THE SAME DETECTOR, TWO DIFFERENT SETTINGS

Choosing an Operating Point — the Cost Asymmetry Decides

Recall-weighted

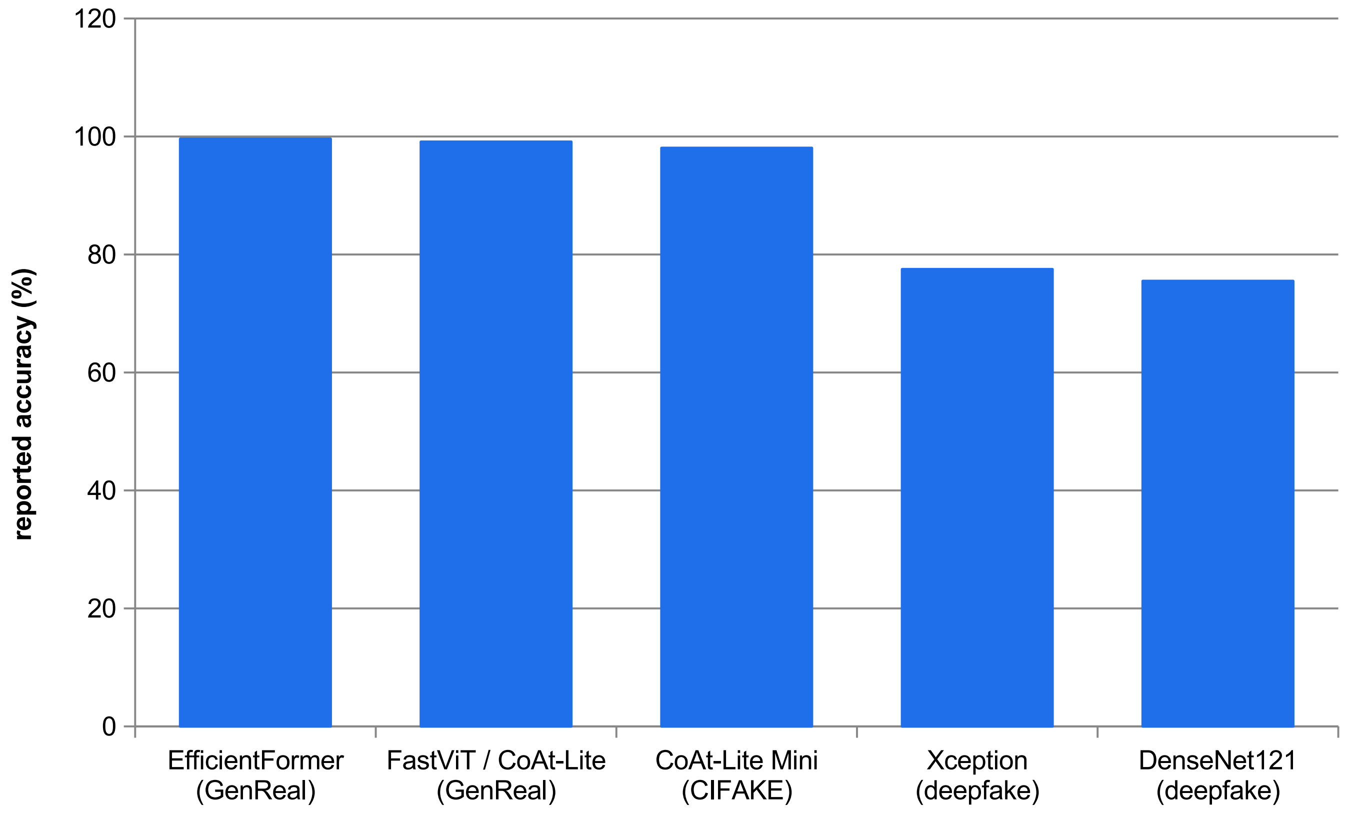
- **Human review queue**
  - Rule: maximise recall subject to precision ≥ 0.60.
  - A false positive costs one reviewer glance.
  - A false negative never reaches a human at all.
  - So you tolerate noise to avoid misses.
  - Lower minNeighbors; lower the SVM weight threshold.

Precision-weighted

- **Auto-publish pipeline**
  - Rule: maximise precision subject to recall ≥ 0.40.
  - A false positive reaches a customer.
  - A false negative just means one fewer asset.
  - So you tolerate misses to avoid errors.
  - Raise minNeighbors; raise the SVM weight threshold.

If no parameter value satisfies your stated rule, the verdict is DO NOT DEPLOY. Do not relax the rule to manufacture an operating point.

Published — Detection Accuracy Is Dataset-Specific



99%+ on curated benchmarks
Distinguishing AI-generated from real images on a purpose-built dataset.

Mid-to-high 70s on hard data
Deepfake detection on a shifted, adversarial distribution.

Never quote a bare number
State the metric, the dataset, the sample size and the distribution type, or the number means nothing.

Report the whole set
Both sources report macro precision, recall and F1 alongside accuracy, because accuracy alone hides class-wise behaviour.

PUBLISHED

Sources S16 and S19 · AI-Generated Image and Video Synthesis, Ch.4 pp.51–53 (AI-generated image detection on the GenReal and CIFAKE datasets, test accuracy) and Ch.11 pp.153–159 (deepfake detection backbones, validation accuracy). Different datasets, different tasks — the bars are NOT comparable to each other.

K10 · THE DEFINING PROBLEM OF GENERATIVE VIDEO

Video adds temporal coherence to spatial coherence.

Every frame must look right AND be consistent with its neighbours. Flicker, identity drift and warping backgrounds are temporal failures that no per-frame quality score can see.

How Early Video GANs Failed

1. **Temporal artifacts**
   Flicker: the texture or colour of an object changes erratically between frames, even when each frame is individually plausible.

2. **Inconsistent motion**
   Jittery or physically implausible movement, because nothing in the objective required continuity.

3. **Mode collapse**
   The same limited set of motions or scenes, repeated — the image-GAN failure, extended along the time axis.

4. **Specialised architectures**
   VGAN separated moving foreground from static background; MoCoGAN and TGAN tried to disentangle content from motion.

5. **They still could not hold**
   Long-term consistency remained out of reach, and clips stayed short and low resolution.

6. **The root cause**
   The training signal is indirect — the generator only learns whether the discriminator was fooled by the whole sequence.

From 2-D U-Net to Diffusion Transformer

2-D U-NET

The image-diffusion workhorse. Convolutions over height and width only — it has no way to represent time.

3-D U-NET

Extend the convolutions and attention to a height × width × TIME volume. The model now learns spatiotemporal features directly, so object identity can persist.

PATCHES, NOT PIXELS

Stop treating video as a dense pixel grid. Cut it into small spatiotemporal patches, each carrying both spatial and temporal information.

DIFFUSION TRANSFORMER

Run diffusion over that patch SEQUENCE with self-attention. Long-range dependencies across many seconds become modellable, and the architecture scales.

SO WHAT Diffusion's learning objective is direct and stable at every step — predict the noise. Augmented with temporal attention, the model is forced to learn frame-to-frame relationships in order to do its job. That is the 'consistency bottleneck' argument for diffusion over GANs.

K10 · SOURCE S22 · ai.google.dev/gemini-api/docs/veo

Current Video Generation — Verified 6 September 2026

Model ID	Duration	Aspect	Resolution	Notes
veo-3.1-generate-preview	4, 6 or 8 s	16:9 or 9:16	720p / 1080p / 4k	Native audio; up to 3 reference images; video extension adds 7 s
veo-3.1-fast-generate-preview	4, 6 or 8 s	16:9 or 9:16	720p / 1080p / 4k	Same feature set, faster and cheaper
veo-3.1-lite-generate-preview	4, 6 or 8 s	16:9 or 9:16	720p / 1080p	No 4k; image-to-video supported
veo-3.0-generate-001	—	—	—	DEPRECATED — do not specify

Generation is a LONG-RUNNING OPERATION: you submit, then poll client.operations.get(operation) until operation.done. Model IDs change — re-verify before you rely on one.

K10, A6 · THE CONSTRAINT THAT SHAPES THE WHOLE WORKFLOW

A documented 8-second maximum means 24 seconds is three shots.

Not one clip. And three separately sampled shots will not hold continuity by themselves — you have to specify it, use the documented continuity mechanisms, and then measure whether it held.

Storyboard to Delivered Film

BRIEF → BEATS

Extract the mandatory beats, the duration and the delivery ratios. A beat with no shot is a brief failure, not an editorial choice.

SHOT LIST

One row per shot: seconds, PURPOSE, subject action, camera, location, audio. A shot with no purpose gets cut, not shortened.

CONTINUITY BIBLE

The fields that must not drift, each with a risk clause. Written once, pasted byte-for-byte into every prompt.

PROMPT PACK

Per-shot structured prompt with a per-second audio track and a negative list. Diff the shots and confirm the continuity block is identical.

ANIMATIC

Render the storyboard at the shot-list timings. Review the timing and beat coverage BEFORE any generation spend.

SO WHAT The animatic exists so the brief can be corrected while correction is free. It is a deterministic OpenCV render, labelled ANIMATIC on every frame, and it is never submitted as a generated film.

The Per-Second Audio Specification

```
"audio": {
  "0-1s": "harbour ambience: distant halyards, water against pilings",
  "1-2s": "ambience continues; footfall on timber begins",
  "2-3s": "footfall, one gull call far off",
  "3-4s": "ambience; low sustained string enters underneath",
  "4-5s": "ambience and string; footfall slows",
  "5-6s": "footfall stops; ambience opens out",
  "6-7s": "ambience; string holds",
  "7-8s": "ambience settles, string resolves"
},
"negative": ["no rendered text or logos in frame",
             "no additional people", "no lens flare",
             "no change to the jacket shell colour",
             "no camera shake"]
```

Second by second

The documented way to control native audio is to say what is heard in each second, not to describe a mood.

Ambience, dialogue, music

Three layers. Name each where it enters and where it resolves.

Negatives are not optional

No rendered text or logos in frame: they go on in post so they stay editable and legible.

This is what a shot IS

The pack is the deliverable. The render is downstream of it.

VERIFY Lab 09 diffs the three shot files and confirms every continuity key is character-identical before the pack is submitted.

K10, A6 · SOURCES S17, S22

Documented Continuity Mechanisms — and What Each Buys

1

Continuity block

Copy the locked description into every prompt programmatically. Free, and the single biggest lever.

2

Reference images

Up to three images to guide content. Documented for Veo 3.1 and 3.1 Fast.

3

Image-to-video

Start the shot from a fixed conditioning frame. Supported across versions, and the strongest continuity lever available.

4

Video extension

Adds 7 seconds to an existing clip, so the continuation inherits the original's state rather than being resampled.

5

Retrieval conditioning

Cross-attention to exemplar images for style control, without increasing parameter count.

6

Then measure it

Centroid jitter, scale drift and inter-frame SSIM. Every mechanism above improves the odds; none guarantees the result.

Background Subtraction — Isolating What Moves

```
sub = cv2.createBackgroundSubtractorMOG2(history=200, varThreshold=16,
                                         detectShadows=True)
fg  = sub.apply(frame)

# detectShadows=True labels shadow pixels 127, not 255 —
# threshold at 200 or you count shadows as subject
_, hard = cv2.threshold(fg, 200, 255, cv2.THRESH_BINARY)

# the alternative, a k-nearest-neighbour model of each pixel
knn = cv2.createBackgroundSubtractorKNN(history=200, dist2Threshold=400.0)

# MEASURED in Lab 10, foreground pixel count over the first ten frames:
#   MOG2 : 0, 0, 0, 0, 73, 1229, 834, 799, 1185, 967
#   KNN  : 307200, 307200, 307200, 307199, 2798, 2744, ...
# both need history before they are trustworthy — MOG2 reports nothing,
# KNN reports EVERYTHING. The first frames of any BS pipeline are junk.
```

A per-pixel model of the scene

MOG2 models each pixel as a mixture of Gaussians over recent history.

Deviation means foreground

A pixel that no longer fits its own model is flagged, producing a binary mask.

It requires a static camera

A pan invalidates the model for every pixel at once.

Warm-up is real

Measured above: the two algorithms fail in opposite directions for the first four frames.

VERIFY Lab 10 runs both and asks you to record which stabilises faster and why the first frames cannot be used.

Mean Shift and CAMShift — the Same Climb, One Difference

Fixed window

- **Mean shift**

– Build a hue histogram of the initial window; back-project it onto each frame.

– Move the window to the local maximum of that density. Repeat until it settles.

– The window size NEVER changes.

– So as the subject approaches the camera, the window stops containing it.

Adaptive window

- **CAMShift**

– Continuously Adaptive Mean Shift. Identical back-projection and climb.

– But it recomputes the window size from the zeroth moment of the back-projection each iteration.

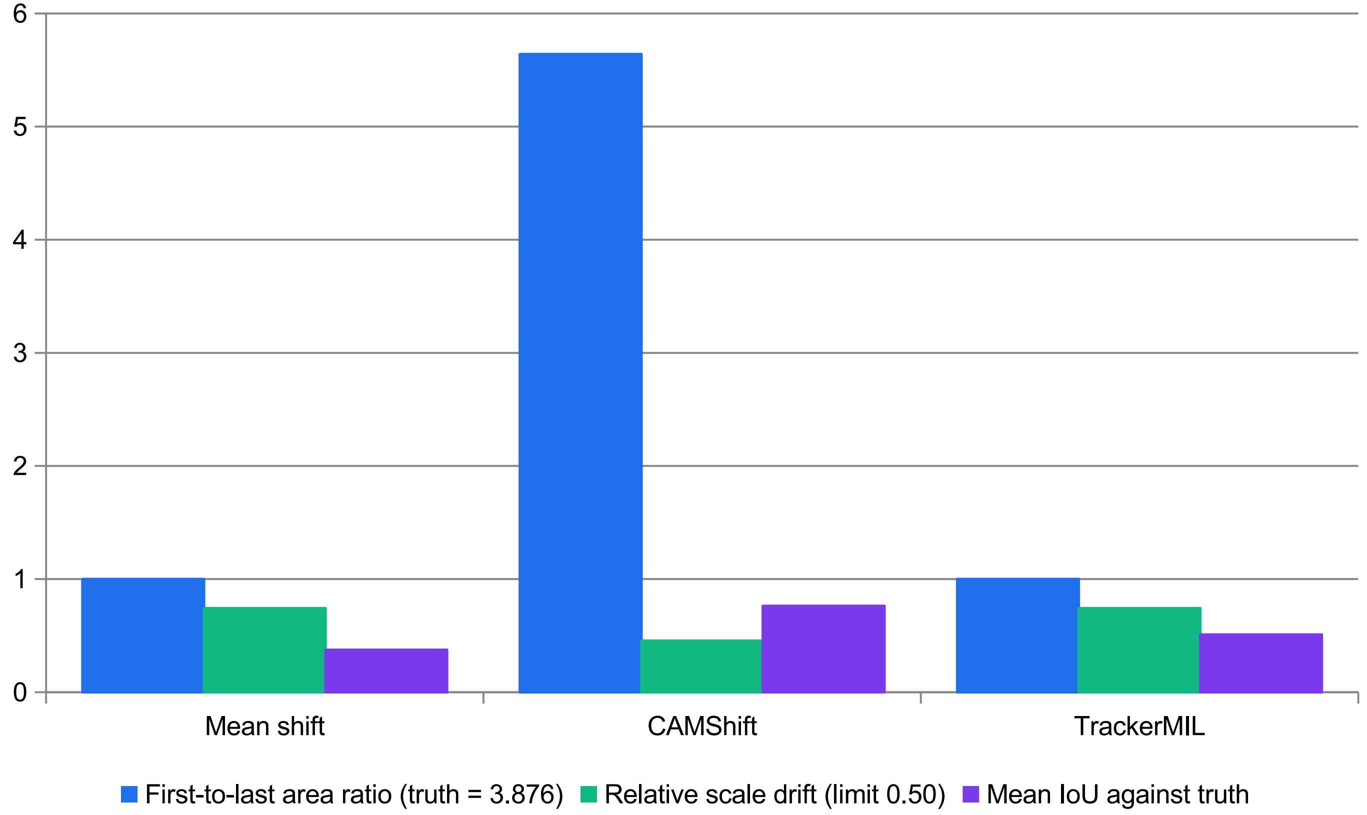
– And it returns an orientation as well as a box.

– So it adapts to a subject that changes apparent size.

Lab 10 measures the difference on a clip whose subject grows by a known factor of 3.876 between the first and last frame.

K10, A6 · LAB 10 · 200 FRAMES, PER-FRAME GROUND TRUTH

Measured — Three Trackers Against Exact Ground Truth



Mean shift: ratio exactly 1.000

The window never resized, against a true growth of 3.876x. This is the defining limitation, measured.

CAMShift: adapts, and overshoots

5.640 against 3.876 — a relative drift of 0.455. It passes the 0.50 limit, but only just.

CAMShift wins on IoU

0.762 against 0.374 and 0.509. For a subject that changes size, it is the right choice.

Report per tracker

The same clip passes with one tracker and fails with another. The failure belongs to the tracker, not the clip.

MEASURED

Measured in Lab 10 with opencv-python 4.13 on the supplied synthetic clip, whose exact per-frame subject box was recorded when the clip was rendered. Relative drift is |measured ratio / truth ratio − 1|.

Dense Optical Flow and Automatic Continuity Checking

```
prev = cv2.cvtColor(frames[0], cv2.COLOR_BGR2GRAY)   # MUST be single channel
mags = []
for frame in frames[1:]:
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    flow = cv2.calcOpticalFlowFarneback(prev, gray, None,
                                        0.5, 3, 15, 3, 5, 1.2, 0)
    mag, ang = cv2.cartToPolar(flow[..., 0], flow[..., 1])
    mags.append(float(mag.mean()))
    prev = gray

# a discontinuity is a spike against the clip's own rolling behaviour
median = float(np.median(mags))
spikes = [i + 1 for i, m in enumerate(mags) if m > 4.0 * median]
print(spikes)      # MEASURED in Lab 10: [81, 100, 102, 121, 162]
#                    transitions100 and102 enter and leave the declared displacement.
#                    The other three are candidates, not defects.
```

Flow is H x W x 2

A (dx, dy) displacement for every pixel. cartToPolar turns it into magnitude and angle.

Expensive

Measured at 54.2 ms per 1280x720 frame in Lab 12 — the single most costly stage in the pipeline.

A declared cut is not a defect

An UNDECLARED one is. That distinction is what the acceptance gate encodes.

Candidates need review

Three of five spikes are outside transitions 100/102; the other spikes need review. Report candidates, not as findings.

VERIFY Lab 10 cross-checks the flow spike against a spike in centroid step size, so a continuity break is confirmed by two independent measurements.

Three Continuity Metrics

Centroid jitter

```
σ( ‖cₜ − cₜ₋₁‖ )
excluding declared
discontinuities
```

Standard deviation of the frame-to-frame centroid step. Turns 'it feels jittery' into a number with a limit. Lab 10's limit is 3.5 px.

Scale drift

$$\left| \frac{\text{ratio\_measured}}{\text{ratio\_truth}} - 1 \right|$$

RELATIVE, not absolute. The relative form normalises the error by true growth; the declared limit is0.50.

Inter-frame SSIM

```
mean SSIM(fₜ₋₁, fₜ)
```

Falls sharply at a cut or a flicker. Measured at 0.9902 on the course clip. State the sampling interval — computing it on every pair of a long clip is expensive.

Compute a metric only on frames where the track is VALID. A jitter figure computed across frames where the tracker had already lost the subject is not a measurement of the clip.

Activity, Scene Understanding and Event Discovery

1

Activity tracking
A track is an identity over time. Once you have a stable track you can measure dwell, speed, direction and trajectory shape.

2

Scene understanding
Beyond 'what is in this frame' to 'what kind of scene is this, and what is the relationship between the things in it'.

3

Event discovery
A rule over a track: crossed a line, dwelled more than N seconds, entered a region, count exceeded a threshold. This is what you actually send upstream.

4

Generative models here
Synthesising rare events for training, reconstructing occluded views, and predicting near-term state — all listed under K10 alongside tracking.

5

Why events, not frames
An event record is a few hundred bytes. A frame is megabytes. Topic 05 turns that into a payload-rate bandwidth comparison.

6

The honest caveat
Every one of these inherits the tracker's errors. A dwell-time statistic built on a track with mean IoU 0.37 is not a statistic.

A6 · MEASURED IN LAB 11: 34 px, 33 px AND 58 px ACHIEVED CAP HEIGHT

Captions That Survive the Delivery Ladder

```
# compute the font scale from the REQUIRED cap height — never hard-code it
(_, h1), _ = cv2.getTextSize('Hg', cv2.FONT_HERSHEY_SIMPLEX, 1.0, 2)
scale      = required_cap_height_px / float(h1)

(tw, th), base = cv2.getTextSize(text, cv2.FONT_HERSHEY_SIMPLEX, scale, 2)

# safe area: inset the frame, and place the caption FROM that rectangle
x0, y0, x1, y1 = safe                  # e.g. 8% inset on a 9:16 rendition
x   = x0 + max(0, ((x1 - x0) - tw) // 2)
y   = y1 - base - 10
pad = max(6, int(cap_h * 0.25))

# an opaque plate FIRST, then the text — never bare text over picture
cv2.rectangle(frame, (x-pad, y-th-pad), (x+tw+pad, y+base+pad), (18,18,22), -1)
cv2.putText(frame, text, (x, y), cv2.FONT_HERSHEY_SIMPLEX, scale,
            (245, 245, 245), 2, cv2.LINE_AA)
```

Derive the scale

A fixed font scale that is legible at 1080p is unreadable at 1080x1080 after a crop.

Plate before text

Text over picture is unreadable on a bright frame and there is no fix in post.

Position from the safe area

Never from a constant offset. That is what puts a caption off-frame in one rendition.

Measure the contrast

Plate-to-text luminance ratio, measured, not judged. Lab 11 records it per profile.

VERIFY Lab 11 verifies every cue in every profile against the safe rectangle and writes out/safe-area-check.csv with a PASS or FAIL per cue.

The Export Ladder — Crop or Letterbox Is a Content Decision

Profile	Output	Reframe	What it costs	Measured size
16x9	1920 x 1080	none	Nothing — the master ratio	2.41 MB
1x1	1080 x 1080	centre crop	The sides; the subject sits centrally in every shot	1.59 MB
9x16	1080 x 1920	centre crop	31.64% of the source width; a letterbox would leave 68.36% of the portrait frame as bars	2.35 MB

Measured in Lab 11 with opencv-python 4.13 on the 12-second synthetic source. Every rendition inherits the source's SIMULATED provenance label — the label travels to every derivative.

Responsible Practice for Synthetic Video

1. **Deepfakes are the named risk**
   Synthetic video of a real person saying something they did not say is the headline harm in every responsible-AI framework.

2. **Voice cloning too**
   Native audio generation extends the same risk to speech — fraud and defamation are the cited misuse cases.

3. **C2PA Content Credentials**
   Tamper-evident metadata recording origin, creator and edit history, so a viewer can verify a file's provenance.

4. **Robust watermarking**
   An invisible signal embedded in the content itself, designed to survive compression, cropping and format change. Current Gemini image outputs carry SynthID.

5. **Model cards**
   A 'nutrition label for AI': version, architecture, developer, intended use and known limitations, published with the model.

6. **Test the watermark, do not assume it**
   Detectability must be evaluated under the edits your pipeline actually applies, not on the pristine file.

K10 · THE RULE THAT APPLIES TO EVERY ARTEFACT YOU PRODUCE

Never present a deterministic render as generative-model output.

The animatic in Lab 09 carries a burned-in label on every frame for exactly this reason. Mislabelling provenance is a governance failure, not a presentation choice — and it is the fastest way to lose an audit.

Deep-Learning Detector Families — Know the Shape of Each

Family	How it works	Trade-off
Two-stage (R-CNN, Fast, Faster R-CNN)	A region proposal network suggests candidate boxes; a second stage classifies and refines each one	Highest accuracy, slowest — two passes per image
Single-shot (SSD)	No proposal stage. Predicts boxes and classes directly from feature maps at several scales, using default boxes per cell	Real-time, weaker on small objects
YOLO family	Divides the image into a grid; each cell predicts a fixed number of boxes with a class and an objectness score, in one pass	Very fast; the design point for live video and embedded use
Transformer detectors	Treat detection as set prediction with attention; no hand-designed anchors	Simpler pipeline, heavier training

This course runs the two detectors that ship inside OpenCV, so that detection is taught with no download and no accelerator. The families above are the landscape those two sit in — each output is still a list of boxes and is still evaluated with IoU.

Non-Maximum Suppression — Turning Many Boxes Into One

The problem

```
one object
-> many boxes
```

Every detector fires at several nearby positions and scales for the same object. Raw output is a cloud of overlapping boxes, not a detection.

The algorithm

```
sort by score
take the best
drop any box with
IoU > t against it
repeat
```

Greedy suppression by overlap. The same IoU you use for evaluation is doing the work here — a threshold around 0.4 to 0.5 is typical.

Where you have met it

```
minNeighbors
in the cascade
```

The cascade's minNeighbors parameter is a simpler version of the same idea: it requires several overlapping hits before emitting one detection.

Set the NMS threshold too high and you keep duplicates, inflating false positives; too low and you suppress genuinely adjacent objects, inflating false negatives. Either way, precision and recall move.

Semantic Segmentation — Classification at the Pixel Level

ENCODER
The same convolutional stack as a classifier, progressively reducing spatial resolution while increasing semantic depth.

BOTTLENECK
The smallest, deepest representation. It knows WHAT is in the image and has largely lost WHERE.

DECODER
Upsamples back to full resolution, recovering spatial detail. An encoder-decoder architecture of exactly this shape underlies multi-class pixelwise segmentation.

SKIP CONNECTIONS
Carry high-resolution detail from encoder to decoder so the boundaries are sharp. This is the U in U-Net — and the same backbone image diffusion uses.

Output is a label map
One class per pixel, not one per image and not a box.

Metric is per-class IoU
Then mean IoU across classes, or the Dice coefficient.

This is what masking wants
A segmentation model is what produces the semantic masks that Topic 02's edit modes consume.

Same U-Net, two jobs
Segment an image, or denoise a latent. The architecture does not care.

Notice the loop closing: the encoder-decoder you met as a VAE in Topic 01 is the segmentation backbone here, and the diffusion backbone in Topic 04.

K9, K10 · A DISTINCTION THE ASSESSMENT EXPECTS YOU TO MAKE

Detection Is Not Tracking

Where is it now?

- **Detection**

– Per frame, independently. No memory of the previous frame.

– Output: a set of boxes with labels and scores.

– Identity is not preserved — box 1 in frame 10 and box 1 in frame 11 may be different objects.

– Evaluated with IoU, precision, recall, mAP.

Is that the same thing?

- **Tracking**

– Across frames. Maintains an identity over time.

– Output: a trajectory per object.

– Identity IS the deliverable — that is what lets you measure dwell time, speed and continuity.

– Evaluated with track-level measures: IoU over time, identity switches, centroid jitter, scale drift.

Tracking-by-detection combines them: detect every frame, then associate detections across frames by IoU or appearance. Every association error becomes an identity switch.

Frame Differencing — the Cheapest Motion Detector

```
prev = cv2.cvtColor(frames[0], cv2.COLOR_BGR2GRAY)
for frame in frames[1:]:
    gray  = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    diff  = cv2.absdiff(prev, gray)
    _, mv = cv2.threshold(diff, 25, 255, cv2.THRESH_BINARY)
    mv    = cv2.morphologyEx(mv, cv2.MORPH_OPEN, kernel)   # drop specks
    moving_px = int(cv2.countNonZero(mv))
    prev = gray

# vs background subtraction:
#   frame differencing  -> detects CHANGE between two frames. An object that
#                          stops moving disappears immediately.
#   background subtract -> models the SCENE over history. A stopped object stays
#                          foreground until the model absorbs it.
```

Two frames, one subtraction

No model, no history, no warm-up. Runs on anything.

It sees change, not objects

A stationary person vanishes. That is sometimes exactly what you want, and sometimes fatal.

Threshold plus morphology

Raw absdiff is noisy. Threshold, then open to remove single-pixel specks.

Choose by the question

'Did anything move?' — difference. 'What is not part of the scene?' — background subtraction.

VERIFY Lab 10 runs MOG2 and KNN background subtraction and shows both need history before they can be trusted; frame differencing needs none.

From a Track to an Event Worth Sending

Event rule	Computed from the track	Typical payload
Line crossing	The centroid trajectory crosses a defined segment	timestamp, direction, track id
Dwell	The centroid stays inside a polygon for more than N seconds	timestamp, region, duration
Region occupancy	The count of distinct active tracks inside a polygon	timestamp, region, count
Shelf gap	A region's foreground fraction falls below a reference for N consecutive frames	timestamp, shelf id, fraction
Continuity break	Mean optical-flow magnitude exceeds 4x the rolling median	frame index, magnitude, declared or not

This is the payload at split point D in Topic 05: measured at 0.00032 Mbit/s per store against 553 Mbit/s for raw frames. Every event rule inherits the tracker's errors — a dwell statistic built on a track with mean IoU 0.37 is not a statistic.

Detecting Synthetic Media — and Reading the Numbers Honestly

Method

- **How detection is framed**

– Supervised binary classification: real against generated, trained on a labelled dataset.

– Backbones reported include EfficientFormer, FastViT, CoAt-Lite, CoAtNet and ConvNeXt.

– Grad-CAM is used to show WHICH pixels drove the decision — explainability, not just a score.

– Macro precision, recall and F1 are reported alongside accuracy.

Reading the result

- **Why the numbers diverge**

– 99.6% on GenReal and 98.04% on CIFAKE — curated benchmarks for AI-generated image detection.

– 77.5% validation accuracy for the best deepfake backbone on a harder, shifted distribution.

– Same metric name, different problem. The bars are not comparable.

– Detection performance does not transfer to a generator the detector has never seen.

This matters practically: a client asking 'can you detect AI images?' is asking a question whose answer depends entirely on which generator, which dataset and which distribution.

Naming Your Video Deliverables Correctly

1. **ANIMATIC**
   A deterministic render of storyboard frames at shot-list timings. Labelled on every frame. Exists to fix timing before spend.

2. **PREVIS**
   A rough moving version of a shot for blocking and camera. Still not a deliverable.

3. **GENERATED CLIP**
   Model output. Record the model ID, the prompt version, the seed policy where one exists, and the watermark expectation.

4. **EDITED CUT**
   An assembly of clips. Its provenance is the union of its sources — if any source is simulated, the cut is a derivative of a simulated asset.

5. **RENDITION**
   A reframed, captioned export of the cut. Inherits everything above. The label travels to every derivative.

6. **The rule**
   Every file carries what it is. A deterministic render is never presented as model output, and a simulated stand-in is never presented as a photograph.

TOPIC 04 · ELO4 · A5

Lab 08 — Object Detection and Quantitative Evaluation

INPUTS

Twelve 640x480 synthetic scene frames containing schematic face-like and person-like targets at known pixel positions (deterministically rendered with OpenCV — SIMULATED classroom assets, not photographs). Because the ground truth is generated with the frames, every box is exact.

TECHNICAL QUESTION

How do confidence and IoU thresholds change false positives and missed objects?

EVIDENCE TO PRODUCE

`out/detections.json` with every predicted box, `out/evaluation.csv` with TP/FP/FN, precision, recall and F1 at three IoU thresholds, and a stated operating point with the parameter values that produce it.

LAB 08 · 60 MIN · Python 3 · opencv-python · NumPy — both detectors ship inside the base wheel, so no model weights are downloaded and no paid service is used

Detailed procedures and prompts: Learner Guide / labs/lab-08-object-detection-evaluation/README.md

TOPIC 04 · ELO5 · A6

Lab 09 — Storyboard to Video Prompt Pack and Deterministic Animatic

INPUTS

Nine 1280x720 storyboard frames, three per shot (deterministically rendered with OpenCV — SIMULATED storyboard art, not generative-model output)

TECHNICAL QUESTION

Do shot durations and storyboard intent agree with the 24-second animatic?

EVIDENCE TO PRODUCE

`out/shot-list.csv`, `out/continuity-bible.md`, `out/video-prompt-pack.json` (three shots with per-second audio), and `out/animatic.mp4` — a deterministic OpenCV render of the storyboard frames, explicitly labelled as an animatic and NOT as generative-model output.

LAB 09 · 30 MIN · Text editor · Python 3 · opencv-python · OPTIONAL free-tier video model. The lab completes fully offline; no paid service is required.

Detailed procedures and prompts: Learner Guide / labs/lab-09-storyboard-video-prompting/README.md

TOPIC 04 · ELO5 · A6

Lab 10 — Motion, Tracking and Temporal Continuity Measurement

INPUTS

Synthetic 8-second 640x480 25 fps clip: a subject block translates across a static shelf background while growing in scale, with a deliberate two-frame position discontinuity at t=4.0 s (deterministically rendered with OpenCV — SIMULATED, not generative-model output)

TECHNICAL QUESTION

Do tracked growth, centroid jitter and frame similarity tell the same story?

EVIDENCE TO PRODUCE

`out/tracking.csv` with per-frame centroid and box size for three trackers, `out/continuity-report.json` with jitter, drift and inter-frame SSIM statistics, and annotated overlay videos for each tracker.

LAB 10 · 60 MIN · Python 3 · opencv-python · NumPy · no network, no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-10-motion-tracking-continuity/README.md

Lab 11 — Captions, Overlays, Safe Areas and Export Profiles

INPUTS

Synthetic 12-second 1920x1080 25 fps approved cut (deterministically rendered with OpenCV — SIMULATED, not generative-model output)

TECHNICAL QUESTION

Do every caption and overlay satisfy safe areas, line limits and export dimensions?

EVIDENCE TO PRODUCE

Three exported renditions (16:9, 1:1, 9:16) with burned-in captions and overlay, `out/export-report.csv` with per-rendition dimensions, duration, file size and bit-rate, and `out/safe-area-check.csv` proving the caption and logo stay inside the safe area in all three.

LAB 11 · 30 MIN · Python 3 · opencv-python · NumPy · no network, no paid service; silent fixture (OpenCV export does not preserve audio)

Detailed procedures and prompts: Learner Guide / labs/lab-11-captions-overlays-export/README.md

TOPIC 04 · ELO4 + ELO5 · K8, K9, K10, A5, A6

Recap — Topic 04

- Classification, detection and segmentation differ by output shape, and the evaluation metric changes with the shape.
- Viola-Jones is genuine machine learning — Haar features, integral image, AdaBoost selection, cascaded stages — and it ships inside OpenCV with no download.
- IoU is the contract; a prediction is a true positive only above the agreed threshold, and each ground-truth box may be matched once.
- Measured on the course dataset, HOG collapses from 0.94 to 0.00 between IoU 0.5 and 0.7 — a localisation finding, not a verdict on the detector.
- Video adds temporal coherence; 3-D U-Nets and diffusion transformers over spatiotemporal patches are what bought it.
- A documented 8-second maximum makes a 24-second film three shots, held together by a continuity block, reference images or image-to-video — and then measured.
- Mean shift never resizes its window (measured ratio 1.000 against a true 3.876); CAMShift adapts and won on mean IoU 0.762.
- Captions are computed from the safe rectangle and the required cap height, and every export inherits the source's provenance label. (A5, A6, K8, K9, K10)

05

TOPIC 05

Evaluating Cloud and Edge AI Creative Workflows

Vision system architecture — the four-layer sensors-to-synthesis stack · vision communication protocols · real-world design constraints — latency, bandwidth, privacy, cost and energy · measured cloud-versus-edge trade-offs

LU6 · ELO6 · K11, K12, K13, A7, A8

Key Concepts — Topic 05

1

A generative visual system has four layers

IoT sensing (cameras, drones, wearables) -> edge intelligence (pre-processing, lightweight inference, federated learning) -> generative visual modelling (VAE/GAN/diffusion, split computing) -> application (dashboards, AR/VR, digital twins).

2

Split computing is the real architecture decision

Latency-critical stages run at the edge; training and heavy sampling run in the cloud. You choose the split point by measuring per-stage latency and the payload size that crosses it.

3

Protocols are chosen by payload and latency class

MQTT and CoAP for small telemetry and events, RTSP/WebRTC for live video, HTTPS/REST and gRPC for model calls, S3-style object storage for renders. Illustrative payload-only example: 100 kB/frame x 10 fps = 1000 kB/s, versus 1 kB/event x 1 event/s = 1 kB/s, a 1000:1 ratio before protocol overhead. Measure actual encoded sizes and event rates before choosing an architecture.

4

Sampling steps and frames influence compute cost

At fixed model, resolution and hardware, S x F is a rough workload heuristic. Measured latency also depends on temporal attention, batching and scheduling. Distillation can reduce sampling steps; latent modelling reduces spatial representation cost, not necessarily the frame count.

5

Edge fits through quantisation plus adapters

Moving weights from 32-bit to b-bit scales parameter memory by b/32; low-rank adapters train only r(d+m) parameters instead of d*m. Keep sensitive activations at higher precision for stability.

6

Energy and carbon are reportable engineering numbers

Energy per sample is roughly the datacentre overhead factor times the energy per FLOP times the FLOPs per sample. Report hardware, kWh, overhead, grid carbon intensity and joules per sample alongside quality.

The Sensors-to-Synthesis Architecture

1 IoT SENSING

Cameras, drones, wearables, industrial vision rigs. The 'eyes' of the system. Issues here are energy, device heterogeneity and data security.

2 EDGE INTELLIGENCE

Pre-processing, noise reduction, feature extraction and lightweight inference on a gateway or embedded computer. Also where federated updates happen.

3 GENERATIVE VISUAL MODELLING

VAE, GAN or diffusion, running at the edge, in the cloud, or split between them. Compression, quantisation and split computing live here.

4 APPLICATION

Review queues, asset stores, dashboards, AR/VR surfaces, digital twins. Not just presentation — this is where a decision or an alert is produced.

Why four layers

Each layer has a different constraint: energy, latency, compute, and human attention.

Split computing

Latency-critical stages at the edge; heavy sampling and training in the cloud.

Federated learning

Devices improve a shared model without sending raw data anywhere.

It is a template

Map YOUR stages onto it. Lab 12 asks you to place every Lab 02–11 stage in a layer.

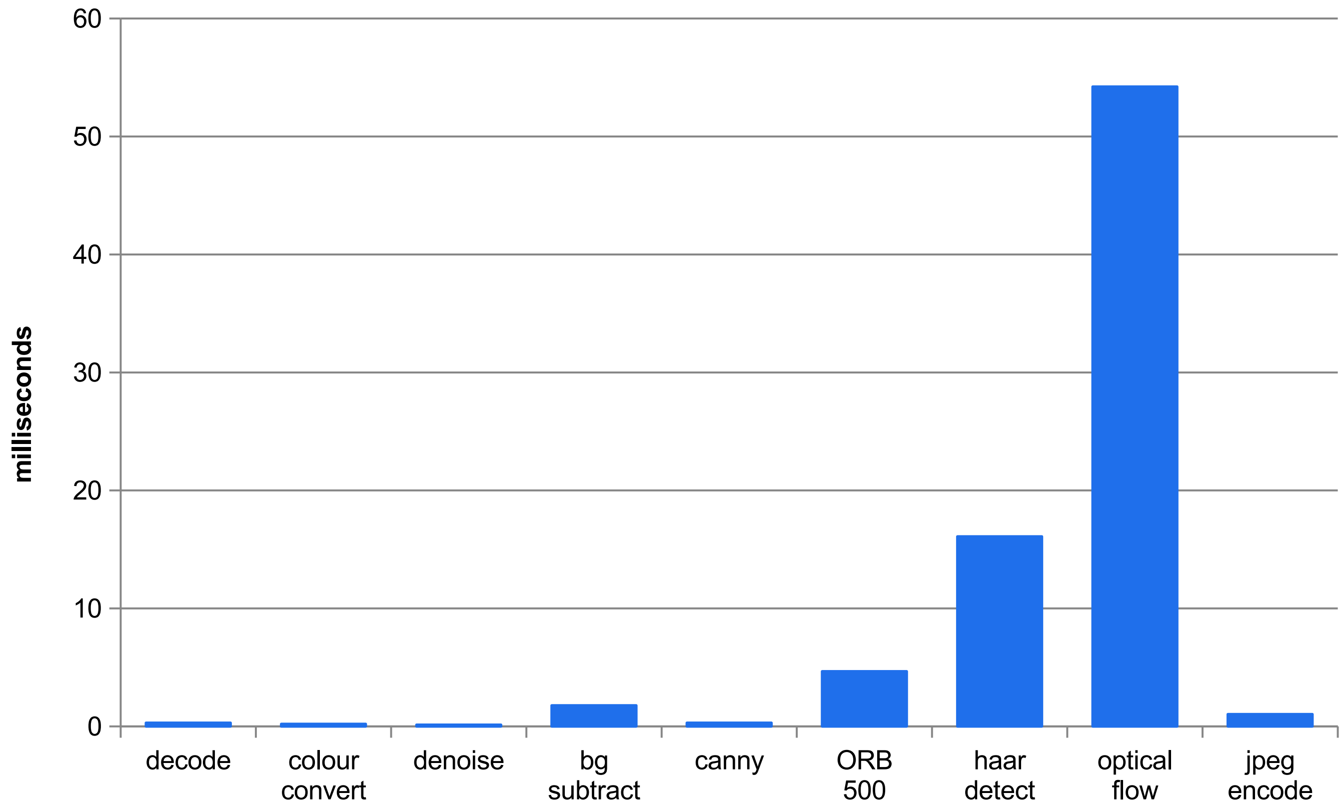
This is the architecture you draw BEFORE measuring anything, and then annotate with the measurements.

Placing Your Own Pipeline Stages

Stage from Labs 02–11	Layer	Why
Decode, colour convert, denoise	2 Edge	Cheap, and it reduces everything downstream
Background subtraction, edges, ORB features	2 Edge	Milliseconds per frame; turns pixels into small payloads
Haar cascade / HOG detection	2 Edge, or 3 if the model is large	Measured at 16.1 ms per frame here; a large detector may not fit the device
Tracking and continuity metrics	2 Edge	Needs every frame, so it must sit where the frames are
Image generation, inpainting, outpainting	3 Cloud	Sampling cost scales with denoising steps; placement depends on measured device capacity
Video generation	3 Cloud	Cost scales with steps × frames — modelled at 200x the image cost
Consistency gate, review queue, asset store	4 Application	Where a human or a policy makes the decision

The placement is a hypothesis until you measure it. Lab 12 measures the edge stages on the machine in front of you and records the CPU alongside the numbers.

Measured — Per-Stage Latency on the Course Pipeline



One stage dominates

Dense optical flow at 54.2 ms is 69% of the 78.6 ms sum of isolated stage medians.

Detection is second

16.1 ms for the Haar cascade — an order of magnitude above the cheap stages.

The cheap stages cost less

Decode, colour convert, denoise and Canny together are under 1 ms.

Budget met, with margin

Isolated medians sum to 78.6 ms versus a 400 ms budget — verify the directly timed end-to-end row and 40 ms frame interval.

MEASURED

Measured in Lab 12 with opencv-python 4.13 on the course development machine, 250 frames, median reported. A latency number without its hardware is not a measurement — your store device will differ, and the harness records the CPU string in the CSV header.

Where Generative Latency Comes From

Image

```
lat ∝ S · c_step
```

S denoising steps, each costing c_step. Under fixed per-step cost, halving steps approximately halves latency — which is exactly what step distillation sells.

Video

```
lat ∝ S · F · c_step
```

S steps for each of F frames. A 25-step, 200-frame clip is 200x the sampling work of a 25-step image. This simplified ratio does not decide edge feasibility.

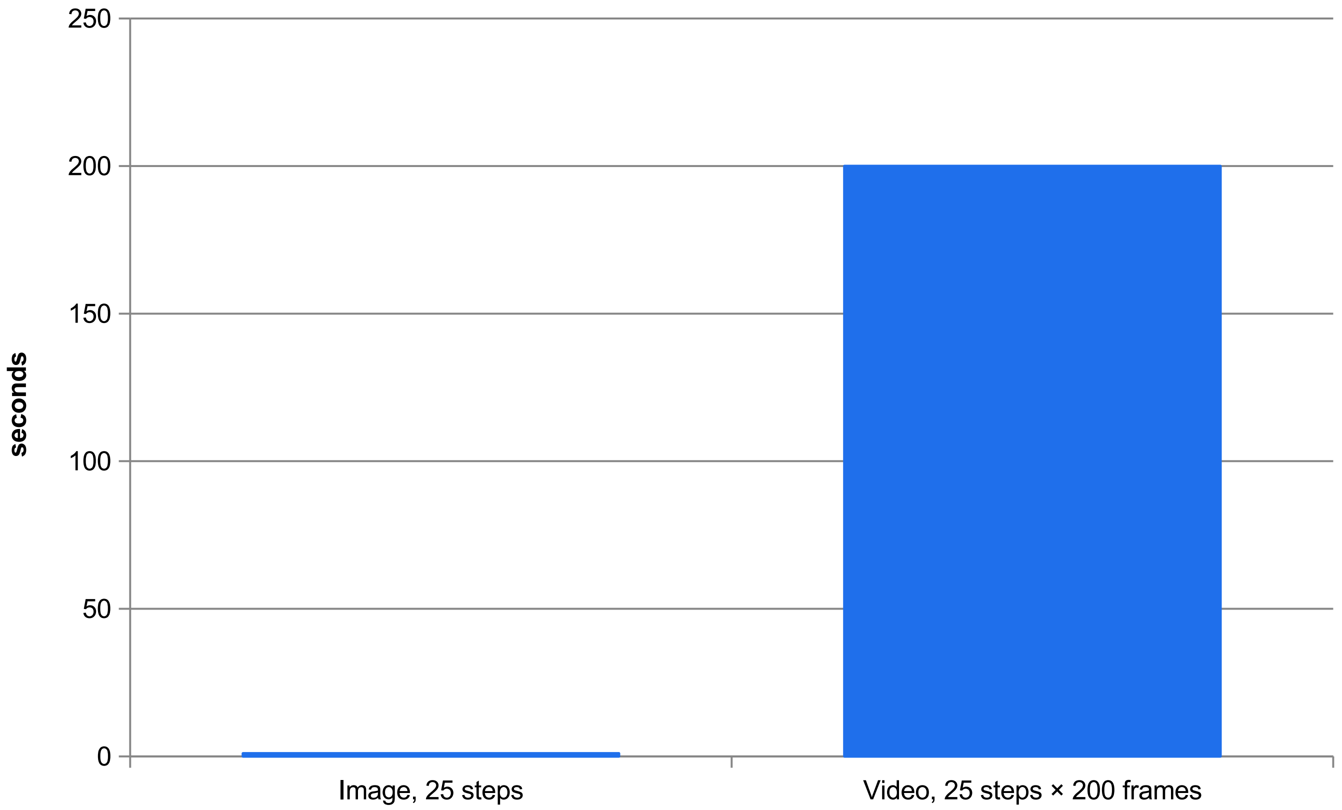
Peak memory

```
M ≈ M_param
  + Σ activations
  (+ optimiser)
```

Parameter memory plus the activation footprint at your batch size and resolution. Optimiser state is present only during training.

Latent diffusion attacks c_step by working on a compressed representation; distillation attacks S; quantisation attacks M_param. Three different levers on three different terms.

Modelled — Image Against Video Sampling Cost



The ratio is F

200 frames means 200x the sampling work at the same step count and step cost.

Steps are the other lever

Distillation reduces S while preserving FID or FVD at fixed guidance.

Temporal modules must stay

For video, the temporal attention or 3-D blocks are retained in the distilled student, or motion falls apart.

Hence the split

Analysis at the edge, generation in the cloud — an arithmetic conclusion, not a preference.

MODELLED

Derived from the documented relationship lat $\propto$ S for images and lat $\propto$ S × F for video, using the classroom parameters in Lab 12's deployment-constraints.json (S = 25, F = 200, 40 ms per step). The RATIO is the finding; the absolute seconds are illustrative.

Vision Communication Protocols — Chosen by Payload and Latency

Protocol	Payload class	Latency class	Typical link
RTSP	Live video	Real-time	Camera → edge node; the standard IP-camera transport
WebRTC	Live video	Interactive, sub-second	Camera or edge → browser preview; NAT traversal built in
MQTT	Small events and telemetry	Seconds	Edge → cloud; publish/subscribe, tiny header, survives a poor link
CoAP	Tiny events	Seconds	Constrained low-power sensor → edge node; UDP-based
HTTPS / REST	Request and reply	Seconds	Application → model endpoint; simple, cacheable, universal
gRPC	Request/reply and streaming	Sub-second	Service → service; binary, typed contracts
HTTPS object storage	Large blobs	Batch	Rendered assets → storage; resumable upload

Every choice must state a payload class AND a latency class. A protocol chosen without reference to payload size is a preference, not a decision. MQTT and CoAP are named in the source as the lightweight sensor-to-edge transports.

The Four Candidate Split Points

A — RAW FRAMES

Cut above decode. The edge does nothing but capture. Payload = W × H × 3 × fps bytes per second, per camera.

→

B — ENCODED VIDEO

Cut above analysis. The edge captures and encodes. Payload = the encoded bit-rate.

→

C — FEATURE VECTORS

Cut above detection. The edge decodes, pre-processes and extracts features. Payload = keypoints × descriptor bytes × fps.

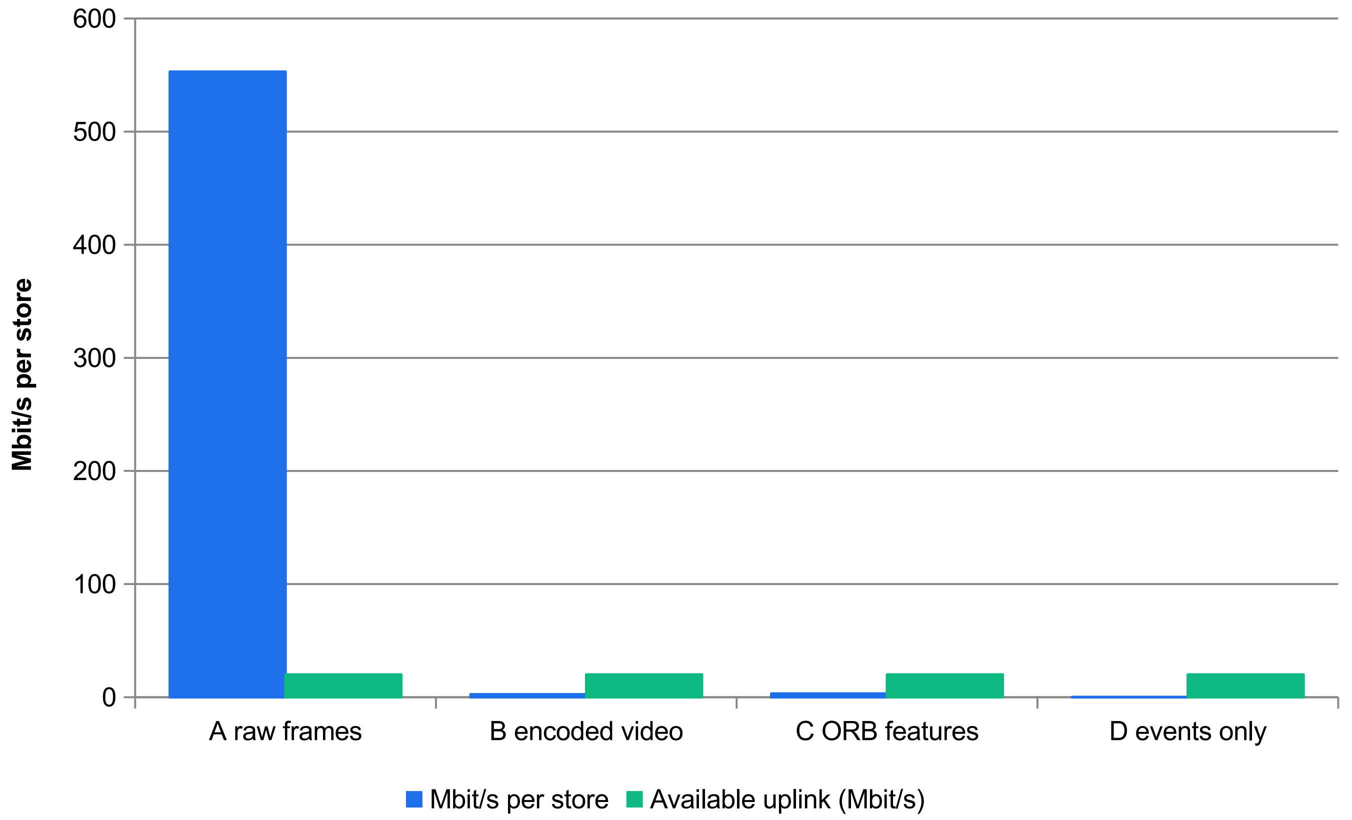
→

D — EVENTS ONLY

Cut above the application layer. The edge runs the whole analytic pipeline. Payload = a few hundred bytes when something happens.

SO WHAT The further right you cut, the less crosses the link and the more the edge device has to do. Split A is included precisely so it can be shown to be arithmetically impossible.

Modelled — Bandwidth per Store at Each Split Point



Split A is impossible

553 Mbit/s against a 20 Mbit/s uplink — 27x over, per camera. It is eliminated by arithmetic, not by opinion.

B and C are comparable

2.5 and 3.2 Mbit/s. Feature vectors are not automatically cheaper than encoded video.

D has much lower payload

0.00032 Mbit/s. Sending events instead of encoded video is about 7,813x lower under these assumptions.

Then apply the privacy gate

The camera feed is classified 'personal', which eliminates A, B and C before cost is even discussed.

MODELLED

Computed in Lab 12 from the measured clip geometry (1280 × 720 × 3 × 25) and the classroom constraint file. The uplink figure is the stated site constraint. Geometry is measured; encoded rate, feature count and event rate are assumptions. Check bits versus bytes yourself.

K13, A8 · THE ORDER OF OPERATIONS

Eliminate on constraint first. Optimise on cost among what survives.

A privacy classification is a gate. A stated uplink is a gate. Only after both have been applied does a latency or cost comparison mean anything — and in this case they leave exactly one candidate.

The Split Decision, Written Down

Split	Mbit/s per store	Feasible on uplink	Privacy gate	Verdict
A raw frames	552.96	NO — 27x over	FAILS — raw frames leave the network	Eliminated twice
B encoded video	2.50	YES	FAILS — identifiable customers leave the network	Eliminated on privacy
C ORB features	3.20	YES	MARGINAL — descriptors are derived, but re-identification risk must be argued	Hold pending assessment
D events only	0.00032	YES	PASSES — no image leaves the store network	SELECTED

Runner-up: split C. The deciding number is the privacy classification, not the 0.7 Mbit/s bandwidth difference — and saying so is what makes the decision defensible six months later.

K13 · SOURCE S15 · CH.3 P.43, TABLE 3.4

Real-World Design Constraints

1

LATENCY

Delay-sensitive applications — AR/VR, autonomous systems, live alerting — set a hard budget. Measure against it; do not assume it.

2

BANDWIDTH

The uplink is fixed and shared. Payload size at the split point is the single biggest lever you control.

3

COMPUTE AND ENERGY

Generative models, especially GANs and diffusion, are computationally heavy and memory-hungry against edge hardware and battery life.

4

PRIVACY AND SECURITY

Adversarial attacks, model and data leakage. Federated learning, differential privacy and secure multiparty computation are the named mitigations.

5

SCALABILITY AND INTEROP

Heterogeneous devices and no standards. Standardised APIs, containerised deployment and trust frameworks are the named directions.

6

ETHICS AND GOVERNANCE

Misuse of generative visuals and lack of explainability. Responsible-AI frameworks, interpretability and transparency.

Scaling Strategies and What Each Targets

Strategy	Target	Typical gain
Trajectory distillation	Diffusion steps S	Fewer sampling steps at similar FID / FVD
Logit distillation	Autoregressive decoding	A smaller decoder at matched negative log-likelihood
Quantisation (8 / 4-bit)	Weights and activations	Reduced latency and memory footprint
Low-rank adapters	Task transfer	Only a small percentage of parameters trainable
Latent diffusion	Resolution scaling	Lower bandwidth and memory cost at high resolution
Retrieval conditioning	Style and domain control	Higher faithfulness without increasing model size

Each of these reports a quality delta as well as a speed-up in the source. A compression lever quoted without its measured quality cost is half a result.

The Two Edge-Fit Levers, With Arithmetic

Quantisation

```
M_param = P · b/8
ratio = b / 32
```

Moving weights from 32-bit to b-bit scales parameter memory by b/32. For a 2.6-billion-parameter model: 10.40 GB at FP32 becomes 2.60 GB at INT8, a ratio of 0.25.

Low-rank adapters

```
ΔW = A·B
A ∈ ℝ^(d×r)
B ∈ ℝ^(r×m)
trainable = r(d+m)
```

Instead of d×m trainable parameters. For a 4096×4096 layer at rank 16: 131,072 instead of 16,777,216 — 0.78%, and the base weights are not duplicated per task.

The stability caveat

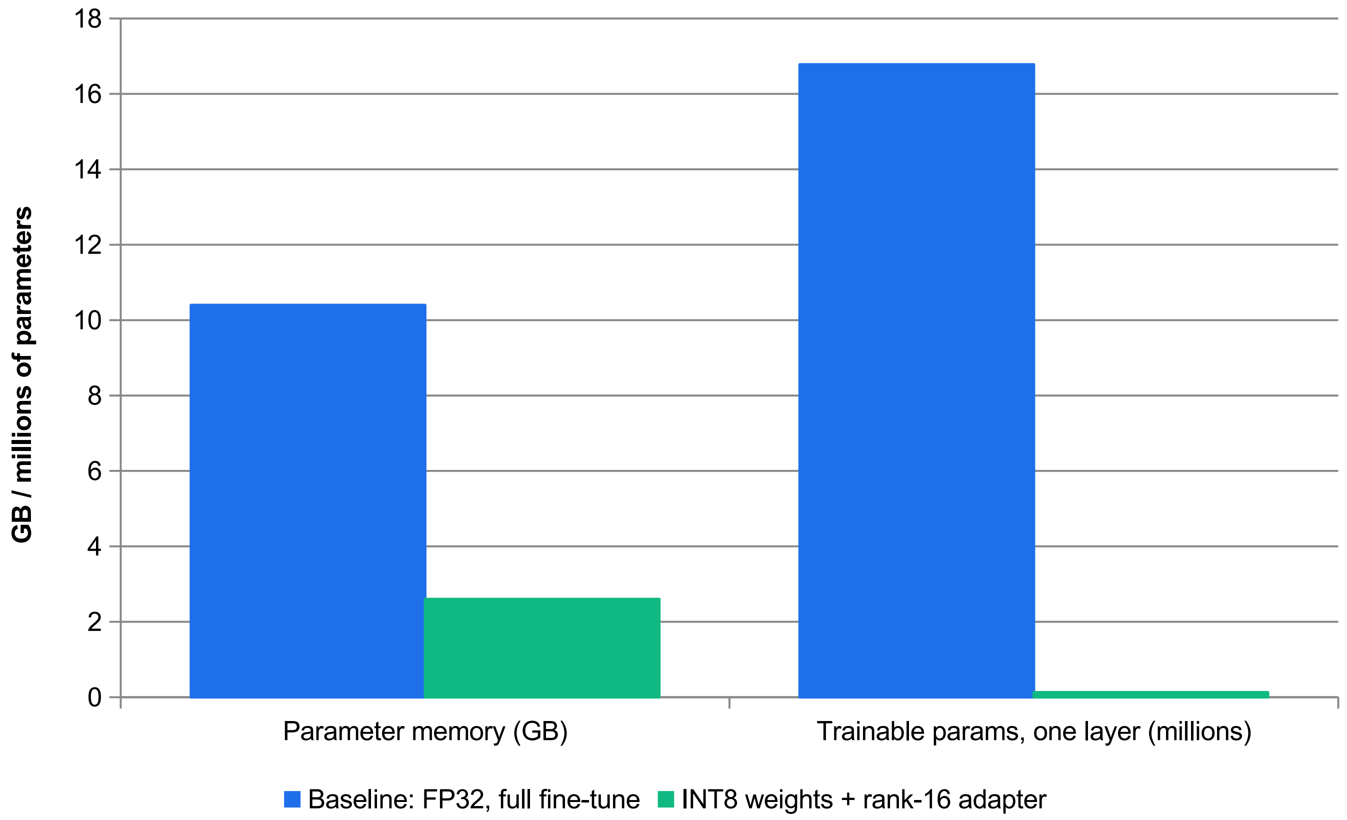
```
keep sensitive
activations at
FP16 / BF16
```

End-to-end INT8 for diffusion is sensitive to guidance and timestep schedules. Practical systems quantise weights and keep a subset of activations at higher precision at the attention bottlenecks.

Both figures are computed in Lab 12 from the classroom constraint file, and both are labelled MODELLED, not measured.

K13 · A 2.6B-PARAMETER MODEL, ONE 4096x4096 LAYER

Modelled — What the Edge-Fit Levers Actually Buy



Memory: 4x smaller

10.40 GB to 2.60 GB. That is the difference between fitting an edge device and not.

Trainable params: 128x fewer

16.78 M to 0.13 M — 0.78% of the full layer.

Per-task loading

Adapters let you swap task specialisation without duplicating the base weights.

Measure the quality cost

Neither number says what accuracy you lost. Report the before/after quality delta or the lever is unproven.

MODELLED

Computed in Lab 12 from the documented relationships: parameter memory scales by b/32, and a rank-r adapter on a d×m layer trains r(d+m) parameters. Model size and rank are classroom values in deployment-constraints.json.

A8 · SOURCE S18 · CH.7 P.100, TABLE 7.4

The Deployment Decision Matrix

Constraint	Recommended stack	Note
Low latency, image	Distilled sampler + INT8 weights	Keep key activations at FP16 for stability
Edge device	INT8 + parameter-efficient adapters	Low-rank adapters on convolution blocks for lightweight adaptation
High resolution	Latent diffusion + few-step sampler	Reduces memory traffic at high output resolutions
Short video	Temporal attention + distilled steps	Check FVD and temporal consistency across frames, not just per-frame quality
Brand / style control	Retrieval conditioning + filters	Cross-attention to exemplar images for style control
Tight privacy	Local open model + provenance tracking	On-premises logging and verifiable provenance

Apply this to each of your accepted use cases from Lab 01. The matrix is a starting point that still has to be justified against your measurements.

Energy and Carbon Are Reportable Engineering Numbers

Energy per sample

```
E ≈ κ · η · FLOPs

κ = overhead
η = J per FLOP
```

Datacentre overhead factor times energy per FLOP times FLOPs per sample. The Lab 12 classroom values give 9,800 J per generated sample — labelled ILLUSTRATIVE, and not a figure to quote.

What a real figure needs

```
hardware inventory
count + utilisation
total kWh
PUE or κ
grid CI (kgCO₂e/kWh)
joules per sample
```

Six items. Publishing energy per sample without them is a number, not a measurement.

What reduces it

```
↓ FLOPs/sample
↓ memory traffic
```

Sampler distillation, quantisation and latent-space modelling all reduce one or both, so they reduce E as a side effect of reducing cost.

Also report the sampler configuration (S, guidance), the precision and the adapter rank alongside the energy figure — otherwise it cannot be reproduced.

Security Risks Specific to Multimodal Systems

1 **Multimodal prompt injection**
Instructions hidden in an IMAGE rather than in the text field, reaching a model that reads both.

2 **Steganography**
A payload concealed inside image data, invisible to a human reviewer and carried through the pipeline intact.

3 **Side-channel attacks**
Inferring protected information from timing, size or other observable behaviour rather than from the output.

4 **Data leakage into a vendor**
The named real-world example is the Samsung leak: confidential material pasted into an external assistant.

5 **Tailor the architecture**
Match the deployment to your data-protection obligations. That is an architecture decision, not a policy document.

6 **Sanitise and filter**
Input filters, retrieval sanitation, policy-constrained decoding and output classifiers, re-audited after every model or guardrail change.

K11, K13 · SOURCES S12, S17

Governance That Travels With the Asset

1

C2PA Content Credentials

Tamper-evident metadata attached to the file: origin, creator and edit history, verifiable by a consumer or a platform.

2

Robust watermarking

An invisible signal in the content itself, designed to survive compression, resizing and format change. Test detectability under YOUR edits.

3

Model cards

A standardised 'nutrition label for AI': version, architecture, developer, intended use, known limitations.

4

Dataset governance

Licences, consent and attribution recorded for every training and conditioning input. Outputs must respect source licences.

5

Red-teaming

Adversarial prompt sets probing toxicity, bias, impersonation and unsafe content — re-run after every model, data or guardrail change.

6

Provenance in the pipeline

Every asset carries model ID, prompt version, watermark expectation and reviewer. This is the same discipline the labs enforce.

A7 · THE OUTPUT OF LAB 12, ANNOTATED WITH MEASUREMENTS

The Reference Architecture You Will Defend

STORE (edge)	Camera → decode → pre-process → detect → track → EVENT. Nothing leaves the store network but a few hundred bytes per event.
REGIONAL GATEWAY	MQTT broker aggregating events from 41 stores; buffers through link outages; no image payload.
CLOUD — MODELLING	Generation, heavy editing and training. Consumes approved catalogue assets, never store camera frames.
CLOUD — APPLICATION	Consistency gate, review queue, asset store with provenance records, and the dashboards the business actually looks at.

Measured
78.6 ms end-to-end edge latency against a 400 ms budget.

Computed
0.00032 Mbit/s per store; 0.0000131 Gbit/s across the estate.

Gated
Privacy classification eliminates every split that moves a frame.

Residual risk
Measured on developer hardware; event rate assumed; the model may be replaced.

Every figure on this slide is labelled MEASURED, COMPUTED or ASSUMED. An unlabelled figure in an architecture document is a defect.

A8 · A DECISION WITHOUT ITS RESIDUAL RISKS IS NOT FINISHED

Residual Risks You Must State

1 **Wrong hardware**
The latency was measured on a developer laptop, not on the store device. Close it by re-running the harness on the target.

2 **Assumed event rate**
The bandwidth model uses an assumed 6 events per minute. Close it with a two-week measurement in three representative stores.

3 **The model will change**
Latency and memory terms are model-specific. Close it by re-running the arithmetic whenever the model is swapped.

4 **Illustrative cost**
Built on a classroom per-GB rate. Close it with current vendor pricing and a measured egress volume.

5 **Re-identification risk**
Feature vectors are derived, but a re-identification argument must be made, not assumed. Close it with a privacy assessment.

6 **Synthetic evaluation**
The detector numbers were measured on synthetic frames. Close it by labelling a real sample and re-measuring.

K11, K13 · SOURCE S15 · CH.3 PP.38–39, 43

Federated Learning — Improve the Model, Keep the Data

LOCAL TRAINING

Each edge node trains on its own data. The frames never leave the store network.

SEND UPDATES

Only model updates — gradients or weight deltas — cross the link. Orders of magnitude smaller than the data that produced them.

AGGREGATE

The server averages the updates into an improved shared model, without ever seeing an individual example.

REDISTRIBUTE

The improved model goes back to every node, and the cycle repeats.

SO WHAT It is a privacy-preserving architecture, not a privacy guarantee. Updates can leak information about the data that produced them, which is why differential privacy and secure multiparty computation are named alongside it rather than instead of it.

Edge-AI Generative Systems Across Domains

Domain	Use case	Models applied	The edge advantage
Healthcare	Synthetic medical images, denoising scans, anomaly detection	GANs, VAEs	Privacy-preserving; patient data never reaches a third-party cloud
Smart cities	Traffic prediction, surveillance synthesis, digital twins	GANs, diffusion	Real-time insight and urban simulation without shipping every frame
AR / VR	Realistic textures, avatars, immersive environments	GANs, diffusion	Low-latency rendering that keeps up with user interaction
Industrial IoT	Predictive maintenance, defect synthesis, inspection	VAEs, GANs	Faster decisions and bandwidth-efficient operation
Agriculture	Crop monitoring, climate prediction, image super-resolution	VAEs, diffusion	Precision farming with localised analytics

Every row is the same architectural argument in a different industry: the generative stage sits where the constraint — privacy, latency or bandwidth — says it must.

K11, K13 · SOURCE S15 · CH.3 PP.43–44, TABLE 3.4

Scalability and Interoperability — the Unglamorous Half

1 **Standardised APIs**
Thousands of heterogeneous devices need one contract. Without it, every device type becomes its own integration project.

2 **Containerised deployment**
Ship the model and its runtime together. This is what makes a model update a deployment rather than a site visit.

3 **Device synchronisation**
Time, model version and configuration must agree across the estate, or your measurements are not comparable between stores.

4 **Trust frameworks**
Named directions include blockchain-based approaches for verifying which device produced which artefact.

5 **Hardware-software co-design**
Neuromorphic processors and FPGA accelerators are the named routes to latency without losing generative quality.

6 **Version everything**
Model ID, adapter, sampler configuration and prompt version. A measurement without them cannot be reproduced next quarter.

K13, A8 · 41 STORES, 340 OPERATING HOURS/MONTH, S$0.16/GB

An Illustrative Cost Model, and Why It Is Labelled

Split	Bytes/s per camera	Monthly GB per camera	Estate monthly cost
A raw frames	69,120,000	84,602.9	S$554,798 — and it does not fit the uplink anyway
B encoded video	312,500	382.5	S$2,509
C ORB features	400,000	489.6	S$3,212
D events only	40	0.049	S$0.32

ILLUSTRATIVE. Built on the classroom unit rate in deployment-constraints.json and an assumed event rate. Not a quotation and not a forecast. A real figure requires current vendor pricing and a measured event rate — which is one of the residual risks you must state.

K13, A8 · A BUDGET IS A SUM OF TERMS, NOT A SINGLE NUMBER

Decomposing a Latency Budget

The terms

- **Where the milliseconds go**

– Capture and decode: fixed by the camera and codec.

– Edge processing: COMPUTED — 78.6 ms sum of isolated stage medians; time end-to-end separately.

– Network transit: round-trip time plus serialisation of the payload.

– Queueing: the part everyone forgets, and the part that dominates under load.

– Model inference: S x c_step for an image; S x F x c_step for video.

– Application and display: rendering the alert a human sees.

The discipline

- **How to use the decomposition**

– Measure what you can measure, and label the rest MODELLED.

– Attack the largest term first. Here that is dense optical flow at 54.2 ms — 69% of the edge total.

– Report a percentile, not a mean. A p95 of 81.7 ms is the number the user experiences on a bad frame.

– State the hardware. The same code on a store device will not produce these numbers.

Lab 12's harness discards the first five frames, reports median and p95, and writes the CPU string into the CSV header — because a latency number without its hardware and its percentile is not a measurement.

A7, A8 · THE FORMAT YOUR DECISION HAS TO SURVIVE IN

The Architecture Decision Record

1

CONTEXT

The estate, the uplink, the privacy classification and the latency budget. Written before any option is discussed.

2

CONSTRAINT GATES

Applied FIRST, not weighted. Which options does privacy eliminate? Which are arithmetically infeasible on the uplink?

3

MEASURED / MODELLED / ILLUSTRATIVE

Three separate sections. Mixing them in one table is the mistake this whole topic exists to prevent.

4

DECISION AND RUNNER-UP

Name the chosen split, name the runner-up, and state the single number that decided between them.

5

CONSEQUENCES

What becomes easy, what becomes hard, and what now has to be built that would not otherwise exist.

6

RESIDUAL RISK

At least three, each with the specific action that would close it. A decision without its residual risks is not finished.

TOPIC 05 · ELO6 · A7, A8

Lab 12 — Cloud-Edge Architecture Design and Measured Trade-offs

INPUTS

Synthetic 10-second 1280x720 25 fps clip used as the pipeline input for timing (deterministically rendered with OpenCV — SIMULATED, not generative-model output)

TECHNICAL QUESTION

Can the architecture meet both latency and throughput targets within the uplink budget?

EVIDENCE TO PRODUCE

`out/stage-latency.csv` (measured, not estimated), `out/bandwidth-model.csv` for four upload strategies at 41 stores, `out/split-decision.md` naming the chosen split point with its latency, bandwidth, privacy and cost consequences, and a labelled four-layer architecture diagram.

LAB 12 · 60 MIN · Python 3 · opencv-python · NumPy · offline with prepared dependencies; no paid service

Detailed procedures and prompts: Learner Guide / labs/lab-12-cloud-edge-architecture-tradeoffs/README.md

TOPIC 05 · ELO6 · K11, K12, K13, A7, A8

Recap — Topic 05

- A generative visual system has four layers: IoT sensing, edge intelligence, generative visual modelling, and application. Place your own stages on it.
- Split computing is the real architecture decision, and you choose the split point by measuring per-stage latency and the payload that crosses it.
- Protocols are chosen by payload class and latency class: RTSP and WebRTC for live video, MQTT and CoAP for events, HTTPS/REST and gRPC for model calls.
- Diffusion latency is proportional to S for images and to S x F for video — modelled at 200x for a 200-frame clip, which is why video generation is not an edge workload.
- Measured: raw frames need 553 Mbit/s per store against a 20 Mbit/s uplink, and events need 0.00032 — a decision of three orders of magnitude.
- INT8 quantisation cuts parameter memory to a quarter; a rank-16 adapter trains 0.78% of a layer's parameters. Both are modelled, and both need a measured quality cost.
- Eliminate on constraint first — privacy is a gate, not a weight — then optimise on cost among what survives, and state your residual risks. (A7, A8, K11, K12, K13)

IMAGEGEN · ACTUAL BEFORE / AFTER

A colour edit can change more than the bottle

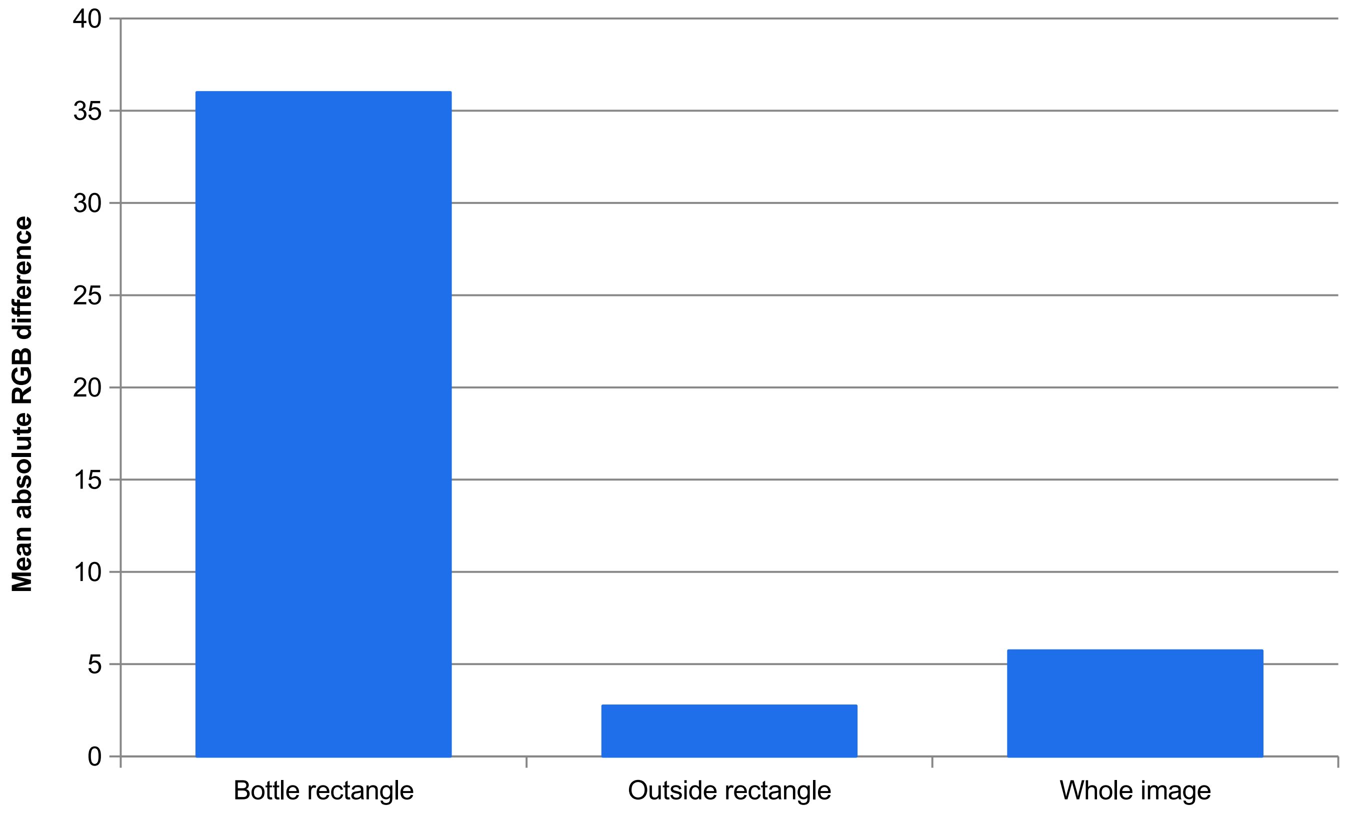
Original generated reference

Edited: bottle body requested in teal

Compare the cap, leaf, shadows and background as well as the requested colour. A correct-looking subject does not prove preservation elsewhere.

IMAGEGEN · PRESERVATION MEASUREMENT

Where did the edited pixels change?



35.99 inside
The requested colour change dominates the bottle rectangle.

2.74 outside
Other pixels also changed. Inspect their visual importance.

A limited metric
Pixel error measures change, not semantic or perceptual quality.

MEASURED

Same-size original/edit pair; RGB difference on a 0–255 scale. Rectangle is approximate, not a segmentation mask.

A realistic reference tests the detector differently

2 fictional adults

Green boxes: approximate manual face annotations.

Evaluate class, IoU, unmatched boxes and misses. Compare with the schematic reference set.

One synthetic image cannot establish real-world accuracy.

Lab08 positive-control folder · AI-generated people · manual truth is separate from detector predictions

WRAP-UP

What You Achieved, and What Comes Next

LEARNING OUTCOMES DELIVERED

What You Achieved

1

LO1 · Vision system concepts and applications

Scored six candidate use cases against measurable need criteria and specified the five-stage pipeline for the survivors. (Lab 01)

2

LO2 · Image processing

Read image and video data correctly, built masks, ran controlled edits, and ranked five filters with PSNR and SSIM. (Labs 02–05)

3

LO3 · Feature extraction

Extracted colour, edge and keypoint features, and audited a 12-asset campaign with global descriptors and template matching. (Labs 06–07)

4

LO4 · Machine-learning computer vision

Ran two built-in ML detectors, implemented IoU, and reported precision, recall and F1 at three thresholds. (Lab 08)

5

LO5 · Video analytics

Specified a three-shot film with a continuity bible, measured tracking jitter and scale drift, and shipped a verified export ladder. (Labs 09–11)

6

LO6 · Edge versus cloud evaluation

Measured per-stage latency, modelled four split points and defended one with bandwidth, privacy and cost numbers. (Lab 12)

EVERY EXTERNAL CLAIM IN THIS DECK IS TRACEABLE

Source Register — Primary Sources Used (1 of 2)

Ref	Source	Locator
S1	Course Proposal Tertiary Infotech CA-WSQ-2020-013290-v2 (SSG-approved)	Part 4 Section E, Curriculum Key Features table
S2	Assessment Plan_OpenCV_TGS-2020505925_V1.0 (28 Dec 2020)	Assessment Duration table; WA and PP specification table
S3	Competency Mapping.docx (TSC ICT-DIT-4022-1.1)	Course Mapping section and ELO/A-code matrix
S4	Computer Vision Technology.docx (Skills Framework TSC extract)	TSC Proficiency Level 4 (ICT-DIT-4022-1.1) knowledge and
S5	Tertiary Courses public registration page	https://www.tertiarycourses.com.sg/wsq-generative-ai-for
S6	M. Tschochohei and F. Schenker, 'Multimodal Generative AI in the Enterpris	Ch.5 pp.56-62
S7	Tschochohei and Schenker (2026), as above	Ch.5 pp.65-67
S8	Tschochohei and Schenker (2026), as above	Ch.5 pp.68-75
S9	Tschochohei and Schenker (2026), as above	Ch.5 pp.76-77 and pp.82-83
S10	Tschochohei and Schenker (2026), as above	Ch.7 pp.106-109
S11	Tschochohei and Schenker (2026), as above	Ch.7 pp.110-119
S12	Tschochohei and Schenker (2026), as above	Ch.8 pp.127-132 and pp.140-143

The full register, including exactly what each source anchors and where it is used, is reproduced in the Learner Guide.

INCLUDING THE VENDOR DOCUMENTATION VERIFIED ON 6 SEPTEMBER 2026

Source Register — Primary Sources Used (2 of 2)

Ref	Source	Locator
S13	Tschochohei and Schenker (2026), as above	Ch.9 pp.150-152
S14	A. Mewada, M. A. Ansari, S. Ahmad and N. Singh (eds), 'AI-Generated Image	Ch.1 pp.1-11
S15	Mewada et al. (2026), as above	Ch.3 pp.35-44 (Vishnoi, Sisodia, Khare and Yadav, 'Senso
S16	Mewada et al. (2026), as above	Ch.4 pp.47-54 (Sibi, Pantola and Gupta, 'Detecting AI-ge
S17	Mewada et al. (2026), as above	Ch.7 pp.94-98 (Dubey, Pinheiro, Singh, Ansari and Kumar,
S18	Mewada et al. (2026), as above	Ch.7 pp.98-100
S19	Mewada et al. (2026), as above	Ch.11 pp.147-160 (Dewang, Mewada, Omkar, Jaiswal and Sin
S20	Google AI for Developers — Gemini API image generation documentation	https://ai.google.dev/gemini-api/docs/image-generation (
S21	Google AI for Developers — Gemini API Imagen documentation	https://ai.google.dev/gemini-api/docs/imagen (verified 6
S22	Google AI for Developers — Veo video generation documentation	https://ai.google.dev/gemini-api/docs/veo (verified 6 Se
S23	OpenCV 4.13 Python package, verified locally on the delivery image	cv2 4.13.0 (opencv-python 4.13.0.92)

Model identifiers change. S20, S21, S22 and S23 record what the official documentation said on 6 September 2026, including which identifiers are deprecated.

CONTINUE THE JOURNEY

Recommended Follow-On Courses

1. **Python Fundamental Course for Beginners**
   WSQ funded · Tertiary Infotech Academy

2. **Basic Machine Learning with ScikitLearn Course**
   WSQ funded · Tertiary Infotech Academy

3. **Deep Learning with Tensorflow Keras**
   WSQ funded · Tertiary Infotech Academy

4. **Generative AI for Business Productivity**
   WSQ funded · Tertiary Infotech Academy

5. **Data Visualisation and Storytelling**
   WSQ funded · Tertiary Infotech Academy

WE'RE HERE TO HELP

Support

- If you have any enquiries during or after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Telephone: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- Course page: https://www.tertiarycourses.com.sg/wsq-generative-ai-for-image-and-video-creation.html

FINAL ASSESSMENT

Assessment

- Written Assessment WA(Q&A) — 13 open-ended questions covering K1 to K13, 60 minutes.
- Practical Performance (PP) — 5 scenario-based tasks covering A1 to A8, 90 minutes.
- Open book: slides, Learner Guide and approved course materials only.
- Remember to take the Assessment digital attendance before you start.
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/
- Sign the Assessment Summary Record when your assessor has completed it.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA(Q&A) 60 min, then PP 90 min — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator displays the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

SEE YOU IN THE NEXT COURSE

Thank You!

You can now specify a generative image or video job, run the underlying computer vision that verifies it, and defend where it should run.